

VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS



FINAL PROJECT REPORT
ENTERPRISE RESOURCE PLANNING

TOPIC:

ERP APPLICATION INTO
ELECTRONICX BUSINESS MODEL

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HO CHI MINH CITY, MAY 2024

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the page.

Ho Chi Minh City, June 12, 2024

ASSIGNMENT AND MEMBER EVALUATION TABLE

Table 1: Assignment and member evaluation table

Name	Student ID	Assignment	Evaluation
Pham Thi Thuy Trang	21522697	• Make a meeting to choose and scope the project. Divide tasks for teammates • Check the data after teammate crawl • Research the contents of project to report • Do the report • Do the PowerPoint • Presenter	100%
Pham Quoc Anh	20521077	• Design Odoo • Do the report • Do the PowerPoint • Present the demo	100%
Nguyen Quoc Tuan	19522476	• Crawl and clean the data • Do the report • Do the PowerPoint • Prepare the question	100%

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HISTORY CHANGE

Actor	Day implement	Contents	Headings
Pham Thi Thuy Trang	20/05/2024	• Init the report • Format the report	
Pham Thi Thuy Trang	20/05/2024	• Project Summary	
Pham Thi Thuy Trang	21/05/2024	• ERP project overview • Project scope • Project objectives • Report layout	1.1 1.2 1.3 1.6
Pham Quoc Anh	21/05/2024	• Master data	1.4
Nguyen Quoc Tuan	21/05/2024	• Stakeholders and key personnel	1.5
Pham Thi Thuy Trang	25/5/2024	• ERP introduction	2.1
Nguyen Quoc Tuan	25/05/2024	• Deployment platform introduction	2.2
Pham Quoc Anh	25/05/2024	• Description of the problem applying ERP to the project topic (or Case study)	2.3
Pham Thi Thuy Trang	27/05/2024	• System overview • Inventory receiving process • Customer debts on time handling process	3.1 3.4 3.9
Nguyen Quoc Tuan	27/05/2024	• Procurement process • Sales process	3.2 3.3

		• Inventory shipping process • Customer debts delay handling process	3.5 3.10
Pham Quoc Anh	27/05/2024	• Financial and Accounting management process • Customer service process • Human resources management process • Supplier debts handling process	3.6 3.7 3.8 3.11
Pham Thi Thuy Trang	31/05/2024	• Advantages • Limitations • Future directions	4.1 4.2 4.3

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LIST OF ABBREVIATIONS

No.	Abbreviation	Content
1	ERP	Enterprise Resource Planning
2	CRM	Customer Relationship Management
3	RFQ	Request for Quotation
4	HR	Human Resources
5	BPMN	Business Process Model Notation
6	SCM	Supply Chain Management
7	FAM	Financials Management
8	MES	Manufacturing Execution System
9	HRM	Human Resource Management

PROJECT SUMMARY

First of all, our team would like to express our deepest thanks and gratitude to lecturer Dr. Tran Van Hai Trieu and Engineer. Le Vo Dinh Kha who taught and shared a lot of knowledge as well as practical examples in lectures. He guided us on topics, corrected and contributed many valuable ideas to help us successfully complete our subject reports.

Enterprise Resource Planning is an interesting, extremely useful and practical subject. However, due to limited professional knowledge and the ability to absorb reality, there are many surprises. Although we have tried our best, it is certainly difficult for the report to avoid shortcomings and many inaccuracies, we look forward to receiving your suggestions and further guidance to improve our knowledge so that our team can use it as a luggage to carry out other topics in the future as well as in later study and work.

Once again, our team would like to send our teachers and friends our sincere and best thanks!

Ho Chi Minh City, June 03, 2024

Student Implementation Team

Pham Thi Thuy Trang

Pham Quoc Anh

Nguyen Quoc Tuan

CHAPTER 1. GENERAL INTRODUCTION

In the first chapter, a brief overview of the ERP project using Odoo is presented. This includes setting the objectives and proposing solutions for the identified problems. The target audience and stakeholders involved in the project are also identified.

1.1. ERP project overview

The company information:

- Store name: Electronics Store – Kitchen Furniture Golden Star
- Head Office: 01 Han Thuyen Street, Quarter 6, Linh Trung Ward, Thu Duc City, Ho Chi Minh City.
- Director: Pham Thi Thuy Trang
- Telephone: 0968666888
- Year of establishment: 2021
- Fanpage: Electronics Store – Kitchen Furniture Golden Star

Introduction:

Golden Star Electronics is a wholesaler and retailer in Ho Chi Minh. Golden Star Electronics specializes in trading consumer electronics products (electronics, refrigeration and home appliances) of many big brands such as SONY, PANASONIC, TOSHIBA, SAMSUNG,... With a source of high quality products – diverse designs – designs, reasonable prices to satisfy the requirements of customers inside and outside the province.

Golden Star Electronics specializes in distributing televisions, air conditioners, stoves, refrigerators, washing machines, ... with 23 different product categories.

1.2. Project scope

Table 1: Project scope

No	Service	The scope of the project
1	Manage sales and distribution of construction materials	Yes
2	Management of purchasing construction materials	Yes
3	Accounting management and financial analysis	Yes

4	Warehouse management and logistics operations	Yes
5	Track potential management and opportunity closure (CRM) processes	Yes
6	Production Management	No
7	Employee management and evaluation	No
8	Control employee working time	No
9	Monitor equipment and manage maintenance requests	No

1.3. Project objectives

- Build a complete technology business model and apply it to practical needs.
- Understand the overall role of ERP system in business administration, know how to implement an ERP project.
- Understand the arising and interactions between components in the system and the professional meaning of each component.
- Improve the ability to organize, manage, supervise, and execute projects as well as soft skills such as teamwork, presentation, communication, information search and report writing skills.

1.4. Master Data

- About more 1000 products

ACCT5123 - Enterprise resource planning

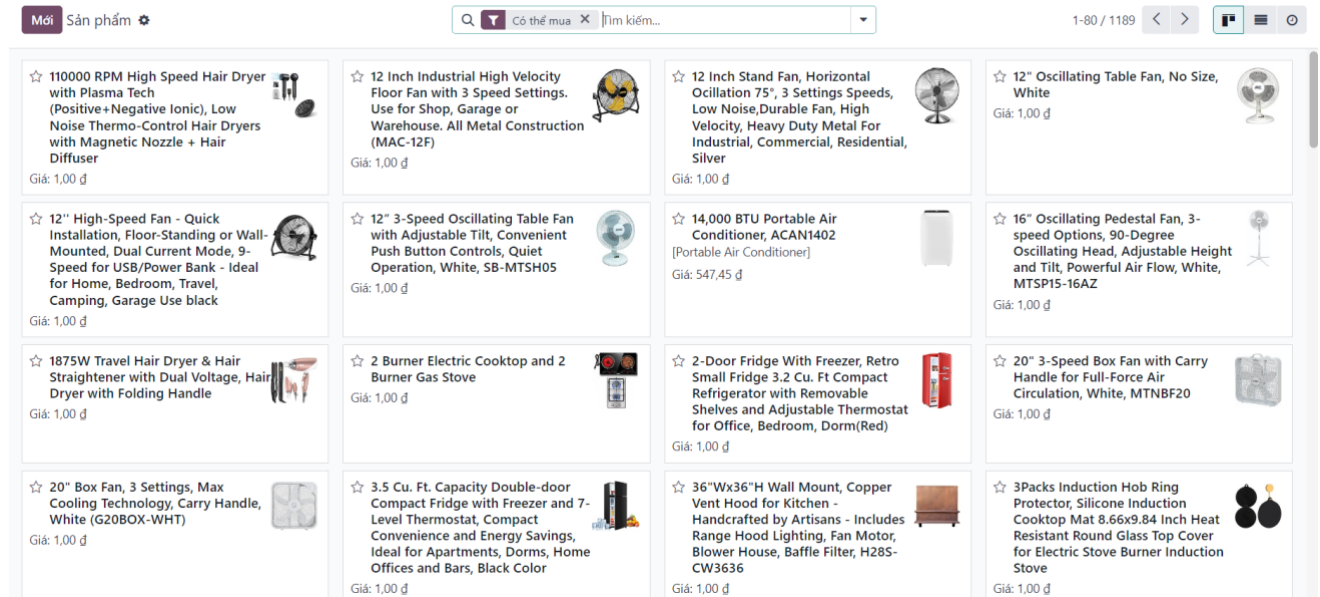


Figure 1: Image 1.4.1: Master data 1000 products

- About more 200 suppliers

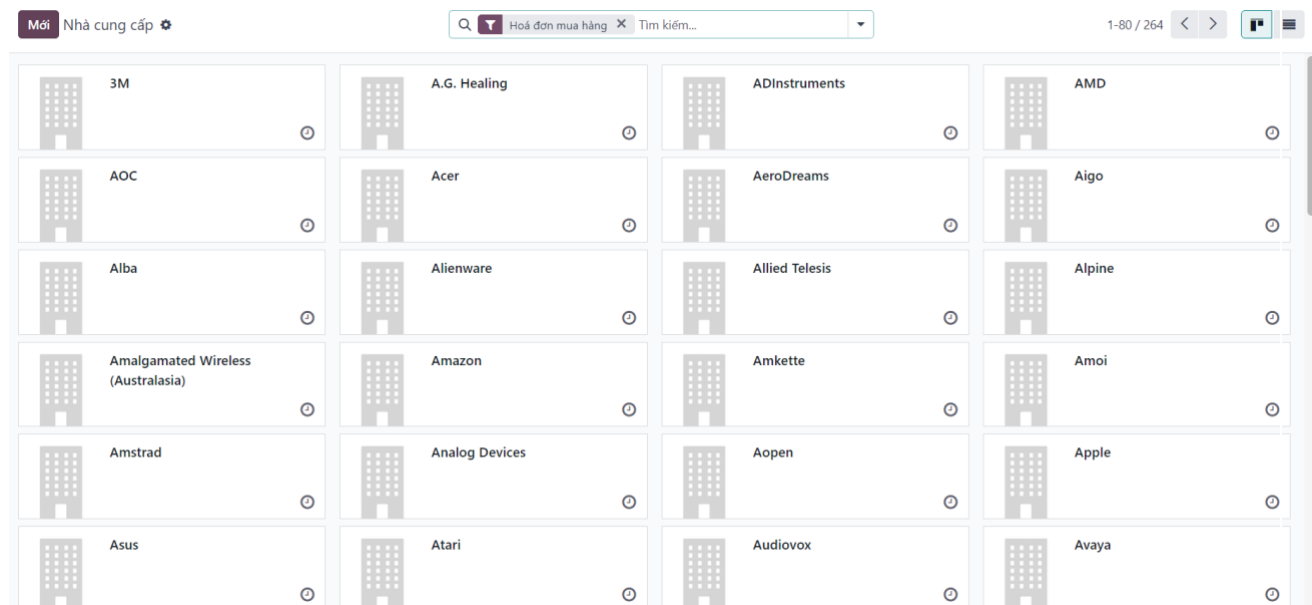


Figure 2: Image 1.4.2: Master data 200 suppliers.

- About 1000 customers

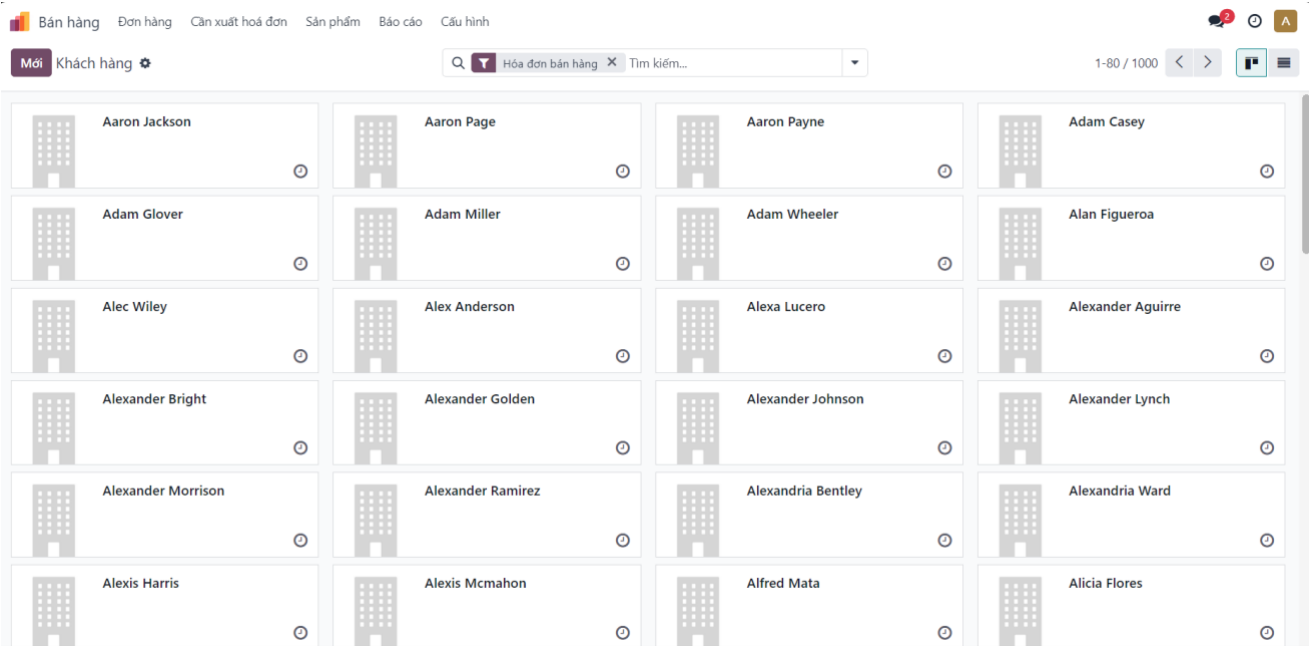


Figure 3: Image 1.4.3: Master data 1000 customers.

- With 5 users

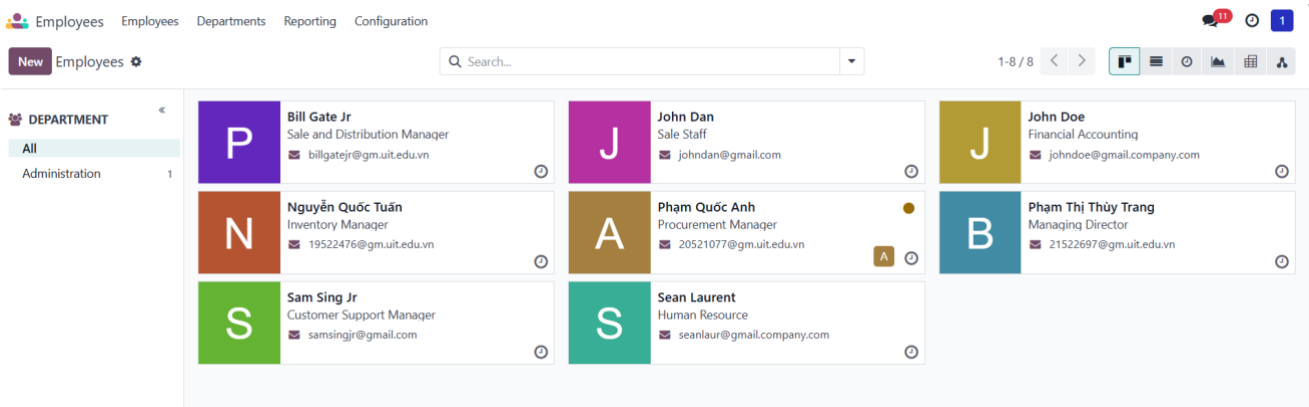


Figure 4: Image 1.4.4: Master data 5 users.

1.5. Stakeholders and key personnel

Table 2: Stakeholders and key components

No.	Full name	Department	Roles and responsibilities	Contact
1	Pham Thi Thuy Trang	Director	• Title: Managing Director. • Develop and implement the company’s long-term strategy. • Monitor and ensure company’s operation and finance.	21522697@gm.uit.edu.vn

			• Provide leadership to managers, leaders, and all staffs.	
2	Bill Gate Jr	Sale and Distribution	• Title: Sales and Distribution Manager. • Design and implement effective sales strategies to achieve sales target. • Manage the sales team and ensure the team's performance to achieve target revenue.	bgjr@gmail.company.com
3	Pham Quoc Anh	Procurement	• Title: Procurement Manager. • Ensure the timely availability, best possible cost, and quality of goods. • Manage and assess suppliers.	20521077@gm.uit.edu.vn
4	Nguyen Quoc Tuan	Inventory	• Title: Inventory Manager. • Ensure that production runs efficiently and succeeds in meeting customer demand and delivery deadlines. • Collaborate with procurement department to plan and allocate resources efficiently.	19522476@gm.uit.edu.vn
5	John Doe	Financial Accounting	• Title: Financial Accounting Manager.	johndoe@gmail.com

			• Prepare and review financial statements, including balance sheets, income statements, and cash flow statements, ensuring accuracy and compliance with accounting standards. • Conduct financial analysis to provide insights about company's state. • Ensuring tax compliance.	
	Sam Sing Jr	Customer Support	• Title: Customer Support Manager. • Handling customer complaints. • Maintaining relationship with customers.	
6	Sean Laurent	Human Resources	• Title: Human Resources Manager. • Managing recruitment and employment process. • Provide training programs to enhance employee skills and performance. • Develop and maintain HR policies and regulations within company.	seanlaurent@gmail.com

1.6. Report layout.

The report is divided into 04 chapters as follows:

- **Chapter 1: General Introduction:** In the first chapter, a brief overview of the ERP project using Odoo is presented. This includes setting the objectives and proposing solutions for the identified problems. The target audience and stakeholders involved in the project are also identified.
- **Chapter 2: Theoretical Overview:** Chapter 2 provides a comprehensive discussion on the theoretical aspects of ERP systems. This includes the definition, outstanding features, software modules, benefits, and limitations of ERP systems. Additionally, it gives an overview of the deployment platform, Odoo, including its history, advantages, and popular modules.
- **Chapter 3: Business Process System:** In this chapter, the focus is on various business processes and how they are managed within Odoo. It includes detailed descriptions and illustrations of processes such as procurement, sales, inventory receiving and shipping, financial and accounting management, customer service, human resources management, and handling customer and supplier debts. Each process is discussed in terms of BPMN (Business Process Model and Notation), specifications, and practical illustrations on Odoo.
- **Chapter 4: Conclusions and Future Directions:** The final chapter summarizes the advantages and limitations of implementing an ERP system using Odoo. It also provides insights into potential future directions for further improving the system and expanding its capabilities.

By following this layout, the report systematically addresses the essential aspects of ERP implementation with Odoo, providing both theoretical background and practical insights.

CHAPTER 2: THEORETICAL OVERVIEW

Chapter 2 provides a comprehensive discussion on the theoretical aspects of ERP systems. This includes the definition, outstanding features, software modules, benefits, and limitations of ERP systems. Additionally, it gives an overview of the deployment platform, Odoo, including its history, advantages, and popular modules.

2.1. ERP introduction

2.1.1. Definition:

ERP abbreviated as Enterprise Resource Planning is an enterprise resource planning system, which is an enterprise resource management software, allowing businesses to control their own resources. From there, come up with plans to exploit resources reasonably from the business processes built into the system.

- **E (Enterprise):** Enterprise is the goal of ERP, how to combine all departments, all business functions of the business into a single computer system that can meet all the different management needs of the departments.
- **R (Resource):** Resource means resources such as finance, personnel, technology, hardware, data, information, ... Therefore, when applying ERP, it is necessary to turn these resources into high-value resources for businesses.
- **P (Planning):** We must calculate, plan, and report the possibilities arising in the process of operating, producing, and doing business, in the use of resources of the enterprise. For example, when applying ERP, it will accurately calculate the material supply plan for the factory to fully supply the supplier's orders. It is necessary to plan the production plan so that it is reasonable, neither lacking nor superfluous to bring the highest profit to the business. Plan business strategies, pricing policies, discounts, ...

The term ERP first came about in 1990 when Gartner used it to extend to MRP, which is also a management software focused on producing goods. By the mid-1990s, ERP was being applied to almost every part of the business, not just the manufacturing side. Government agencies and NGOs have also started ERP applications. Until the year 2000, the term "ERP II" was born and refers to ERP software capable of using a web interface for access and use. ERP II not only allows the company but also customers and suppliers in the supply chain to view

information. In other words, this new generation of ERP supports collaboration between companies, not just internally.

ERP is referred to as a typical business management software, which integrates all tools and functions in just one compact software. Thanks to ERP, businesses can more easily collect, store, manage information and data, support in the process of making strategic decisions and evaluating business performance.

As a software that fully integrates the functions that a business needs, constantly updating processes and business activities of departments immediately, ERP helps managers easily track cash flow, materials, inventory, ERP Project Report number 23 number of orders and customer relationship status for all the Modules and functions are all linked together into an overall system, so different departments (from accounting, purchasing, sales, production, customer care, human resources,...) easily receive data circulating in a stream, never interrupted. Typically, software operates on an integrated platform using common data that operates on a single database.

2.1.2. Outstanding features of ERP system

Unified Administration:

- ERP allows the expansion and development of various functions in the future without affecting the existing program structure.
- ERP systems create a unified management environment, from production to business operations, helping to optimize work efficiency. This helps connect every member of the business to functional departments and creates an orderly workflow.

Automatic management:

- An ERP system is an automated management software that helps organize and automate business processes in an enterprise, allowing departments and departments within the company to share information and interact automatically and seamlessly.
- However, ERP software systems do not completely replace human power and are not an automated production line. The system provides tools and features to automate and

manage business processes, but still requires human involvement and intervention in handling complex tasks, making strategic decisions, and solving unforeseen problems.

- With the combination of ERP and human power management systems, businesses can optimize operations and improve work efficiency, while enabling employees to focus on analytical, creative, and interactive tasks with customers.

Management has clear rules and plans:

- ERP systems have clear rules and processes in business management activities of enterprises. Each employee has specific duties and works according to consistent and strict regulations.
- This system helps organize and define workflows, from order management, warehouse management, financial management to personnel management. Employees are guided and adhere to previously defined rules and procedures, ensuring consistency and accuracy of work.
- Moreover, ERP systems also support production and business planning at regular intervals. Managers can plan production, manage resources, budget materials and time required for business activities. This helps ensure the efficiency and progress of work, while increasing the ability to predict and respond to fluctuations in business operations.

Departmental affiliation:

- ERP systems help create links and cohesion between departments in the company, creating a working environment where all employees participate, exchange information, and collaborate with each other instead of operating independently.
- Through ERP systems, data and information are shared automatically and seamlessly across departments. For example, information from the accounting department can be accessed by the sales department to determine the financial status of the customer. Or order information from sales can automatically transfer to the production department for production planning.

2.1.3. Software modules of ERP system:

ERP software is a software system that integrates many functions. ERP does not limit the number of modules, businesses can expand and add arbitrarily based on the actual needs of the organization and the deployment capacity of the technical team.

However, the key modules of ERP software are:

- Human Resource Management – HRM
- Financial Management – Accounting – FAM
- Supply Chain Management – SCM
- Procurement Management – MES
- Marketing Automation
- Customer Relationship Management – CRM
- Inventory Management – Stock Management
- Production Management

In addition to the above modules, depending on the characteristics and specific requirements, businesses can expand to many other modules: Sales Management, Purchase Management, Goods Management, Transportation Management,...

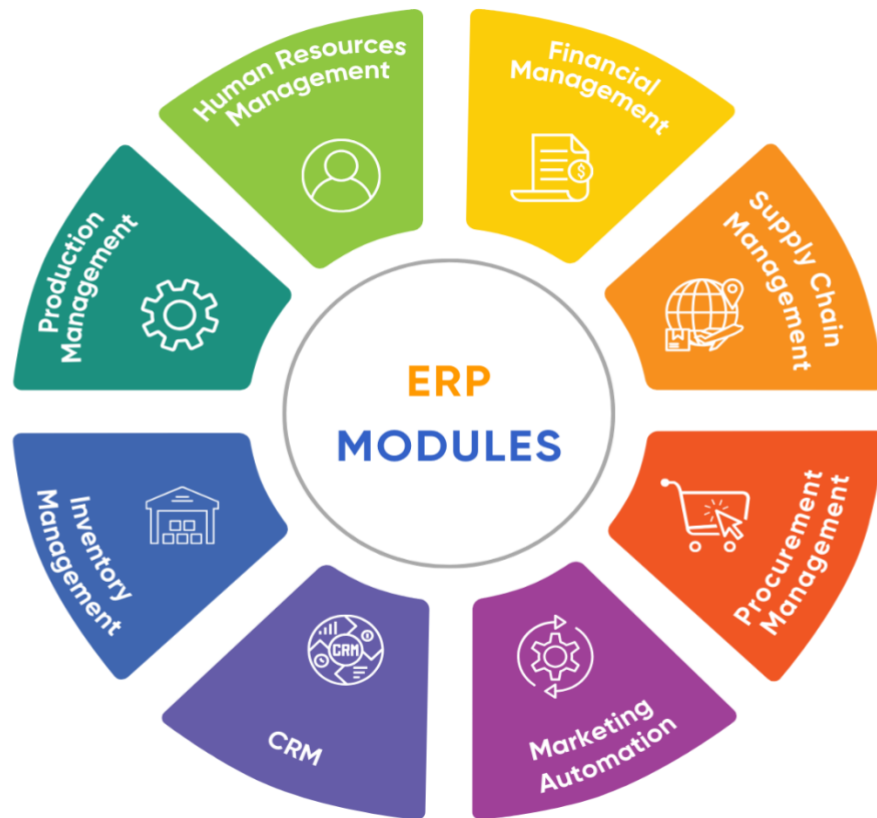


Figure 5: Image 2.1.3: Core modules of ERP.

2.1.4. Benefits ERP brings to businesses:

ERP brings a lot of benefits to businesses, some of the benefits we can see most clearly are:

- ERP provides information management and work orientation.
- ERP helps businesses reduce costs.
- ERP helps increase the quality of finished products.
- ERP helps accuracy in the transportation process – Logistics operations.
- ERP helps boost work efficiency.
- Businesses can store data more efficiently.
- ERP helps improve coordination between departments.

With the ability to comprehensively manage, ERP system is an effective support tool in business administration.

2.1.5. Limitations of ERP software:

The first drawback of ERP software is the high initial investment cost:

- Implementing an ERP system requires a large investment, including the cost of installation, customization, and staff training. This can be a burden for SMEs with limited resources.
- In Vietnam, although technology has developed rapidly recently, because of cost issues, many businesses are not willing to invest in ERP software, even cheap and manufactured by domestic technology companies. Instead, they still use Excel or free software.

ERP requires large deployment time:

- An integrated on-premises ERP software will take about 10-12 months to build and develop, while this figure with cloud ERP is 2-6 months. Currently, most businesses will choose cloud ERP solutions due to shorter deployment time. However, this time is still

quite long for some small and medium enterprises when they cannot have enough resources to manage the stages of software development.

- Large deployment times can be difficult and affect the day-to-day business operations of businesses. It can be costly in terms of time, resources, and money, especially for small and medium-sized businesses with limited resources.

ERP has a high level of complexity:

- ERP certainly has a much higher level of complexity than conventional ways of managing businesses through Excel. Because of the comprehensiveness and high integration of ERP software, implementing and managing it can require significant in-depth understanding and skills. This level of complexity can make it difficult for businesses to implement and use ERP software effectively. To be able to take full advantage of the potential that ERP software brings, businesses need to invest a lot of budget and time in training employees and improving the usability and specific skills related to this software.

ERP has a level of risk when it may not be suitable for business processes of the enterprise:

- Although ERP software is comprehensive and integrated, it may not fully meet the unique business requirements and processes of each business. Some businesses may feel the need to customize ERP software to fit their own business processes. However, excessive customization can result in a loss of comprehensiveness and integration of the software, posing risks to future upgradeability and maintenance.
- In addition, using ERP software requires adaptation on the part of employees. Without adequate support and training, employees can find it difficult to adapt to new software and changing business processes.

ERP takes effort and time to train employees:

- Given the comprehensive and complex nature of ERP software, employees need to be trained to understand the process, features, and how to use the software.

- Employee training can require considerable effort and time on the part of the business. Employees need to learn new concepts, become familiar with software interfaces, and master related business processes.

ERP is prone to disruption when updating and upgrading software:

- When a new version of ERP software is released or when improvements and patches are applied, the update and upgrade process can cause disruptions in the operations of the business. The process of updating and upgrading ERP software requires a significant amount of time and effort on the part of employees and IT departments.
- During ERP software updates and upgrades, there is a risk of crashes and data loss. Failure to implement the update and upgrade process carefully can result in the loss of critical data and impact business operations.

ERP is difficult to succeed when businesses do not have a clear implementation strategy:

- To achieve success with ERP software, businesses need a complete and clear implementation strategy, including specific steps and plans. A clear implementation strategy helps ensure coordination between departments and employees during ERP software implementation. Lack of coordination can cause inconsistencies and difficulties in software integration and adoption.

2.2. Deployment platform introduction

Deployment software: Odoo

2.2.1. Overview of Odoo

Starting in 2005, Odoo was developed by Fabien Pinckaers and his team in Belgium. Initially, it was a small ERP software called TinyERP. Through its development, it was renamed OpenERP, and starting from version 8.0 in 2014, the official name Odoo became widely used. Odoo was written using Python 2.7, JavaScript, XML, HTML5, and the PostgreSQL database by Fabien Pinckaers and his team.

Odoo is an integrated ERP and CRM software solution. Although there had been several open-source ERP software solutions like OpenBravo, Apache OFBiz, Compiere, etc., each had its own shortcomings that made them unsuitable for the specific operational needs of

businesses, leading to their short-lived presence in the market. Only Odoo has consistently maintained its stability and demonstrated its growth trend in the market up to now.

Odoo Community is an open-source software that does not require licensing fees. It is predicted to grow even more robustly and capture a larger market share among businesses worldwide, with over 4 million users currently, thanks to its diverse features and high security, like well-known ERP software like Oracle, Microsoft, SAP, and others.

2.2.2. History of development

Founding and Early Development (2005-2010)

- **Foundation (2005):** Odoo was founded by Fabien Pinckaers in Belgium. Initially, the software was called TinyERP. The goal was to provide an open-source ERP solution for small to medium-sized enterprises.
- **OpenERP Rebranding (2009):** TinyERP was rebranded to OpenERP. The name change reflected the broader scope of the software, which had grown beyond just a tiny enterprise resource planning (ERP) system to a full-featured ERP solution.
- **Community and Growth:** During this period, OpenERP gained significant traction in the open-source community, with many contributors helping to expand its functionality.

Expansion and Evolution (2010-2014)

- **Version 6.0 (2010):** The release of OpenERP 6.0 marked a significant milestone with major enhancements in usability and new features like a web client.
- **Version 7.0 (2012):** This version further improved the user interface and introduced a more modular architecture, allowing users to easily add new functionalities.
- **Odoo Rebranding (2014):** OpenERP was rebranded to Odoo. This change was made to reflect the broader suite of applications beyond ERP, including CRM, e-commerce, and CMS.

Maturity and Market Penetration (2014-Present)

- **Odoo 8.0 (2014):** The first release under the new name, Odoo 8.0, included major new applications like a website builder and an e-commerce module.

- **Odoo 9.0 (2015):** Introduced an enterprise version, offering additional proprietary features and services, while maintaining a community version.
- **Odoo 10.0 to 13.0 (2016-2019):** Each new version introduced numerous enhancements, such as improved performance, new modules, better integration capabilities, and a more refined user interface.
- **Odoo 14.0 (2020):** Brought significant improvements in productivity tools, user experience, and additional new modules.
- **Odoo 15.0 (2021):** Focused on performance enhancements, new features, and increased integration options.
- **Odoo 16.0 (2022):** Continued the trend of adding new modules, enhancing existing functionalities, and improving overall performance.
- **Odoo 17.0 (2023):** Brings numerous enhancements and new features designed to improve user experience and functionality across various business operations.

2.2.3. Advantages of Odoo

Odoo benefits in business administration:

- Comprehensive management solution for businesses.
- Increase performance, reduce costs.
- Improve the competitive ability.

Implementation benefits:

- Optimize deployment costs.
- Flexible customization, expansion, and easy integration
- Quick deployment time
- Large community

Benefits for corporate personnel:

- Easy to use.

- Manage and execute work effectively.
- Easy to interact and connect.

Benefits for customers and partners:

- Add convenience, enhance experience.
- Provide suitable products and services.

2.2.4. Popular Odoo modules

Currently, Odoo has more than 50 different subsystems. Below are some of the most seen and used subsystems:

- Customer Relationship Management Module – CRM
- Sales Management module - Sales Management
- Project Management Subsystem – Project Management
- Manufacturing Resource Planning & Management Subsystem – MRP
- Purchase Management module - Purchase Management
- Retail Point of Sale – POS module
- Live chat module – Live
- Schedule module – Calendar
- Marketing subsystem
- Human Resource Management Subsystem – HRM

And there are many other subsystems...

2.3. Description of the problem applying ERP to the project topic (or Case study)

2.3.1. Case study 1

ElectroTech Inc. is a mid-sized electronics manufacturing company specializing in consumer electronics such as smartphones, tablets, and wearable devices. The company has experienced rapid growth over the past decade and needed to integrate new systems to support

their growing-scale business as their existing system had not been meet the requirement anymore.

ElectroTech was using multiple standalone systems for different functions such as inventory management, sales, customer service, and accounting. This led to issues like data redundancy, inconsistent information across departments, and inefficient processes. The primary objectives for implementing an ERP system were:

- Improving operational efficiency by integrating all business processes.
- Enhancing decision-making with real-time data visibility.
- Reducing operational costs through process standardization and automation.
- Supporting scalable growth to handle increased production and sales volumes.

Technological upgradation can be beneficial; however, integrating ERP with existing legacy systems can be complex, costly and take a long time to complete. Moreover, data migration from existing systems to the new ERP system must be done carefully because any loss in data can disrupt business operations.

2.3.2. Case study 2:

Dien May Xanh is one of Vietnam's leading electronics retailers, part of the The Gioi Di Dong Corporation. The company specializes in selling a wide range of electronic products, including home appliances, consumer electronics, and mobile devices, with a vast network of stores across the country.

The Gioi Di Dong (Dien May Xanh) once use Excel as their management platform. However, as we all know, Excel is not effective when it comes to large data sizes (Amis, 2023). Therefore, The Gioi Di Dong spend a lot of money and time to build their own ERP system which involves Sales and Distribution, Financial and Accounting, Human Resources, ...

As the business growing, problems rise, and they continue to add more subsystems into the ERP system making it a big and complex system (Amis, 2023). As a result, maintainance is important to ensure that the system work seamlessly; because when one subsystem works incorrectly, it would affect the other subsystems.

CHAPTER 3: BUSINESS PROCESS SYSTEM

In this chapter, the focus is on various business processes and how they are managed within Odoo. It includes detailed descriptions and illustrations of processes such as procurement, sales, inventory receiving and shipping, financial and accounting management, customer service, human resources management, and handling customer and supplier debts. Each process is discussed in terms of BPMN (Business Process Model and Notation), specifications, and practical illustrations on Odoo.

3.1. System overview

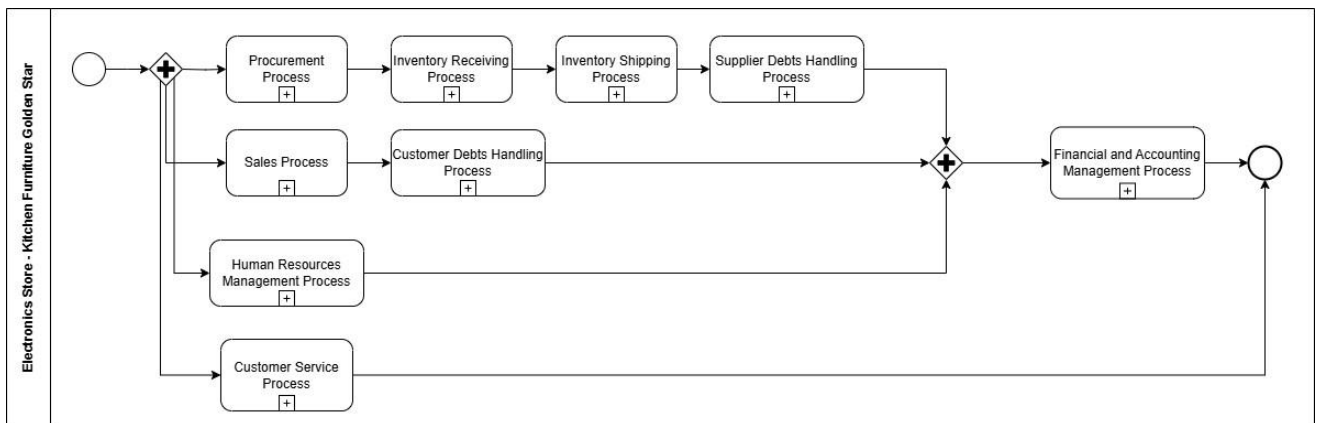


Figure 6. Image 3.1: BPMN model about System Overview

3.2. Procurement process

3.2.1. BPMN

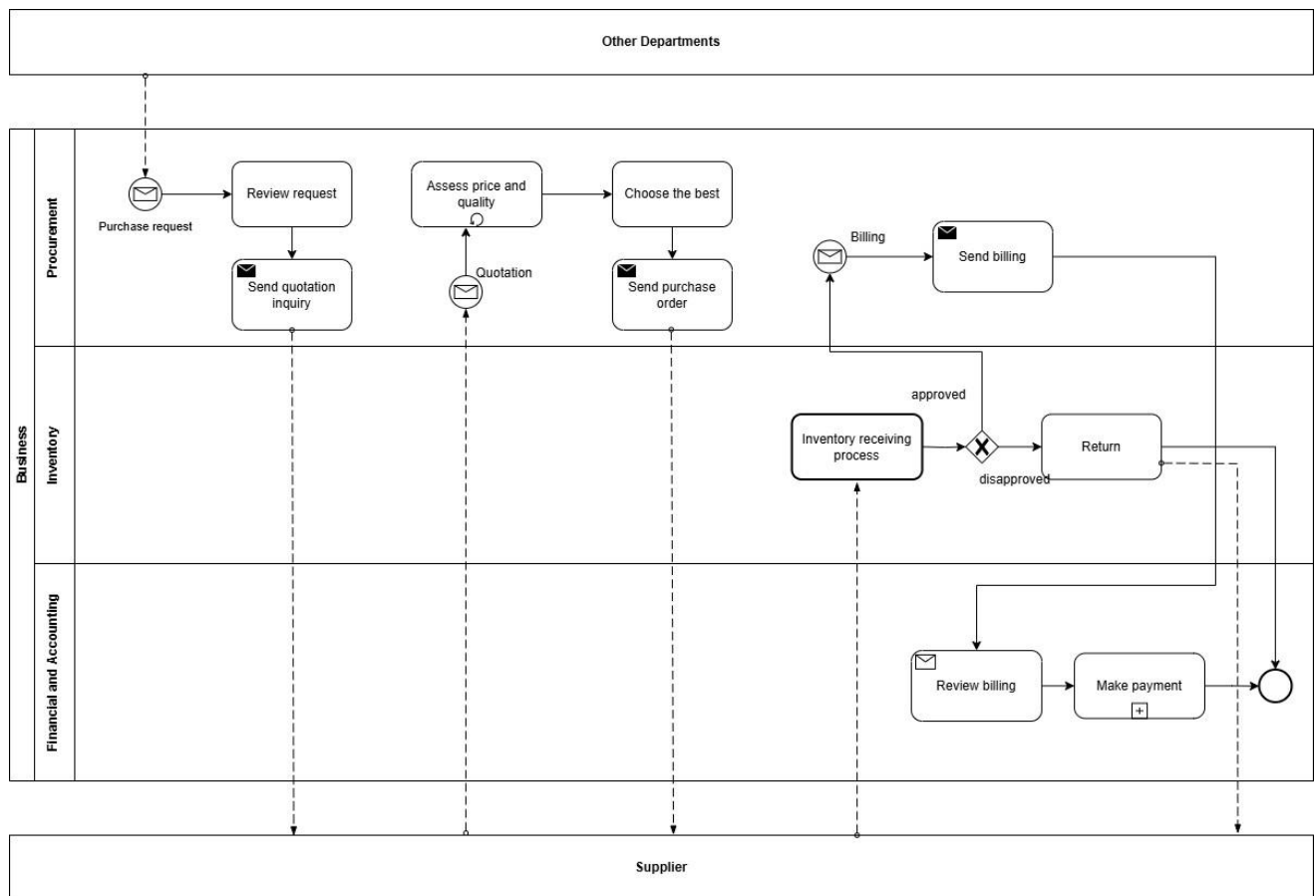


Figure 7. Image 3.2.1.1: BPMN model about Procurement Process



Figure 8. Image 3.2.1.2: BPMN model: Cancel purchase order (only before delivery)



Figure 9. Image 3.2.1.3: BPMN model: Cancel receipt.

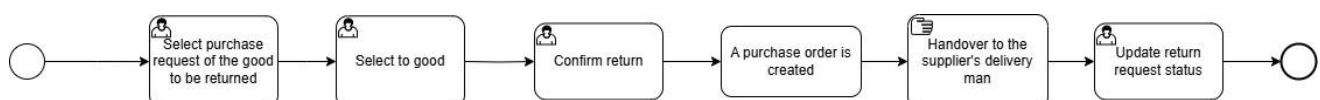


Figure 10. Image 3.2.1.4: BPMN model: Return to supplier.

3.2.2. Specification

3.2.2.1. Specification of the steps of the Procurement process*Table 3: 3.2.2.1. Specification of the steps of the Procurement process*

No	Actor	Action	Description
1	Procurement	Send quotation inquiry	Send quotation inquiry to supplier
2	Procurement	Receive quotation	Receive quotation from supplier
3	Procurement	Compare the price	Compare the price between supplier
4	Procurement	Choose the best	Choose the best price supplier
5	Procurement	Send purchase order	Send the purchase order to supplier
6	Inventory	Inventory receiving	Inventories receive goods from supplier
7	Inventory	Check good's condition	Check good's condition after inventory receiving
8	Inventory	Send billing	Send billing to procurement department
9	Procurement	Review billing	Validate billing information
10	Procurement	Send billing	Send billing to financial and accounting department
11	Financial and accounting	Review billing	Validate billing information
12	Financial and accounting	Make payment	Pay the bill

3.2.2.2. Data library specification:**Master Data**

- Supplier

Table 4: 3.2.2.2. Data library specification: Master data Supplier

No	Property	Description	Note
----	----------	-------------	------

1	Supplier Type	Select supplier type Individual/Company	Required property with one out of two options
2	Supplier Image	Supplier image	Not required
3	Name	Supplier name	Required
4	Address	Supplier location	Not required
5	Tax ID	Unique identifier used by tax authorities to track individuals and businesses for tax purposes	Not required
6	Phone	Phone number	Not required
7	Mobile	Mobile phone number	Not required
8	Email	Supplier email address	Not required
9	Website	Supplier website	Not required
10	Language	All emails and documents being sent to supplier are translated into this language	Required
11	Contact and Addresses	Contacts and locations involved	Not required
12	Sales and Purchase		
12.1	Sales		
12.1.1	Salesperson	User who oversees this supplier	Not required
12.1.2	Payment Terms	Conditions under which a seller will complete a sale and a buyer will make payment	Not required
12.1.3	Pricelist	Use specifically for this supplier instead of default price list	Not required
12.2	Purchase		
12.2.1	Buyer	Who oversees procurement with this supplier	Not required

12.2.2	Payment Terms	Conditions under which a buyer will make payments to suppliers for goods or services procured.	Not required
12.2.3	Receipt Reminder	Reminder to receive the goods before X days from planned delivery	One out of two options: true/false
12.3	Fiscal Information	Fiscal position: Decide types of taxes/account used for this supplier	Not required
12.4	Other information		
12.4.1	Company ID	Company registered ID	Not required
12.4.2	Reference	Reference code	Not required
12.4.3	Industry	Supplier's industry	Not required
13	Accounting		
13.1	Bank accounts	Customer bank accounts	Not required
13.2	Electronic invoicing	Format: format of the electronic invoicing used for this customer	Not required
13.3	Accounting entries	Account Receivable amounts that customers or suppliers owes for goods or services provided on credit. Account Payable: amounts that business owes to customers or suppliers for goods or services purchased on credit	Required
14	Internal Note	Internal notes about this supplier	Not required

- Product

Table 25: 3.2.2.2. Data library specification: Master data Product

No	Property	Description	Note
----	----------	-------------	------

1	Product Name	Product name	Required
2	Product Image	Product display image	Not required
3	General information	Product general information	
3.1	Product Type	One of four types: consumption, service, stock, scheduling fee	Required
3.2	Invoicing Policy	A policy to ensure that invoicing is conducted in a standardized, efficient, and compliant manner. One of two options: quantity ordered; quantity purchased	Required
3.3	Sales Price	Price at which the product is sold to customer. Default is 1,00	Required
3.4	Customer's Taxes	Default tax used when selling products	Required
3.5	Cost	Value of the product (automated calculated). Used to value the product when the purchase cost is not known	Not required
3.6	Product Category	Product category	Not required
3.7	Internal Reference	Unique identifier assigned to the product	Not required
3.8	Barcode	Machine-readable representation of data in a visual form	Not required
3.9	Product Template Tags	Placeholders used within product templates to dynamically insert product-specific information	Not required
3.10	Internal Notes	Internal notes	Not required
4	Sales		

4.1	Upsell and Cross-sell	Optional Products: are suggested whenever customer hits “Add to Cart”	Not required
4.2	Sales Description	Added to sales orders and invoices	Not required
5	Purchase		
5.1	Vendor	General information about vendor	Not required
5.2	Vendor bills		
5.2.1	Vendor taxes	Default taxes used when buying the product	Required
5.2.2	Control policy	On ordered quantities: Control bills based on ordered quantities. On received quantities: Control bills based on received quantities	One out of two options: On ordered quantities/On received quantities
5.3	Purchase description	Added to purchase orders	Not required
6	Inventory		
6.1	Operations		
6.1.1	Routes	Whether it will be bought, manufactured, etc	One out of two options: buy/not buy
6.2	Logistics		
6.2.1	Responsible	The user will be responsible of the next activities related to logistics of the product	Not required
6.2.2	Weight	Product weight	Not required
6.2.3	Volume	Product volume	Not required
6.2.4	Customer lead time	Delivery lead time, in days	Not required

6.3	Description for receipts	Added to receipt orders	Not required
6.4	Description for delivery orders	Added to delivery orders	Not required
6.5	Description for internal transfer	Added to internal transfer orders (where to pick the product in the warehouse)	Not required

3.2.3. Illustration on Odoo

a. Procurement process

Step 1: Go to “**Purchase**” module and click “**New**” to create a new RFQ.

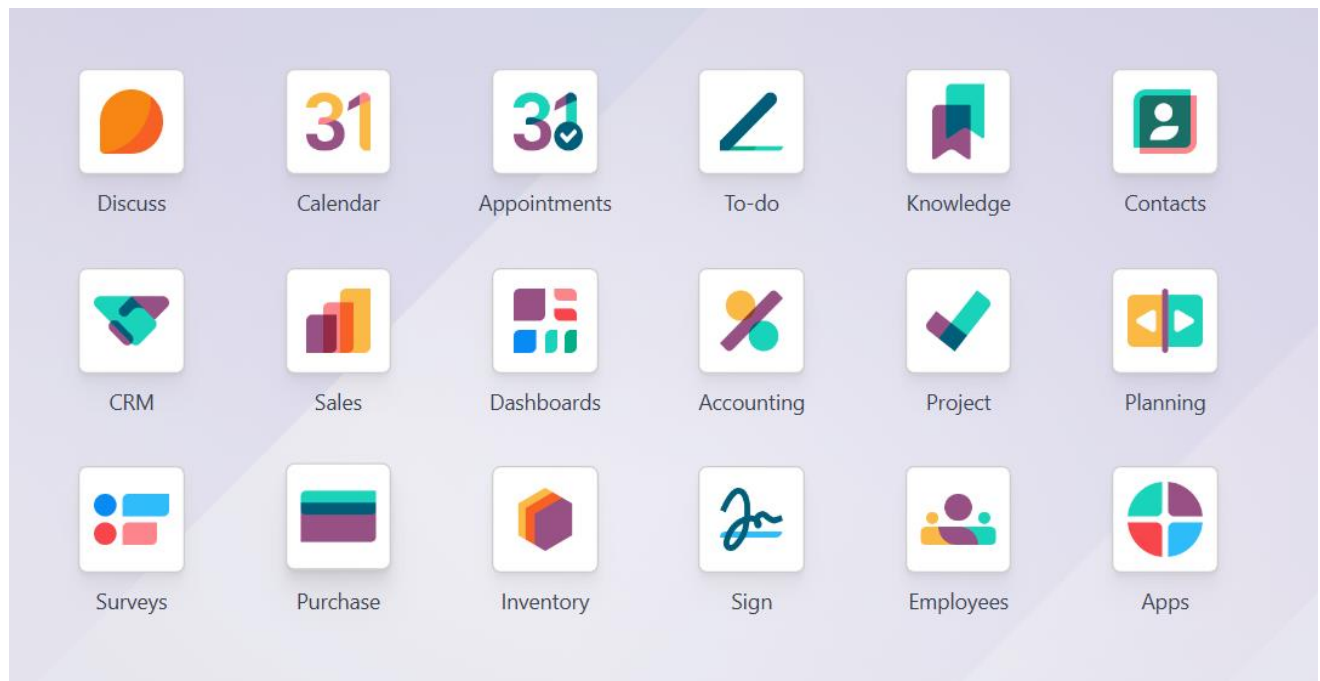


Figure 11. Image 3.2.3.1: Dashboard

Step 2: Input necessary information for RFQ (Vendor, Order Deadline, Expected Arrival, Deliver To, Products)

ACCT5123 - Enterprise resource planning

Request for Quotation

☆ **P00015**

Vendor ? CEA Technologies

Vendor Reference ?

Order Deadline ? 06/05/2024 20:29:37

Expected Arrival ? 06/05/2024 20:29:37

Deliver To ? Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh: Phiếu nhập kho

No On-time Delivery Data

☐ Ask confirmation

Products

Other Information

Product	Description	Quantity	Unit Price	Taxes	Tax excl.
[40S350R] TCL 40-Inch Class S3 1080p LED ...	[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model). Compatible with Alexa, Google Assistant, and Apple HomeKit Compatibility, Streaming FHD Television,Black	10.00	189.99	10%	1,900 đ
[AEPIS2] Klein Tools AEPIS2 Bluetooth Spea...	[AEPIS2] Klein Tools AEPIS2 Bluetooth Speaker With Magnetic Strip and Hook,	10.00	39.97	10%	400 đ

Figure 12. Image 3.2.3.2: Request for Quotations

Step 3: Click “**Send by Email**” to send RFQ to supplier’s email (add supplier’s email address if needed).

Compose Email

×

Recipients

Followers of the document and

CEA Technologies <cea@gmail.s... ✕

Subject

ACCT5123 N22 CTTT Đơn hàng (Mã P00015)

Kính gửi CEA Technologies

Tệp đính kèm là yêu cầu báo giá **P00015** từ ACCT5123 N22 CTTT.

Nếu bạn có bất kỳ thắc mắc nào, xin đừng ngần ngại liên hệ với chúng tôi.

Trân trọng,

--

19522476

 Yêu cầu Báo giá - P00015.pdf

PDF

✓

Attachments

Load template

Purchase: Request For Quotation

Send

Discard

Save Template

Figure 13. Image 3.2.3.3: Compose email to send the Vendor.

Step 4: When vendor replies with quotation, update the unit price in the “**Products**” field.

Products		Other Information						
Product		Description		Quantity	Unit Price		Taxes	Tax excl.
[40S350R]	TCL 40-Inch Class S3 1080p	[40S350R]	TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assistant, and Apple HomeKit Compatibility, Streaming FHD Television,Black	10.00	195.00		10%	1,900
[AEPJS2]	Klein Tools AEPJS2 Bluetooth Speaker With Magnetic Strip and Hook, Rechargeable, Wireless and Aux Capable, Hands Free Capable, Smart Phone Charging, 10 Hr Run Time, IP54 Dust and Water Resistant	[AEPJS2]	Klein Tools AEPJS2 Bluetooth Speaker With Magnetic Strip and Hook, Rechargeable, Wireless and Aux Capable, Hands Free Capable, Smart Phone Charging, 10 Hr Run Time, IP54 Dust and Water Resistant	10.00	39.97		10%	400
Add a product Add a section Add a note Catalog								

Figure 14. Image 3.2.3.4: List of products

Step 5: Click “**Confirm order**” to send purchase order to supplier.

Receive ProductsSend PO by EmailCreate BillCancelLock

RFQRFQ SentPurchase Order

Purchase Order

☆ P00015

Vendor ?CEA Technologies

Vendor Reference ?

Confirmation Date06/01/2024 20:40:00

Expected Arrival ?06/05/2024 20:29:37

☐ Ask confirmation

Deliver To ?Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh: Phiếu nhập kho

Products		Other Information						
Product		Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
[40S350R]	TCL 40-Inch Class S3 10...	[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assistant, and Apple HomeKit Compatibility, Streaming FHD Television,Black	10.00	0.00	0.00	195.00	10%	1,950 đ
[AEPJS2]	Klein Tools AEPJS2 Blueto...	[AEPJS2] Klein Tools	10.00	0.00	0.00	39.97	10%	400 đ

Figure 15. Image 3.2.3.5: Purchase bill

Step 6: Click on “**Receipt**” to navigate to the transfer form.

NewRequests for QuotationP00015

Receipt1

1 / 1 < >

Receive ProductsSend PO by EmailCreate BillCancelLock

RFQRFQ SentPurchase Order

Purchase Order

☆ P00015

Figure 16. Image 3.2.3.6: Request for Quotation

Step 7: When receive goods at inventory, make sure goods’ quantity and quality are checked, then click “**Validate**” to complete the transfer form.

Print LabelsPrintReturn

DraftReadyDone

☆ WH 4/IN/00001

Receive FromCEA Technologies

Scheduled Date ?06/05/2024 20:29:37

Destination LocationWH 4/Tồn kho

Effective Date ?06/01/2024 20:43:57

Source Document ?P00015

OperationsAdditional InfoNote

Product	Demand	Quantity
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assistant, and Apple HomeKit Compatibility, Streaming FHD Tel...	10.00	10.00
[AEPJS2] Klein Tools AEPJS2 Bluetooth Speaker With Magnetic Strip and Hook, Rechargeable, Wireless and Aux Capable, Hands Free Capable, Smart Phone Charging, 10 Hr Run Time, IP54...	10.00	10.00

Figure 17. Image 3.2.3.7: Warehouse information

Step 8: Go back to RFQ page and click “**Create bill**” to navigate to Bill Draft

NewRequests for QuotationP00015

Receipt1

1 / 1 < >

Create BillSend PO by EmailCancelLock

RFQRFQ SentPurchase Order

Purchase Order

☆ P00015

Vendor ?CEA Technologies

Confirmation Date06/01/2024 20:40:00

Vendor Reference ?

Expected Arrival ?06/05/2024 20:29:37

Deliver To ?Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh: Phiếu nhập kho

Arrival ?06/01/2024 20:43:57

ProductsOther Information

Product	Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compati...	[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google	10.00	10.00	0.00	195.00	10%	1,950 d

Figure 18. Image 3.2.3.8: Request for Quotation

Step 9: Review bill information, input necessary information and click “**Confirm**” to create and save the bill to the system.

Confirm Cancel Draft Posted

Vendor Bill

BILL/2024/06/0003

Vendor CEA Technologies

Bill Date 06/01/2024

Bill Reference

Accounting Date 06/01/2024

Auto-Complete ? Select a purchase order or an old bill

Payment Reference ?

Recipient Bank ?

Due Date 06/01/2024 or Payment Terms

Journal Vendor Bills

Invoice Lines Journal Items Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Go	P00015: [40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model),	1561 Purchase costs	10.00	195.00	10%	1,950 d

Figure 19. Image 3.2.3.9: Request for Quotation

Step 10: Click “Register Payment” to make the payment.

Register Payment Credit Note Reset to Draft Draft Posted

Vendor Bill

BILL/2024/06/0003

Vendor CEA Technologies

Bill Date 06/01/2024

Bill Reference

Accounting Date 06/01/2024

Payment Reference ?

Recipient Bank ?

Due Date 06/01/2024

Journal Vendor Bills

Invoice Lines Journal Items Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Go	P00015: [40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model),	1561 Purchase costs	10.00	195.00	10%	1,950 d

Figure 20. Image 3.2.3.10: Request for Quotation

b. Cancel purchase order.

Step 1: Click “Cancel” to cancel purchase order.

ACCT5123 - Enterprise resource planning

New

Requests for Quotation

P00016

Receipt

1 / 1

Receive Products

Send PO by Email

Create Bill

Cancel

Lock

RFQ

RFQ Sent

Purchase Order

Purchase Order

☆ P00016

Vendor ? Beutel

Vendor Reference ?

Confirmation Date 06/01/2024 20:53:07

Expected Arrival ? 06/02/2024 20:51:17

☐ Ask confirmation

Deliver To ? ACCT5123 N22 CTTT: Receipts

Products

Other Information

Product	Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
[43 inch Mirror 4K TV Android OS]...	[43 inch Mirror 4K TV Android OS] Soulaca 43 inches Smart Android Mirror 4K LED TV Big Screen IP65 Waterproof Built-in WiFi Bluetooth SPA ATSC DTV	2.00	0.00	0.00	1,489.00	10%	2,978 d

Add a product

Add a section

Add a note

Catalog

Figure 21. Image 3.2.3.11: Request for Quotation

Step 2: Click “Set to Draft” if you want to re-request for quotation.

New

Requests for Quotation

P00016

Receipt

1 / 1

Set to Draft

RFQ

RFQ Sent

Purchase Order

Cancelled

Purchase Order

☆ P00016

Vendor ? Beutel

Vendor Reference ?

Order Deadline ? 06/02/2024 20:51:17

Expected Arrival ? 06/02/2024 20:51:17

☐ Ask confirmation

Deliver To ? ACCT5123 N22 CTTT: Receipts

No On-time Delivery Data

Products

Other Information

Product	Description	Quantity	Unit Price	Taxes	Tax excl.
[43 inch Mirror 4K TV Android OS] Soulaca ...	[43 inch Mirror 4K TV Android OS] Soulaca 43 inches Smart Android Mirror 4K LED TV Big Screen IP65 Waterproof Built-in WiFi Bluetooth SPA ATSC DTV	2.00	1,489.00	10%	2,978 d

Figure 22. Image 3.2.3.12: Request for Quotation

c. Cancel receipt.

Step 1: Click “Receipt” to navigate to receipt page.

ACCT5123 - Enterprise resource planning

New Requests for Quotation P00017

Receipt 1

1 / 1

Receive Products Send PO by Email Create Bill Cancel Lock RFQ RFQ Sent Purchase Order

Purchase Order
☆ P00017

Vendor ? 3M

Vendor Reference ?

Confirmation Date 06/01/2024 20:57:40

Expected Arrival ? 06/01/2024 20:56:48

☐ Ask confirmation

Deliver To ? ACCT5123 N22 CTTT: Receipts

Products Other Information

Product	Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
[24C MICRO] JBL Professional 24C Micro 4.5-Inch Background/Foreground Ceiling S...	[24C MICRO] JBL Professional 24C Micro 4.5-Inch Background/Foreground Ceiling Speaker, White, Sold as Pair	1.00	0.00	0.00	169.56	10%	170 d

Add a product Add a section Add a note Catalog

Figure 23. Image 3.2.3.13: Request for Quotation

Step 2: Click “Cancel” to cancel receipt (this is equivalent to cancel purchase order).

New Requests for Quotation / P00017 WH/IN/00012

Detailed Operations

1 / 1

Validate Print Print Labels Cancel Draft Ready Done

☆ WH/IN/00012

Receive From 3M

Destination Location WH/Stock

Scheduled Date ? 06/01/2024 20:56:48

Deadline ? 06/01/2024 20:56:48

Source Document ? P00017

Operations Additional Info Note

Product	Demand	Quantity
[24C MICRO] JBL Professional 24C Micro 4.5-Inch Background/Foreground Ceiling Speaker, White, Sold as Pair	1.00	1.00

Add a line

Send message Log note Activities

Follow

Figure 24. Image 3.2.3.14: Warehouse information

d. Return goods to supplier.

Step 1: On Purchase Order page, click “Receipt” to navigate to receipt page.

ACCT5123 - Enterprise resource planning

New / P00015 / BILL/2024/06/0003

Vendor Bills 1 Receipt 1

1 / 1

Send PO by EmailCreate BillCancelLock

RFQRFQ SentPurchase Order

Purchase Order

☆ P00015

Vendor ? CEA Technologies

Vendor Reference ?

Confirmation Date 06/01/2024 20:40:00

Expected Arrival ? 06/05/2024 20:29:37

Deliver To ? Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh: Phiếu nhập kho

Arrival ? 06/01/2024 20:43:57

ProductsOther Information

Product	Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compati...	[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assistant and	10.00	10.00	10.00	195.00	10%	1,950

Figure 25. Image 3.2.3.15: Purchase bill

Step 2: Click “Return” to return goods to supplier.

New / BILL/2024/06/0003 / P00015

Detailed Operations

1 / 1

Print LabelsPrintReturn

DraftReadyDone

☆ WH 4/IN/00001

Receive From CEA Technologies

Destination Location WH 4/Ton kho

Scheduled Date ? 06/05/2024 20:29:37

Effective Date ? 06/01/2024 20:43:57

Source Document ? P00015

OperationsAdditional InfoNote

Product	Demand	Quantity
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assistant, and Apple HomeKit Compatibility. Streaming FHD Tel...	10.00	10.00
[AEPJ52] Klein Tools AEPJ52 Bluetooth Speaker With Magnetic Strip and Hook, Rechargeable, Wireless and Aux Capable, Hands Free Capable, Smart Phone Charging, 10 Hr Run Time, IP54 ...	10.00	10.00

Figure 26. Image 3.2.3.16: Warehouse information

Step 3: Select product and input quantity to be returned and click “Return” to confirm.

Reverse Transfer

Product	Quantity
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assist...	10.00
[AEPJS2] Klein Tools AEPJS2 Bluetooth Speaker With Magnetic Strip and Hook, Rechargeable, Wireless and Aux Capable, Hands F...	10.00

Return Location Partners/Vendors

Return

Cancel

Figure 27. Image 3.2.3.17: List of products

Step 4: When goods is delivered, click “**Validate**” to complete the process.

New / P00015 / WH 4/IN/00001
WH 4/OUT/00001

Detailed Operations

1 / 1

ValidatePrintPrint LabelsCancel

DraftWaitingReadyDone

☆ WH 4/OUT/00001

Delivery Address CEA Technologies

Scheduled Date [?] 06/01/2024 21:06:25

Source Location WH 4/Tôn kho

Product Availability [?] Available

Source Document [?] Return of WH 4/IN/00001

OperationsAdditional InfoNote

Product	Demand	Quantity	
[40S350R] TCL 40-Inch Class S3 1080p LED Smart TV with Roku TV (40S350R, 2023 Model), Compatible with Alexa, Google Assistant, and Apple HomeKit Compatibility, Streaming FHD Television...	10.00	10.00	
[AEPJS2] Klein Tools AEPJS2 Bluetooth Speaker With Magnetic Strip and Hook, Rechargeable, Wireless and Aux Capable, Hands Free Capable, Smart Phone Charging, 10 Hr Run Time, IPS4 Dust ...	10.00	10.00	
Add a line			

Figure 28. Image 3.2.3.18: Warehouse information

3.3. Sales process

3.3.1. BPMN

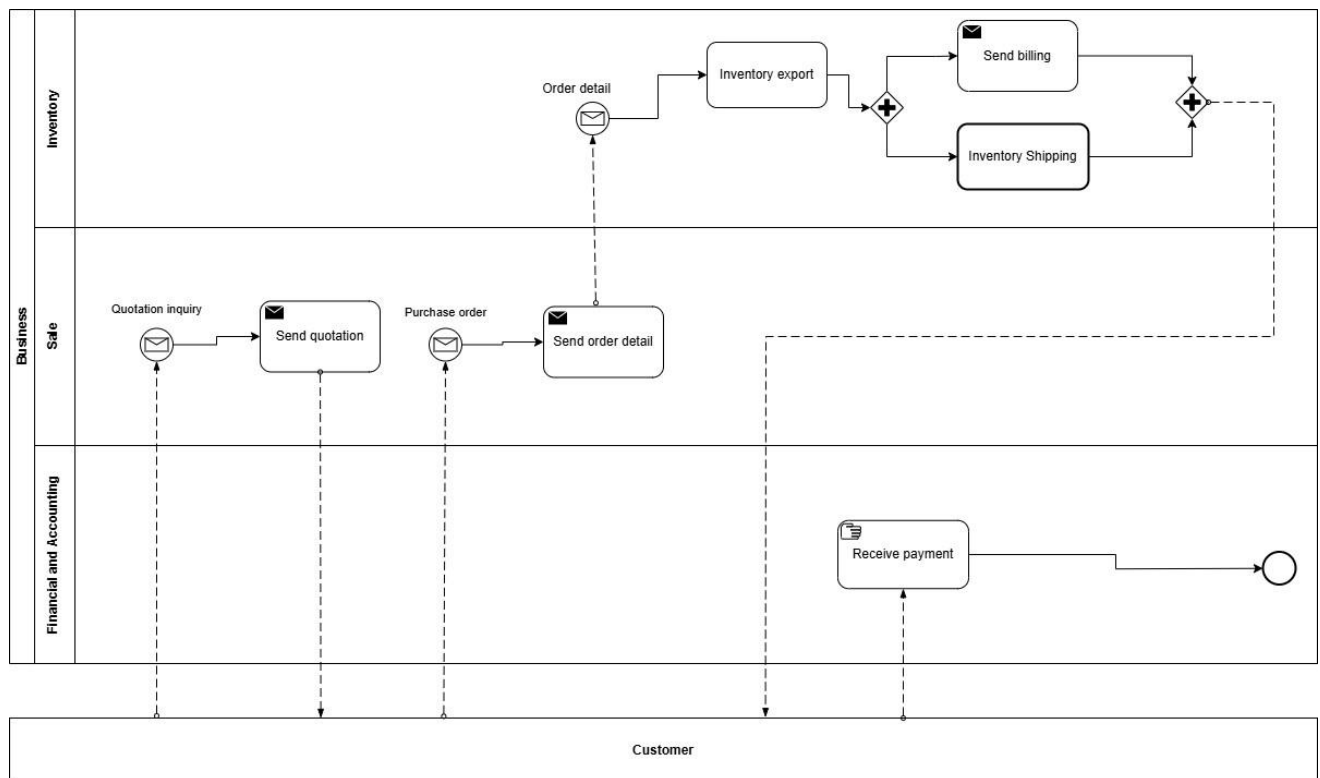


Figure 29. Image 3.3.1.1: BPMN model about Sale Process



Figure 30. Image 3.3.1.2: BPMN model: Cancel purchase order (only before delivery)

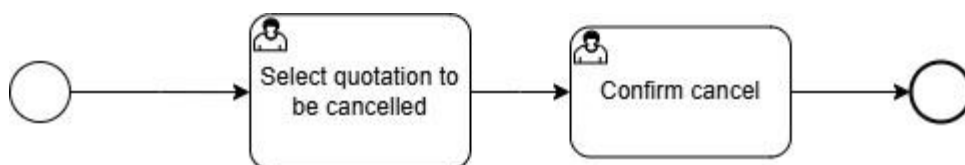


Figure 31. Image 3.3.1.3: BPMN model: Cancel quotation.



Figure 32. Image 3.3.1.4: BPMN model: Cancel receipt (only before payment)

3.3.2. Specification

3.3.2.1. Specification of the steps of the Sales process

Table 6: 3.3.2.1. Specification of the steps of the Sales process

No	Actor	Action	Description
1	Sales	Receive quotation inquiry	Receive quotation inquiry from customer
2	Sales	Send quotation	Send quotation to customer
3	Sales	Receive purchase order	Receive purchase order from customer
4	Sales	Send order details	Send customer order details to inventory
5	Inventory	Receive order details	Receive customer order details from sales department
6	Inventory	Inventory export	Export product from inventory
7	Inventory	Delivery	Delivery product to customer
8	Inventory	Send invoice and delivery information	Send billing and delivery information to customer
9	Financial and Accounting	Receive payment	Receive payment from customer

3.3.2.2. Data library specification:

a. Master data

- Customer

Table 7: 3.3.2.2. Data library specification: Master data Customer

No	Property	Description	Note
1	Customer Type	Select customer type Individual/Company	Required property with one out of two options
2	Customer Image	Customer image	Not required
3	Name	Customer name	Required
4	Address	Customer location	Not required

5	Tax ID	Unique identifier used by tax authorities to track individuals and businesses for tax purposes	Not required
6	Phone	Phone number	Not required
7	Mobile	Mobile phone number	Not required
8	Email	Customer email address	Not required
9	Website	Customer website	Not required
10	Language	All emails and documents being sent to supplier are translated into this language	Required
11	Contact and Location	Contacts and locations involved	Not required
12	Sales and Purchase		
12.1	Sales		
12.1.1	Salesperson	User who oversees this supplier	Not required
12.1.2	Sale payment term	Conditions under which a seller will complete a sale and a buyer will make payment	Not required
12.1.3	Price list	Use specifically for this supplier instead of default price list	Not required
12.2	Purchase		
12.2.1	Purchase person	Who oversees procurement with this supplier	Not required
12.2.2	Purchase payment term	Conditions under which a buyer will make payments to suppliers for goods or services procured.	Not required
12.2.3	Reminder to receive the goods	Reminder to receive the goods before X days from planned delivery	One out of two options: true/false

12.3	Fiscal information	Fiscal position: Decide types of taxes/account used for this supplier	Not required
12.4	Other information		
12.4.1	Company ID	Company registered ID	Not required
12.4.2	Reference code	Reference code	Not required
12.4.3	Field	Supplier's field	Not required
13	Accounting		
13.1	Bank accounts	Customer bank accounts	Not required
13.2	Electronic invoicing	Format: format of the electronic invoicing used for this customer	Not required
13.3	Accounting entries	Account Receivable amounts that customers or suppliers owes for goods or services provided on credit. Account Payable: amounts that business owes to customers or suppliers for goods or services purchased on credit	Required
14	Internal Note	Internal notes about this supplier	Not required

3.3.3. Illustration on Odoo

Step 1: Go to “Sales” module and click “New” to create a new quotation.

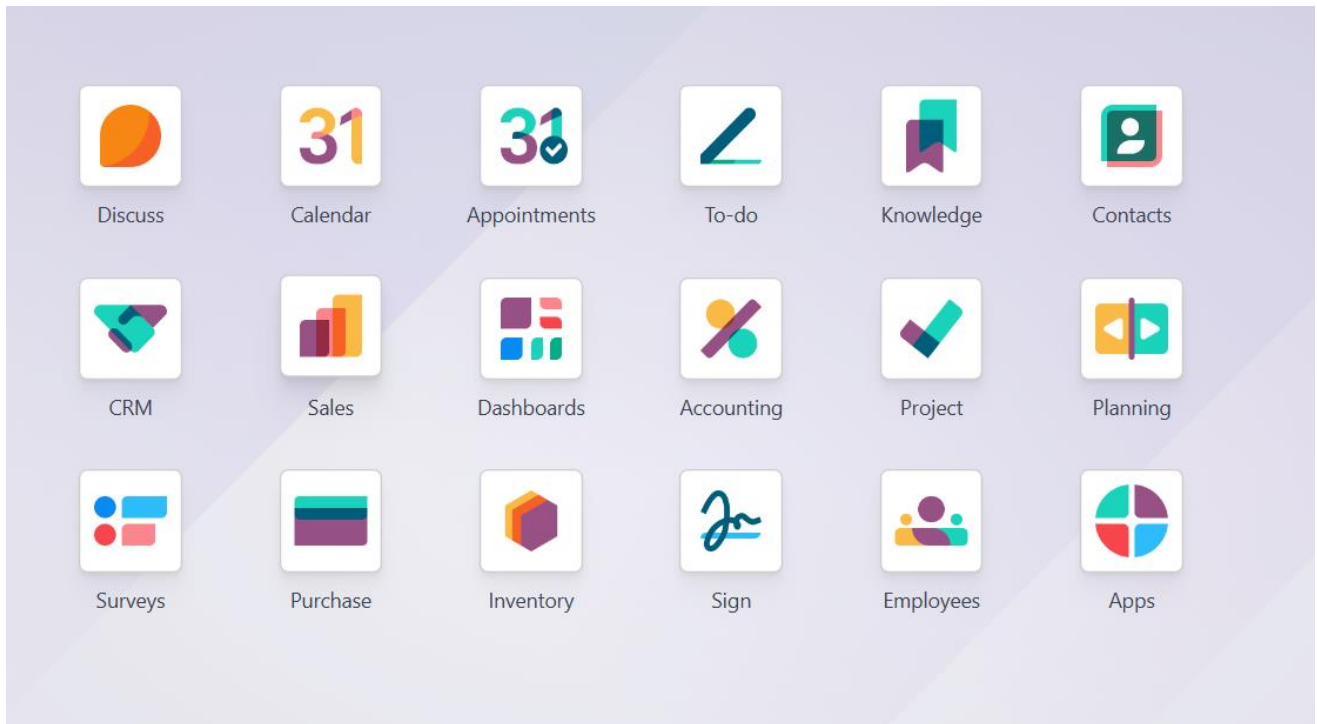


Figure 33. Image3.3.3.1: Dashboard

Step 2: Input necessary information (Customer, Expiration, Pricelist, Payment Terms, Order Lines, ...) and click “Send by Email” to send quotation to customer’s email address.

The image shows the 'New' form for creating a quotation in Odoo. The form includes the following sections:

- Buttons:** 'Send by Email', 'Confirm', 'Preview'.
- Navigation:** 'Quotation' (active), 'Quotation Sent', 'Sales Order'.
- Customer Information:**
 - Customer: Adam Wheeler
 - Expiration: 07/31/2024
 - Pricelist: Default VND pricelist (VND)
 - Payment Terms: Immediate Payment
- Order Lines:**
 - Order Lines (selected), Optional Products, Other Info
 - Table with columns: Product, Description, Quantity, Unit Price, Taxes, Tax excl.
 - Row 1: 12" Oscillating Table Fan, No Size, White | 12" Oscillating Table Fan, No Size, White | 1.00 | 20.00 | 10% | 20 d
 - Buttons: Add a product, Add a section, Add a note, Catalog
- Terms and conditions:** Text area for 'Terms and conditions...'.
- Summary:**
 - Untaxed Amount: 20 d
 - VAT 10%: 2 d
 - Total: 22 d

Figure 34. Image 3.3.3.2: Quotations

Step 3: Click “**Confirm**” to create sales order.

ConfirmSend by EmailPreviewCancel

QuotationQuotation SentSales Order

S00005

CustomerAdam Wheeler

Expiration07/31/2024

Pricelist[?]Default VND pricelist (VND)

Payment TermsImmediate Payment

Order LinesOptional ProductsOther Info

Product	Description	Quantity	Unit Price	Taxes	Tax excl.
12" Oscillating Table Fan, No Size, White	12" Oscillating Table Fan, No Size, White	1.00	20.00	10%	20 đ

Add a productAdd a sectionAdd a noteCatalog

Terms and conditions...

Untaxed Amount:20 đ

VAT 10%:2 đ

Total:22 đ

Figure 35. Image 3.3.3.3: Quotations

Step 4: Click “**Create Invoice**” to create and save the invoice to the system.

NewQuotationsS00005

Delivery1

1 / 1 < >

Create InvoiceSend by EmailPreviewCancel

QuotationQuotation SentSales Order

S00005

CustomerAdam Wheeler

Order Date[?]06/01/2024 21:17:18

Pricelist[?]Default VND pricelist (VND)

Payment TermsImmediate Payment

Order LinesOther Info

Product	Description	Quantity	Invoiced	Unit Price	Taxes	Tax excl.
12" Oscillating Table Fan, No Size, W	12" Oscillating Table Fan, No Size, White	1.00 €	0.00	20.00	10%	20 đ

Add a productAdd a sectionAdd a noteCatalog

Terms and conditions...

Untaxed Amount:20 đ

VAT 10%:2 đ

Total:22 đ

Figure 36. Image 3.3.3.4: Quotations

Step 5: When products are safely delivered to the customer, click “Validate” on Delivery tab.

The screenshot shows the 'Draft Invoice' form for 'WH/OUT/00007'. The top navigation bar includes 'New', 'Draft Invoice (* 14) / S00005', and 'Detailed Operations'. The main header has 'Validate', 'Print', 'Print Labels', and 'Cancel' buttons. The right side shows 'Draft', 'Waiting', 'Ready', and 'Done' tabs. The form displays the following information:

- Delivery Address:** Adam Wheeler
- Source Location:** WH/Stock
- Scheduled Date:** 06/01/2024 21:17:18
- Deadline:** 06/01/2024 21:17:18
- Product Availability:** Available
- Source Document:** S00005

Below this information is a table with columns: Product, Demand, and Quantity. The table contains one row for '12" Oscillating Table Fan, No Size, White' with a demand of 1.00 and a quantity of 1.00. There are also buttons for 'Add a line', 'Operations', 'Additional Info', and 'Note'.

Figure 37. Image: 3.3.3.4: Warehouse

Step 6: When customer make the payment, click “Confirm” on Invoice tab to complete the process.

The screenshot shows the 'Customer Invoice' form for 'Draft'. The top navigation bar includes 'Confirm', 'Preview', and 'Cancel' buttons. The right side shows 'Draft' and 'Posted' tabs. The form displays the following information:

- Customer:** Adam Wheeler
- Invoice Date:**
- Payment Reference:**
- Payment terms:** Immediate Payment
- Journal:** Customer Invoices

Below this information is a table with columns: Product, Label, Account, Quantity, Price, Taxes, and Tax excl. The table contains one row for '12" Oscillating Table Fan, No Size, White' with a quantity of 1.00, a price of 20.00, and a tax of 10%. There are also buttons for 'Add a line', 'Add a section', 'Add a note', and 'Catalog'.

At the bottom right, the summary shows:

- Untaxed Amount: 20 d
- VAT 10%: 2 d
- Total: 22 d

Figure 38. Image 3.3.3.5: Warehouse

3.4. Inventory receiving process.

Warehousing is the first step in the logistics warehouse management process and one of the most important. To properly carry out the receiving process, the warehouse department must be able to verify that the product has been received, in sufficient quantity, in the right conditions and at the time of delivery. When the goods are received, it means that responsibility for the goods will be transferred from the supplier to the warehouse department. At this point,

the warehouse department needs to ensure that the best condition of the goods is maintained until they are shipped again.

3.4.1. BPMN

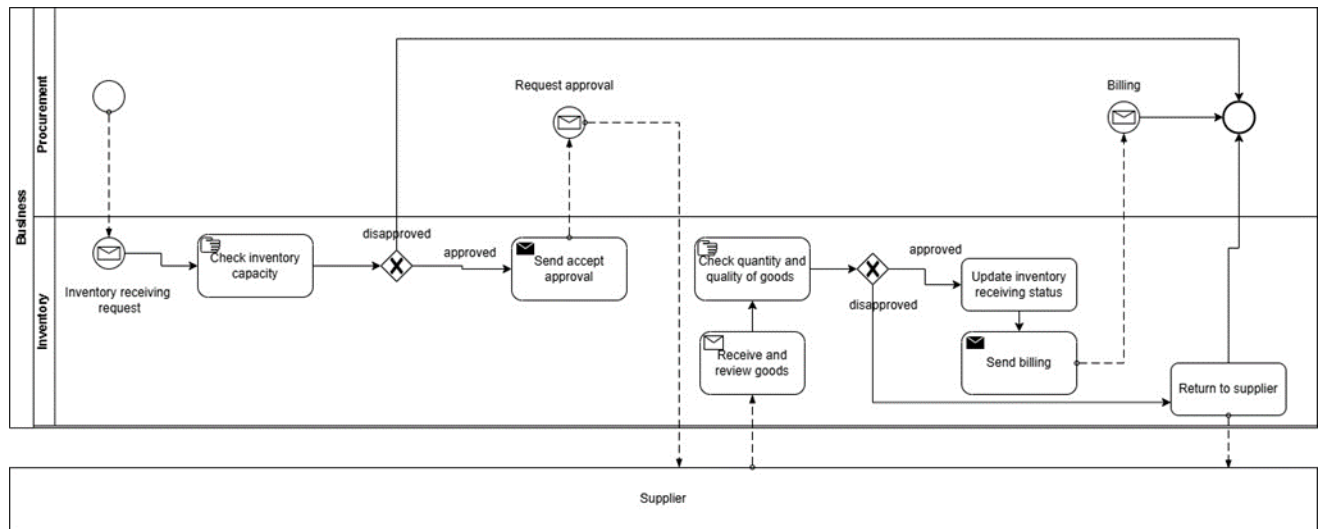


Figure 39: Image 3.4.1: BPMN model about inventory receiving process.

3.4.2. Specification

3.4.2.1. Specification of the steps of the warehousing process

Table 8: 3.4.2.1. Specification of the steps of the warehousing process

No	Actor	Actions	Description
1	Employee	Find the inventory slip on the system	Staff search and access the inventory slip on the Odoo system to view the details and status of the voucher
2	Employee	Check warehouse capacity	Employees check the capacity of the warehouse to ensure that the warehouse has enough space to receive new goods
3	Employee	Send a warehouse request notification	Employees send notifications or information related to the warehousing process to management for assistance or confirmation
4	Employee	Warehousing	Employees carry out the warehousing process by receiving goods from suppliers and confirming relevant information

5	Supplier	Receive requests from staff	The supplier receives the order request from the staff and confirms that the request has been received and is ready for processing
6	Supplier	Handling requests from agents	The supplier handles order requests by preparing goods, packing, and arranging transportation to the delivery address
7	Supplier	Supply of goods	The supplier delivers the goods on request from the staff, ensuring that the goods are delivered in the right quantity and quality
8	Employee	Receiving orders from suppliers	The staff receives orders from suppliers, checks and confirms that the goods are delivered in accordance with the requirements and quantity
9	Employee	Check quantity and quality	Employees check the quantity and quality of goods entering the warehouse to ensure that they meet the requirements and expected quality
10	Employee	Confirm	The employee confirms that the warehousing process is complete and the goods are ready to be used or distributed
11	Employee	Put up the file room	Employees bring information and documents related to the warehousing process to the records room for storage and management
12	Supplier	Receive requests from staff	The supplier receives the order request from the staff and confirms that the request has been received and is ready for processing

3.4.2.2. Data library specification:

PurchaseOrdersProductsReportingConfiguration

NewRequests for QuotationNew

Send by EmailPrint RFQConfirm OrderCancel

RFQRFQ SentPurchase Order

Request for Quotation

☆ New

Vendor ?AOC

Vendor Reference ?

Order Deadline ?06/01/2024 15:10:42

Expected Arrival ?06/01/2024 15:10:42

0% On-Time Delivery

☐ Ask confirmation

Deliver To ?ACCT5123 N22 CTTT: Receipts

ProductsOther Information

Product	Description	Expected Arrival	Quantity	Unit Price	Taxes	Tax excl.
[4.5L Dual Cook Pro Deep Fryer] ...	[4.5L Dual Cook Pro Deep Fryer] Chefman 4.5L Dual Cook Pro Deep Fryer with Basket Strainer and Removable Divider, Jumbo XL Size, Adjustable Temp & Timer, Perfect for Chicken, Fries, Chips and More, Easy to Clean, Stainless Steel	06/01/2024 15:10:42	500.00	69.99	10%	34,995 ₪
			0.00	0.00		0 ₪

[Add a product](#) [Add a section](#) [Add a note](#) [Catalog](#)

Define your terms and conditions ...

Untaxed Amount:34,995 ₪

VAT 10%:3,500 ₪

Total:38,495 ₪

Figure 40: Image 3.4.2.2: Request for Quotation

Table 9: 4.3.2.2. Data library specification:

No	Property	Description	Note
1	Vendor	The vendor data field of the product you want to add	Required field select
2	Vendor Reference	The field describes the vendor's reference path	Optional fields
3	Order deadline	The data field describes the date and time of the order	A field that is self-defined, or automatically defined at the time of order creation
4	Expected Arrival	Data field describing the desired delivery time	A field that is self-defined, or automatically defined at the time of order creation

5	Deliver to	A data field describing the destination inventory you want to import	Required field select
6	Product	The field describes the name of the product you want to enter	The field can be selected or left blank
7	Description	Data fields describing imported product details	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
8	Quantity	The data field describes the quantity to be imported of the selected product	Fields can be entered or left blank (default is 0)
9	Unit price	Data fields describing the price of each product	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
10	Taxes	Product tax description data field	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
11	Tax excl	Data field to product invoice excluding tax	The field is automatically defined when selecting the product name (defined when importing product data), which can be redefined through editing the product unit price

12	Discount	Data field describing the percentage discount for the product	Optional fields
13	Tax incl	Description data field to product invoice after tax is included	The field is automatically defined when selecting the product name (defined when importing product data), which can be redefined through editing unit price, taxes of the product

3.4.3. Illustration on Odoo

Step 1: Go to the "**Purchase**" module to create new order from supplier

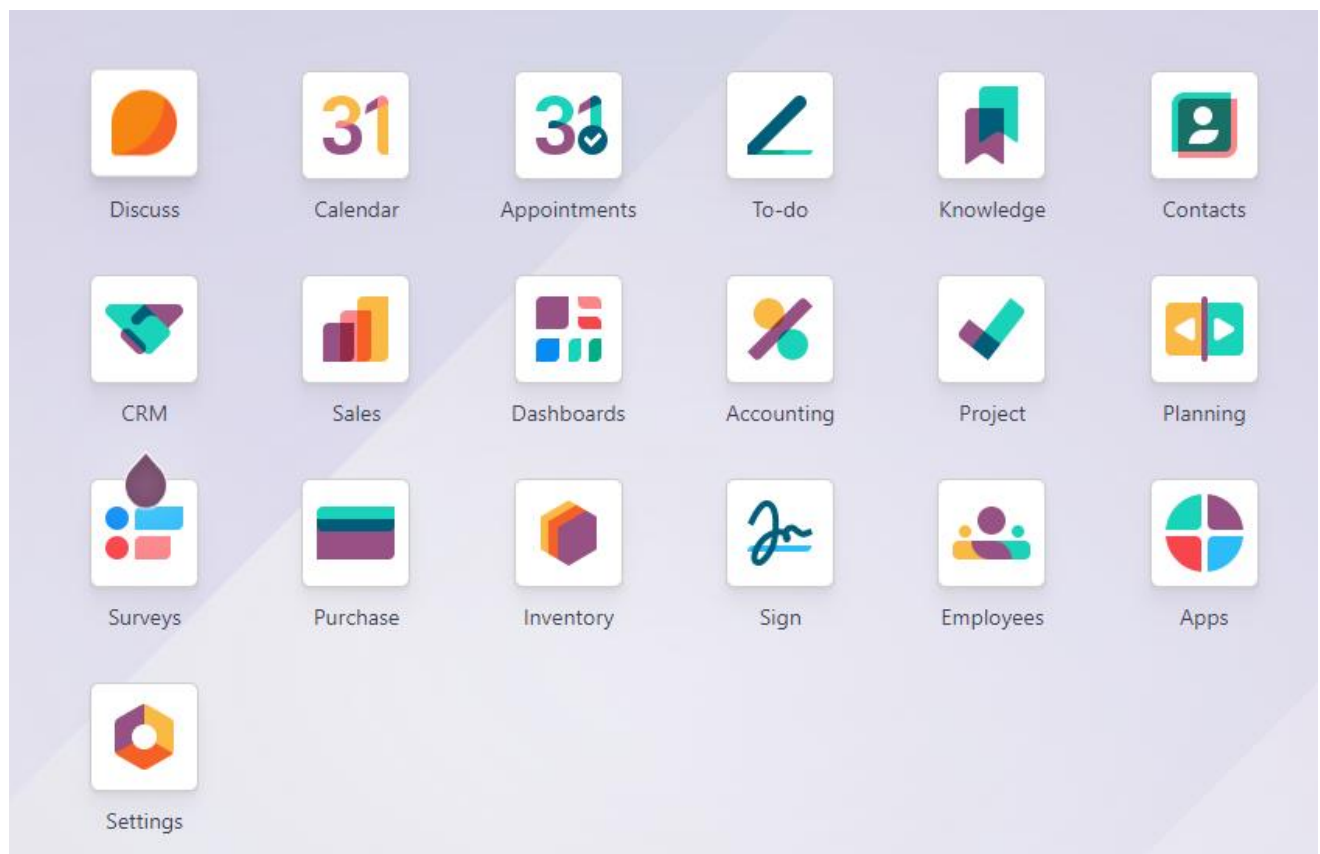


Figure 41: Image 3.4.3.1: Dashboard

Step 2: Click “New” to create order

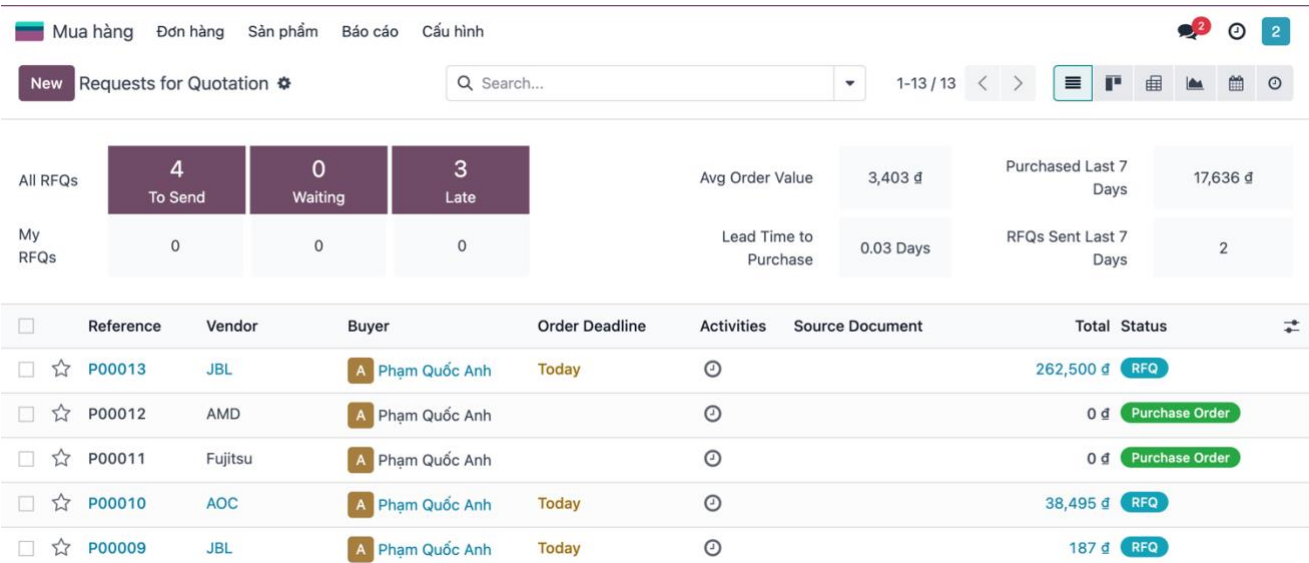


Figure 42: Image 3.4.3.2: Purchase dashboard.

Step 3: Enter field of order, then click “Confirm Order” to confirm order.

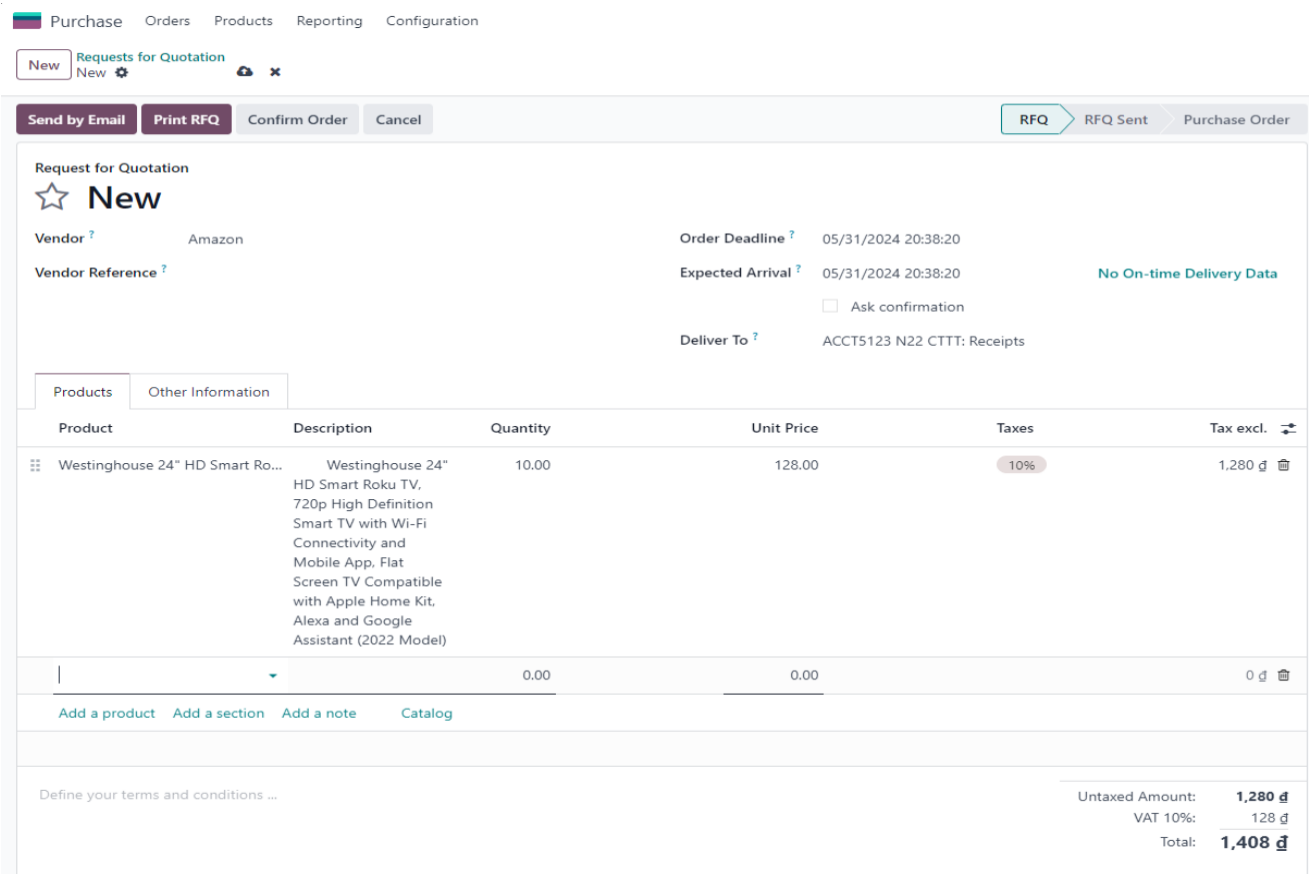


Figure 43: Image 3.4.3.3: Requests for Quotation form.

Step 4: After confirm order, the bill will be generate, then click “Create Bill” to create bill order

PurchaseOrdersProductsReportingConfiguration

NewRequests for QuotationP00007Receipt1

Receive ProductsSend PO by EmailCreate BillCancelLockRFQRFQ SentPurchase Order

Purchase Order☆P00007

Vendor ?ADInstruments

Vendor Reference ?

Confirmation Date06/01/2024 14:23:30

Expected Arrival ?06/01/2024 14:23:10

☐ Ask confirmation

Deliver To ?ACCT5123 N22 CTTT: Receipts

ProductsOther Information

Product	Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
⌵ [32S335] TCL 32-inch 3-S...	[32S335] TCL 32-inch 3-Series 720p Roku Smart TV - 32S335, 2021 Model	10.00	0.00	0.00	205.18	10%	2,052 g

Add a productAdd a sectionAdd a noteCatalog

Define your terms and conditions ...

Untaxed Amount:2,052 gVAT 10%:205 gTotal:2,257 g

Figure 44: Image 3.4.3.4: Requests for Quotation

Step 5: Return to homescreen, click “Inventory” module to check the bill order.

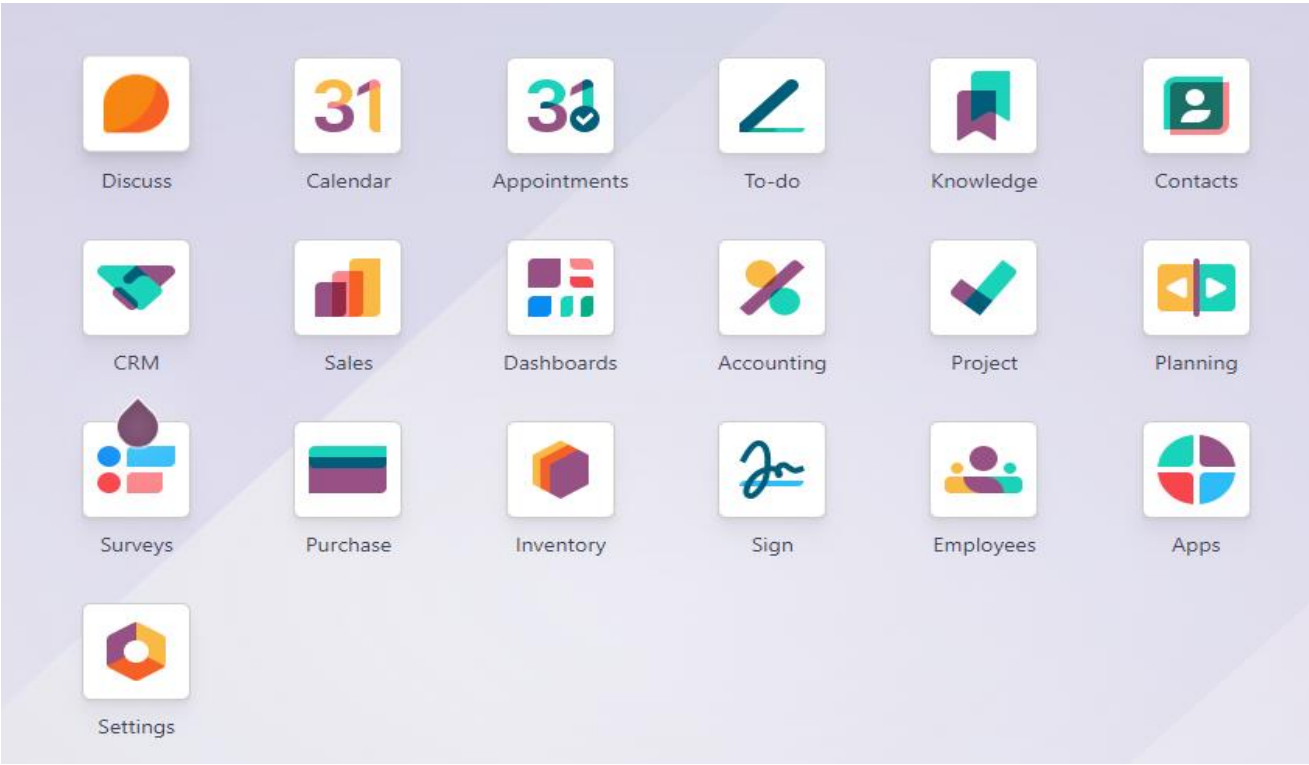


Figure 45: Image 3.4.3.5: Dashboard

As you can see, in receipt of corresponding warehouse will display the the number of process.
So, we click on it, we will see the order is waiting for deliver.

Receipts ACCT5123 N22 CTTT 1 To Process 1 Late	Internal Transfers ACCT5123 N22 CTTT 1 To Process 1 Late	Delivery Orders ACCT5123 N22 CTTT 1 To Process 1 Late	Phiếu nhập kho Warehouse 1 45 Nguyen Tat Thanh, Phuong Ngo May, TP Quy Nhon, Binh Dinh 0 To Process	Điều chuyển nội bộ Warehouse 1 45 Nguyen Tat Thanh, Phuong Ngo May, TP Quy Nhon, Binh Dinh 0 To Process
Phiếu xuất kho Warehouse 1 45 Nguyen Tat Thanh, Phuong Ngo May, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu nhập kho Warehouse 2 10 Le Hong Phong, Phuong Le Hong Phong, TP Quy Nhon, Binh Dinh 0 To Process	Điều chuyển nội bộ Warehouse 2 10 Le Hong Phong, Phuong Le Hong Phong, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu xuất kho Warehouse 2 10 Le Hong Phong, Phuong Le Hong Phong, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu nhập kho Warehouse 3 23 Tran Hung Dao, Phuong Hai Cang, TP Quy Nhon, Binh Dinh 0 To Process
Điều chuyển nội bộ Warehouse 3 23 Tran Hung Dao, Phuong Hai Cang, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu xuất kho Warehouse 3 23 Tran Hung Dao, Phuong Hai Cang, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu nhập kho Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh 0 To Process	Điều chuyển nội bộ Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu xuất kho Warehouse 4 78 Le Loi, Phuong Le Loi, TP Quy Nhon, Binh Dinh 0 To Process
Phiếu nhập kho Warehouse 5 90 Nguyen Hue, Phuong Tran Phu, TP Quy Nhon, Binh Dinh 0 To Process	Điều chuyển nội bộ Warehouse 5 90 Nguyen Hue, Phuong Tran Phu, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu xuất kho Warehouse 5 90 Nguyen Hue, Phuong Tran Phu, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu nhập kho Warehouse 6 12 Le Duan, Phuong Nhon Phu, TP Quy Nhon, Binh Dinh 0 To Process	Điều chuyển nội bộ Warehouse 6 12 Le Duan, Phuong Nhon Phu, TP Quy Nhon, Binh Dinh 0 To Process
Phiếu xuất kho Warehouse 6 12 Le Duan, Phuong Nhon Phu, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu nhập kho Warehouse 7 56 Hai Ba Trung, Phuong Dong Da, TP Quy Nhon, Binh Dinh 0 To Process	Điều chuyển nội bộ Warehouse 7 56 Hai Ba Trung, Phuong Dong Da, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu xuất kho Warehouse 7 56 Hai Ba Trung, Phuong Dong Da, TP Quy Nhon, Binh Dinh 0 To Process	Phiếu nhập kho Warehouse 8 30 Nguyen Van Cu, Phuong Ghenh Rang, TP Quy Nhon, Binh Dinh 0 To Process

Figure 46: Image 3.4.3.6: Inventory dashboard.

This is the screen of process:

Inventory Overview							
ACCT5123 N22 CTTT: Receipts							
Operation Type: ACCT5123 N22 CTTT: Receipts X Ready X Search...							
1-1 / 1							
Reference	From	To	Contact	Scheduled Date	Source Document	Status	
WH/IN/00003	Partners/Vendors	WH/Stock	Amazon	Today	P00004	Ready	

Figure 47: Image 3.4.3.7: Inventory Overview

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Step 6: Click to the order, we will see the details of order that is waiting to deliver to the destination’s warehouse. Click to the button “Validate” to access the order deliver to the warehouse.

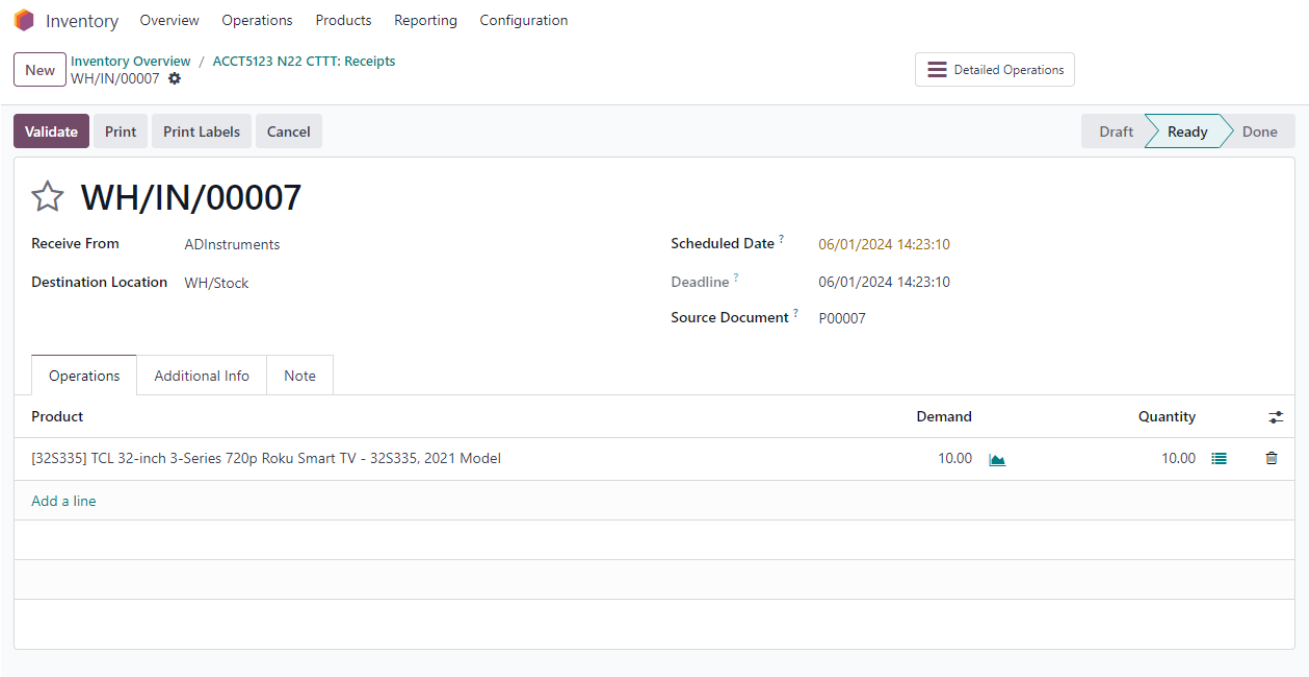


Figure 48: Image 3.4.3.8: Inventory Overview New order.

Step 7: This is the screen after validate. Click to “Product” to check that products have been deliver to the warehouse yet or not.

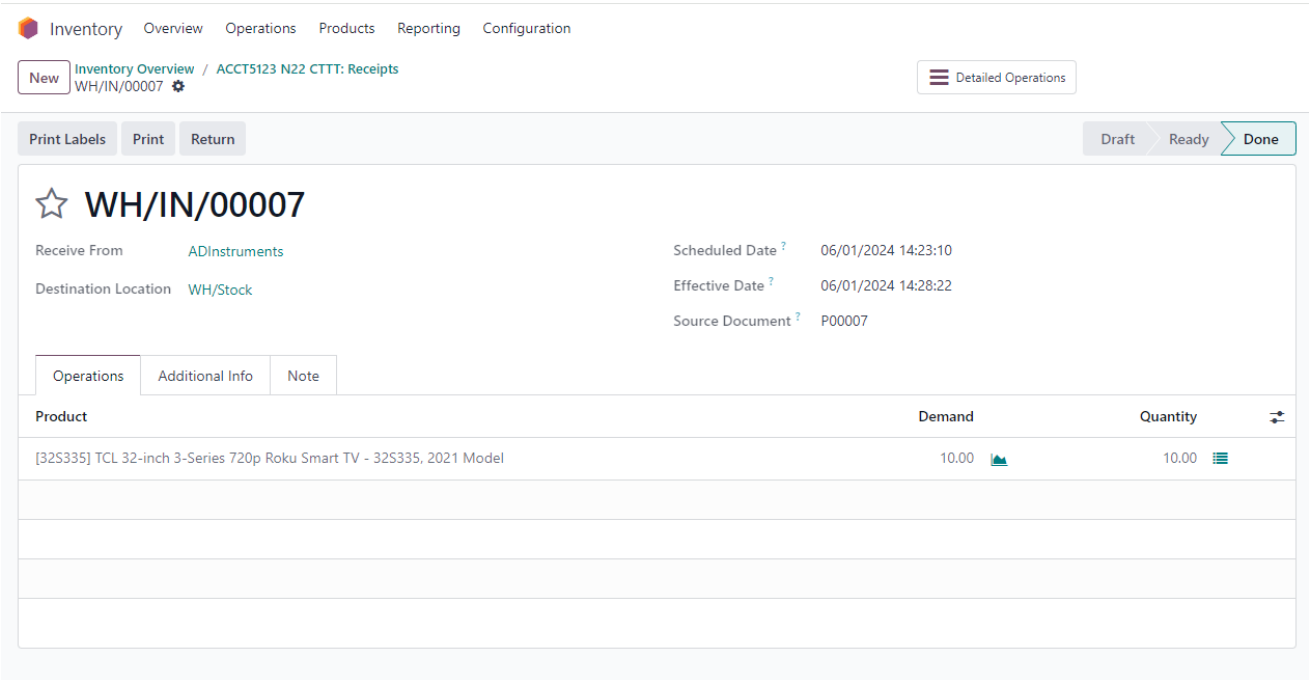


Figure 49: Image 3.4.3.9: Inventory Overview

This is the screen when we click “**Products**” button:

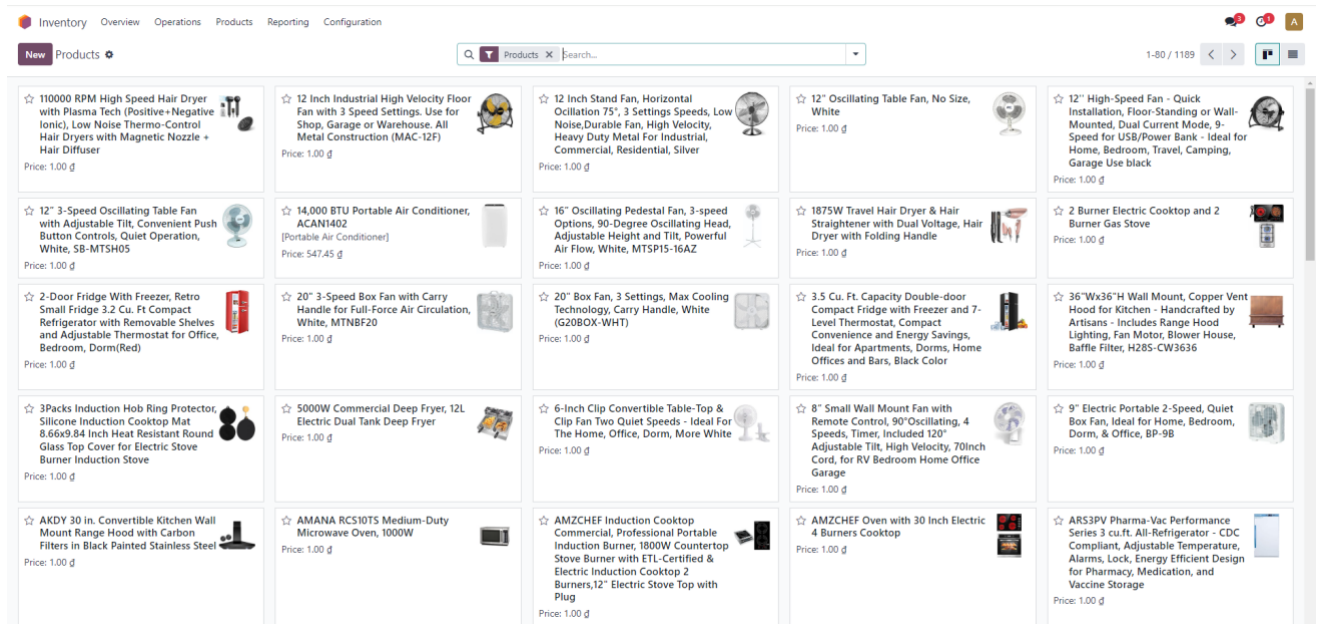


Figure 50: Image 3.4.3.10: Production list.

This is the result after we create new order from supplier to warehouse.

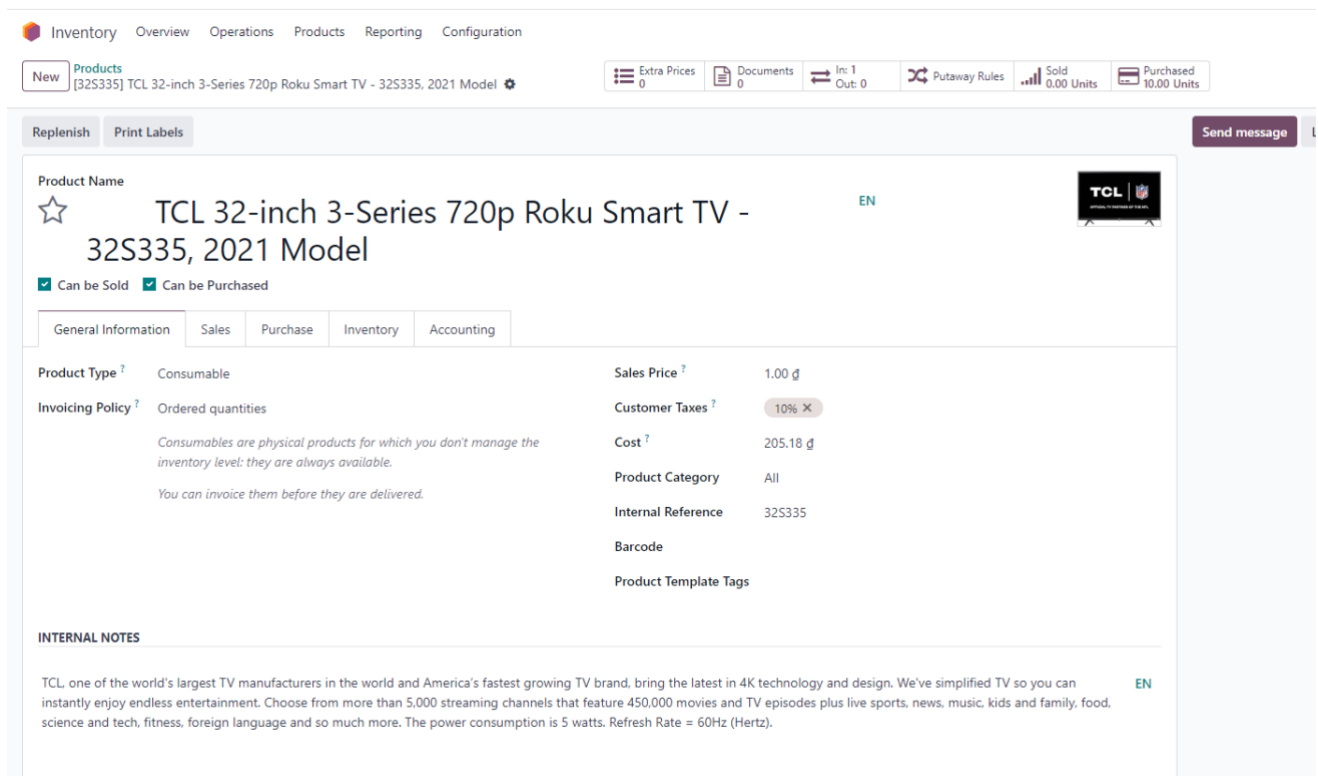


Figure 51: Image 3.4.3.11: Purchase products.

3.5. Inventory shipping process.

3.5.1. BPMN

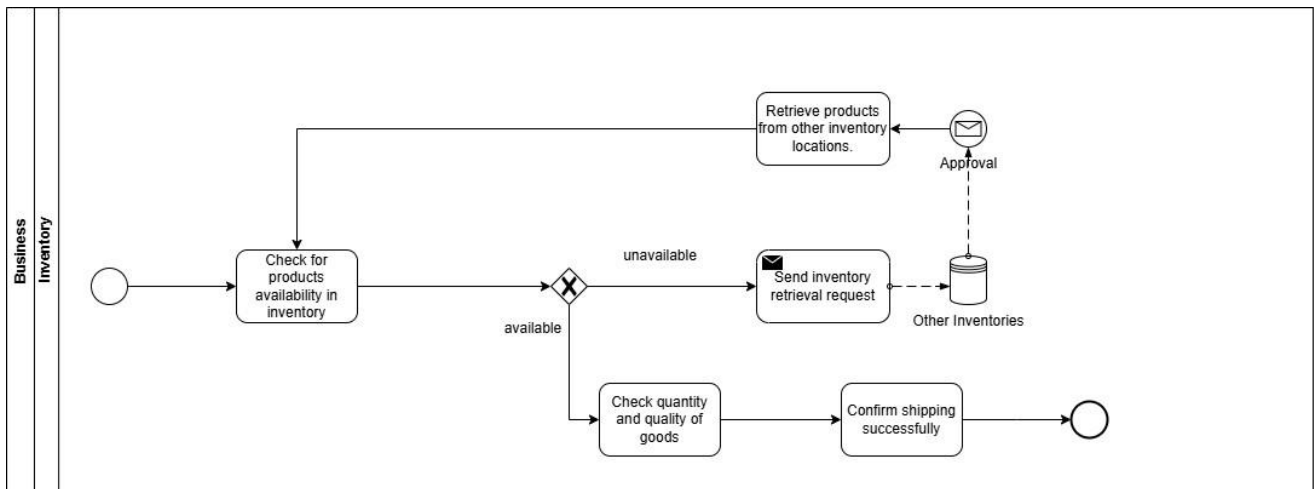


Figure 52: Image 3.5.1.1: BPMN model about inventory shipping process.

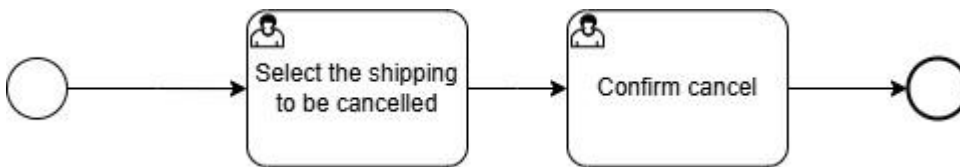


Figure 53: Image 3.5.1.2: BPMN model about inventory shipping process.

3.5.2. Specification

3.5.2.1. Specification of the steps of the Inventory Shipping process

Table 10: 3.5.2.1. Specification of the steps of the Inventory Shipping process

No	Department	Action	Description
1	Inventory	Check for product availability in inventory	Check for product availability in inventory
2	Inventory	Send inventory retrieval request	Send inventory retrieval request to other inventories
3	Inventory	Approve inventory retrieval request	Approve inventory retrieval request from the other inventory
4	Inventory	Retrieve products from the other inventory location	Proceed to retrieve products from the other inventory location

5	Inventory	Receive and check retrieved products	Receive and check retrieved products
6	Inventory	Confirm	Confirm inventory shipping successfully

3.5.2.2. Data library specification:

Table 11: 3.5.2.2. Data library specification:

No	Property	Description	Note
1	Name	Code/name of internal transfer	Required
2	Contact	Contact related to user or supplier who are responsible for this transfer	Not required
3	Source Location	Source inventory location	Required
4	Scheduled Date	Scheduled time for the first part of the shipment to be processed	Required
5	Source Document	Reference to the document	Not required
6	Operations		
6.1	Product	Product name	Required
6.2	Demand	Quantity needs to be shipped	Required
6.3	Quantity	Current number of available products	Automatically display
7	Additional Info		
7.1	Other Information		
7.1.1	Shipping policy	Specify goods to be delivered partially or all at once	Required
7.1.2	Responsible	Internal user who is responsible for this transfer	Automatically assigned to current user
8	Note	Internal note that will be printed on the Picking Operations sheet	Not required

3.5.3. Illustration on Odoo

Step 1: Go to “Inventory” module.

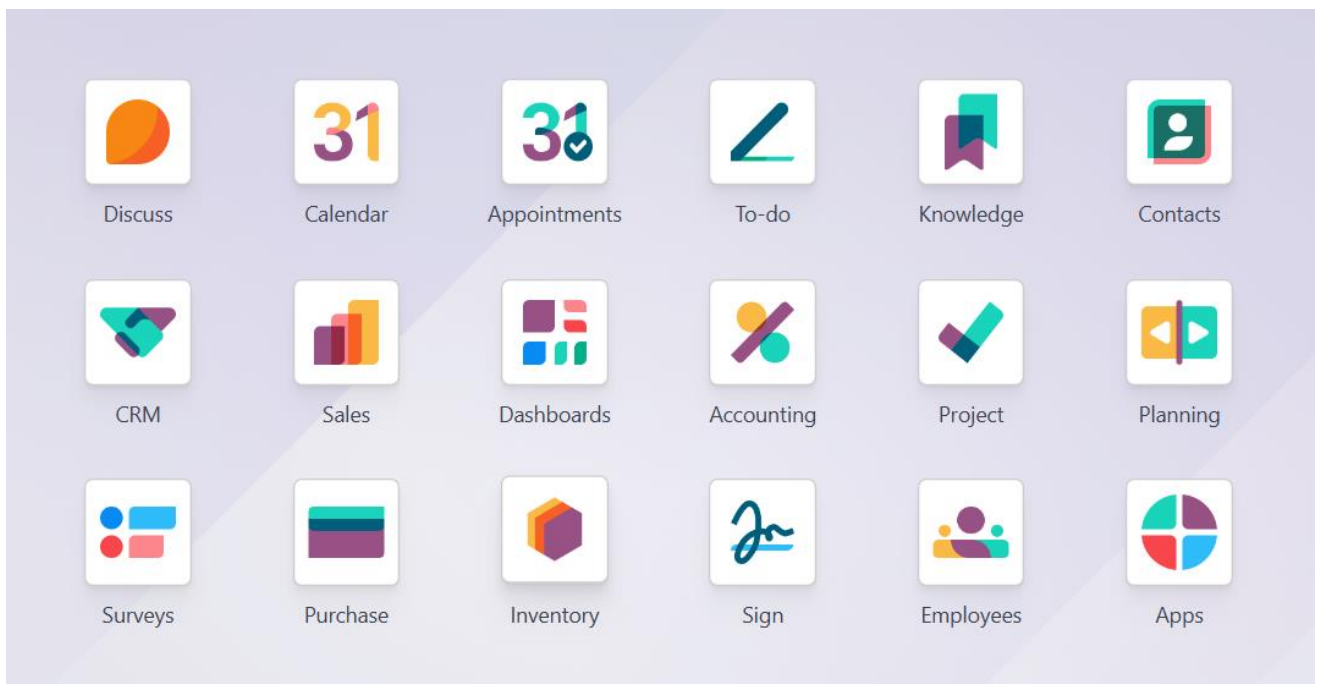


Figure 54: Image 3.5.3.1: Dashboard

Step 2: Click on “Operations” menu and navigate to Internals Transfers page. Then click “New” to create a new Internal Transfer

Inventory Overview Operations Products Reporting Configuration							
New Internal Transfers							
Internal Search...							
1-3 / 3							
☐	Reference	From	To	Contact	Scheduled Date	Source Document	Status
☐	WH/INT/00002	WH/Stock	WH/Stock				Cancelled
☐	WH/INT/00003	WH/Stock	WH/Stock		Today		Ready
☐	WH/INT/00001	WH/Stock	WH 10/Ton kho	19522476			Done

Figure 55: Image 3.5.3.2: Inventory dashboard.

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Step 3: Fill necessary fields to create an internal transfer (Contact, Operation Type, Source Location, Destination Location, Product and Demand). Then click “Mark as Todo” to confirm creating internal transfer.

New Internal Transfers / Detailed Operations

Mark as Todo Validate Print Labels Cancel Draft Waiting Ready Done

Contact 19522476 Scheduled Date 06/02/2024 21:26:35
Operation Type ACCT5123 N22 CTTT: Internal Transfers Source Document e.g. PO0032
Source Location WH 4/Tôn kho
Destination Location WH/Stock

Operations Additional Info Note

Product	Demand
[32S355] TCL 32" Class 3-Series HD 720p LED Smart Roku TV - 32S355, Black	5.00
[43 inch Mirror 4K TV Android OS] Soulaca 43 inches Smart Android Mirror 4K LED TV Big Screen IP65 Waterproof Built-in WiFi Bluetooth SPA ATSC DTV	5.00
Add a line	

Send message Log note Activities

Figure 56: Image 3.5.3.3: Inventory form.

Step 4: When goods are delivered, make sure goods’ quantity and quality are checked, then click “Validate” to complete the process.

New Internal Transfers WH/INT/00004 / Detailed Operations

Validate Print Print Labels Cancel Draft Waiting Ready Done

☆ WH/INT/00004

Contact 19522476 Scheduled Date 06/02/2024 21:26:35
Operation Type ACCT5123 N22 CTTT: Internal Transfers Source Document e.g. PO0032
Source Location WH 4/Tôn kho
Destination Location WH/Stock

Operations Additional Info Note

Product	Demand	Quantity
[32S355] TCL 32" Class 3-Series HD 720p LED Smart Roku TV - 32S355, Black	5.00	5.00
[43 inch Mirror 4K TV Android OS] Soulaca 43 inches Smart Android Mirror 4K LED TV Big Screen IP65 Waterproof Built-in WiFi Bluetooth SPA ATSC DTV	5.00	5.00
Add a line		

Send message Log note Activities

Figure 57: Image 3.5.3.4: Inventory form.

3.6. Financial and Accounting management process

The financial and accounting management process encompasses a range of activities critical to the fiscal health and regulatory compliance of an organization. In an ERP system,

this process is streamlined and integrated, allowing for efficient management of financial transactions, reporting, and analysis.

3.6.1. BPMN

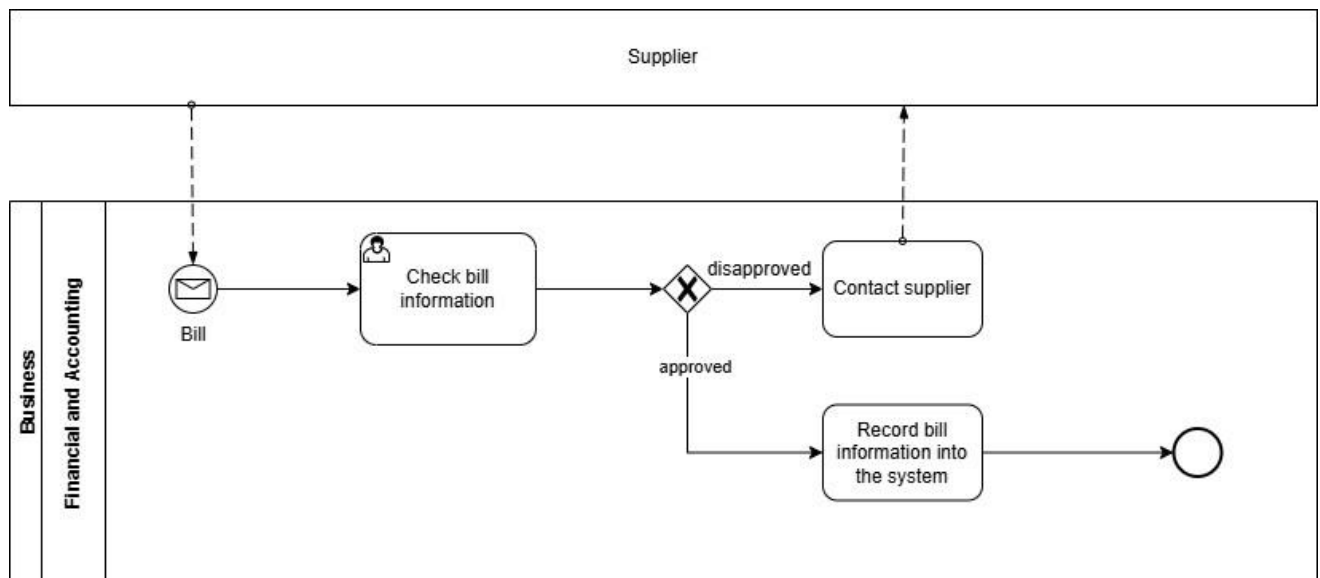


Figure 58. BPMN Purchase Invoice Processing Process

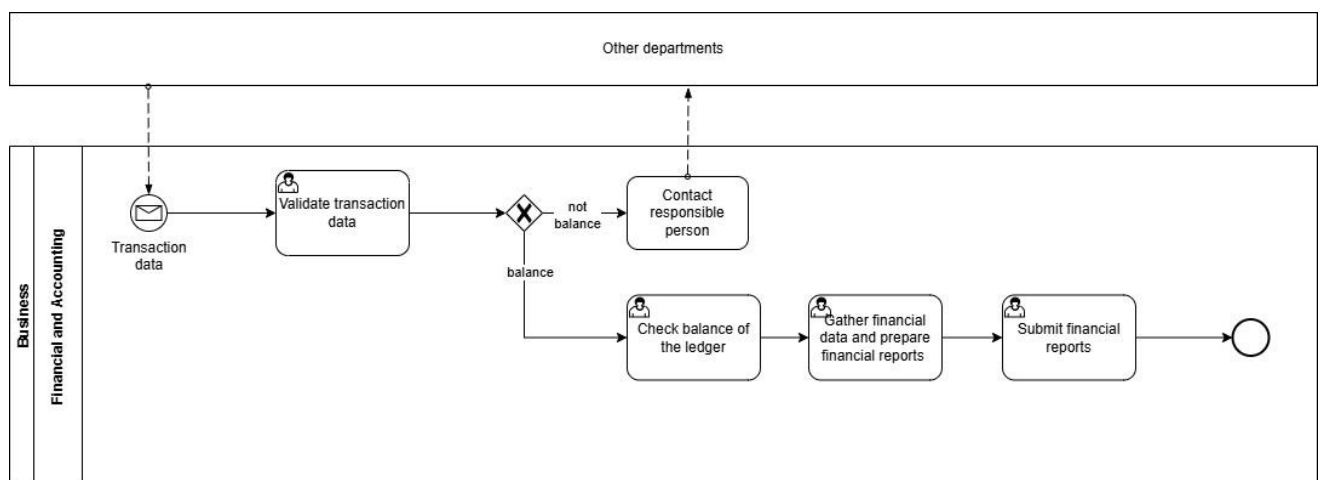


Figure 59. BPMN Accounting Recording Process and Financial Reporting Process

3.6.2. Specification

3.6.2.1. Specification of process steps

Table 12: 3.6.2.1. Specification of process steps

No.	Actor	Action	Description
-----	-------	--------	-------------

1	Supplier	Provide payment invoices	The supplier will send invoices to pay for the products sold to the company
2	Accountant	Receive invoices from suppliers.	The invoice is received from the supplier and entered the processing process.
3	Accountant	Check invoice information.	Information on invoices is checked to ensure accuracy and completeness.
4	Accountant	Verify invoice accuracy	Invoices are verified to ensure there are no errors
5	Accountant	Case: No	If there are errors, the invoice is returned to the supplier
6	Accountant	Case: Yes	If there are no errors, the invoice will continue to be processed
7	Accountant	Record invoices into the system	Invoices are recorded in the company's accounting system
8	Accountant	Pay the bill	The process ends when the invoice is paid in full

Table 13: 3.6.2.1. Specification of process steps

No.	Actor	Action	Description
1	Department	Provide payment invoices	Departments will send transaction data to the accounting department

2	Accountant	Verify and classify transactions	Transaction data from other departments is included in the process
3	Accountant	Record transactions in the ledger	Transactions are verified and classified according to accounting criteria
4	Accountant	Check the balance of the ledger	The ledger is checked to ensure that total debits and total credits are equal
5	Accountant	Case: Balanced	If balanced, the process ends.
6	Accountant	Case: Unbalanced	If unbalanced, the error will be found and corrected.
7	Accountant	Gather financial data	Financial data is gathered from relevant departments
8	Accountant	Prepare financial reports	Financial reports are prepared based on aggregated data
9	Accountant	Check and approve reports	Financial statements are checked for accuracy and completeness, then approved
10	Accountant	Case: No	Repair reports
11	Accountant	Case: Yes	Send report
12	Accountant	Submit financial reports	Financial reports are sent to relevant parties

3.6.2.2. Payment method*Table 14: 3.6.2.2. Payment method*

No.	Actor	Action	Description
1	Journal	Here's your choice of payment methods, including bank and cash	These properties are required
2	Payment Method	Payment method	These properties are required
3	Amount	Total order amount	These properties are required
4	Payment Date	Payment Date	These properties are required
5	Memo	Payment invoice code	These properties are not required
6	Receipt Bank Account	The name of the bank you will use to make the payment	These properties are not required

3.6.3. Illustration on Odoo

- **Import goods from suppliers.**

Step 1: I create a transaction to deliver products to my warehouse, after that i click “**Create bill**”

ACCT5123 - Enterprise resource planning

Purchase Orders Products Reporting Configuration

New

Requests for Quotation

P00019

Receipt 1

Receive Products Send PO by Email Create Bill Cancel Lock

RFQ RFQ Sent Purchase Order

Purchase Order

☆ P00019

Vendor ? ADInstruments

Vendor Reference ?

Confirmation Date 06/01/2024 21:42:18

Expected Arrival ? 06/01/2024 21:41:32

☐ Ask confirmation

Deliver To ? ACCT5123 N22 CTTT: Receipts

Products Other Information

Product	Description	Quantity	Received	Billed	Unit Price	Taxes	Tax excl.
⌵ [FLIP 5] JBL FLIP 5 Portable S...	[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	50.00	0.00	0.00	5,000,000.00	10%	250,000,000 ₪

Add a product Add a section Add a note Catalog

Define your terms and conditions ...

Untaxed Amount: 250,000,000 ₪

VAT 10%: 25,000,000 ₪

Total: 275,000,000 ₪

Figure 60: Image 3.6.3.1: Purchase product.

Then I click “Confirm” to create the bill

Purchase Orders Products Reporting Configuration

New

Purchase Orders / P00019

Draft Bill (* 21)

Purchases 1

Confirm Cancel

Draft Posted

Vendor Bill

Draft

Vendor ADInstruments

Bill Reference

Auto-Complete ? Select a purchase order or an old bill

Bill Date

Accounting Date 06/01/2024

Payment Reference ?

Recipient Bank ?

Due Date 06/01/2024 or Payment Terms

Journal Vendor Bills

Invoice Lines Journal Items Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
⌵ [FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (B	P00019: [FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	1561 Purchase cos...	50.00	5,000,000.00	10%	250,000,000 ₪

Add a line Add a section Add a note Catalog

Terms and Conditions

Untaxed Amount: 250,000,000 ₪

Figure 61: Image 3.6.3.2: Purchase product.

Then click “Register Payment” to pay the bills

Purchase

Orders

Products

Reporting

Configuration

New

Purchase Orders / P00019

BILL/2024/06/0005

Purchases

1

Register Payment

Credit Note

Reset to Draft

Draft

Posted

Vendor Bill

BILL/2024/06/0005

Vendor ADInstruments

Bill Date 06/01/2024

Accounting Date 06/01/2024

Payment Reference ?

Recipient Bank ?

Due Date 06/01/2024

Journal Vendor Bills

Bill Reference

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case	P00019: [FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	1561 Purchase costs	50.00	5,000,000.00	10%	250,000,000 đ

Terms and Conditions

Untaxed Amount: 250,000,000 đ

Figure 62: Image 3.6.3.3: Purchase product.

Then click “Create Payment”

Register Payment

Journal Bank

Amount 5,500,000 đ

Payment Method ? Manual

Payment Date 06/01/2024

Recipient Bank Account

Memo BILL/2024/06/0007

Create Payment

Discard

Figure 63: Image 3.6.3.3: Purchase product.

After paying the invoice, I ship the goods to my warehouse

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PurchaseOrdersProductsReportingConfiguration

New

Requests for Quotation / P00019

BILL/2024/06/0005

Purchases

Credit NoteReset to Draft

DraftPosted

Vendor Bill

BILL/2024/06/0005

VendorADInstruments

Bill Reference

Bill Date06/01/2024

Accounting Date06/01/2024

Payment Reference ?

Recipient Bank ?

Due Date06/01/2024

JournalVendor Bills

IN PAYMENT

Invoice LinesJournal ItemsOther Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (B P00019:	[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterpro... Bundle with gSport Deluxe Hardshell Case (Black)	1561 Purchase cos...	50.00	5,000,000.00	10%	250,000,000 đ

Terms and Conditions

Untaxed Amount:250,000,000 đ

Figure 64: Image 3.6.3.4: Purchase product.

- Sell products to customers.

PurchaseOrdersProductsReportingConfiguration

New

Requests for Quotation / P00019

WH/IN/00015

Detailed Operations

Print LabelsPrintReturn

DraftReadyDone

☆ WH/IN/00015

Receive FromADInstruments

Destination LocationWH/Stock

Scheduled Date ?06/01/2024 21:41:32

Effective Date ?06/01/2024 21:42:43

Source Document ?P00019

OperationsAdditional InfoNote

Product	Demand	Quantity
[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	50.00	50.00

Figure 65: Image 3.6.3.5: Delivery product to warehouse.

Step 1: Click “Create invoice” to start creating sales invoices for customers

SalesOrdersTo InvoiceProductsReportingConfiguration

New

Quotations

S00007

Delivery

1

Create InvoiceSend by EmailPreviewCancel

QuotationQuotation SentSales Order

S00007

Customer

David Cummings

Order Date ?

06/01/2024 21:44:26

Pricelist ?

Default VND pricelist (VND)

Payment Terms

Order LinesOther Info

Product	Description	Quantity	Invoiced	Unit Price	Taxes	Tax excl.
[FLIP 5] JBL FLIP 5 Portable Spe	[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	1.00	0.00	6,500,000.00	10%	6,500,000

Add a product

Add a section

Add a note

Catalog

Terms and conditions...

Untaxed Amount:

6,500,000

VAT 10%:

650,000

Total:

7,150,000

Figure 66: Image 3.6.3.6: Sale product to customer.

Then click “Create Draft”

Create invoice(s)

Create Invoice ?

Regular invoice

Down payment (percentage)

Down payment (fixed amount)

Create Draft

Cancel

Figure 67: Image 3.6.3.7: Create an invoice.

Step 2: We click “Register Payment” to create payment methods for customers

SalesOrdersTo InvoiceProductsReportingConfiguration

NewQuotations / S00007INV/2024/00005

Sale Orders

Send & PrintRegister PaymentPreviewCredit NoteReset to DraftDraftPosted

Customer Invoice

INV/2024/00005

Customer: David Cummings

Invoice Date: 06/01/2024

Payment Reference: INV/2024/00005

Due Date: 06/01/2024

Journal: Customer Invoices

Invoice LinesJournal ItemsOther Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport	[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	5111 Revenue from sales of merchandises	1.00	6,500,000.00	10%	6,500,000 đ

Terms and Conditions

Untaxed Amount: 6,500,000 đ
VAT 10%: 650,000 đ
Total: 7,150,000 đ
Amount Due: 7,150,000 đ

Figure 68: Image 3.6.3.8: Create an invoice.

Then click “Create Payment”

Register Payment

Journal: BankAmount: 7,150,000 đ

Payment Method: ManualPayment Date: 06/01/2024

Recipient Bank AccountMemo: INV/2024/00007

Create PaymentDiscard

Figure 69: Image 3.6.3.9: Payment

After that, we will see "In Payment" appear, which means the customer has paid for the item. We click “Sale Orders” to delivery product to customer

ACCT5123 - Enterprise resource planning

SalesOrdersTo InvoiceProductsReportingConfiguration

New

Quotations / S00007
INV/2024/00005

Sale Orders
1

Send & PrintPreviewCredit NoteReset to Draft

DraftPosted

Customer Invoice

INV/2024/00005

CustomerDavid Cummings

Invoice Date06/01/2024

Payment Reference ?INV/2024/00005

Due Date06/01/2024

JournalCustomer Invoices

Invoice LinesJournal ItemsOther Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport [FLIP 5]	JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	5111 Revenue from sales of merchandises	1.00	6,500,000.00	10%	6,500,000 đ

Terms and Conditions

Untaxed Amount:6,500,000 đ

VAT 10%:650,000 đ

Total:7,150,000 đ

Paid on 06/01/2024

7,150,000 đ

Figure 70: Image 3.6.3.10: Payment

SalesOrdersTo InvoiceProductsReportingConfiguration

New

... / INV/2024/00005 / S00007
WH/OUT/00009

Detailed Operations

Print LabelsPrintReturn

DraftWaitingReadyDone

☆ WH/OUT/00009

Delivery AddressDavid Cummings

Scheduled Date ?06/01/2024 21:44:26

Source LocationWH/Stock

Effective Date ?06/01/2024 21:45:54

Source Document ?S00007

OperationsAdditional InfoNote

Product	Demand	Quantity
[FLIP 5] JBL FLIP 5 Portable Speaker IPX7 Waterproof Bundle with gSport Deluxe Hardshell Case (Black)	1.00	1.00

Figure 71: Image 3.6.3.11: Delivery product to customer.

ACCT5123 - Enterprise resource planning

Step 3: We go to the Accounting, click “Reporting” > “Journal Report” to see the amount we have imported goods from suppliers and the number of sales to customers

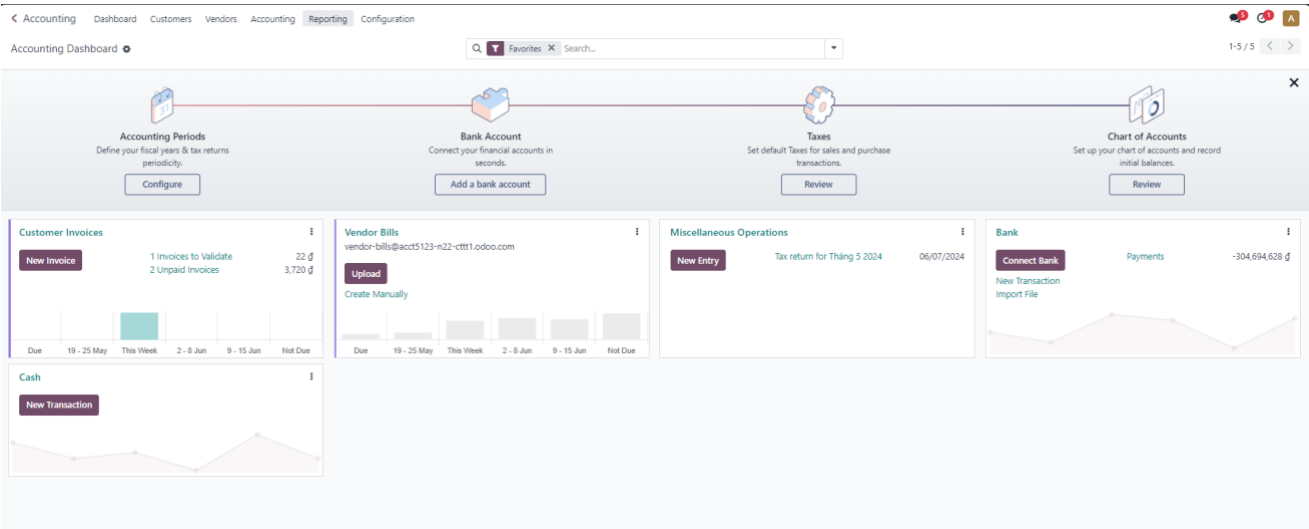


Figure 72: Image 3.6.3.12: Accounting dashboard.

Customer Invoices (INV) Journal Items							
Name	Invoice Date	Account	Label	Debit	Credit	Taxes	Tax Grids
INV/2024/00001	2024-05-24	1310 Aaron Page	INV/2024/00001	220 g	0 g		
		5111 Revenue from sales of merchandises	[FBA_JBLFLUP3GRAY] JBL Flip3 Splashproof B ...	0 g	200 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	20 g	B: 200 g	+VAT on sales of goods and services 10%
INV/2024/00002	2024-06-01	1310 Aaron Jackson	INV/2024/00002	3,535 g	0 g		
		5111 Revenue from sales of merchandises	12 inch Industrial High Velocity Floor Fan ...	0 g	3,214 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	321 g	B: 3,214 g	+VAT on sales of goods and services 10%
INV/2024/00003	2024-06-01	1310 Charles Long	INV/2024/00003	22,000 g	0 g		
		5111 Revenue from sales of merchandises	[325335] TCL 32-inch 3-Series 720p Roku Sm ...	0 g	20,000 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	2,000 g	B: 20,000 g	+VAT on sales of goods and services 10%
INV/2024/00004	2024-06-01	1310 Ethan Harris	INV/2024/00004	11,000,000 g	0 g		
		5111 Revenue from sales of merchandises	[MC18] Portable Swamp Coolers - 1300 CFM M ...	0 g	10,000,000 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	1,000,000 g	B: 10,000,000 g	+VAT on sales of goods and services 10%
INV/2024/00005	2024-06-01	1310 David Cummings	INV/2024/00005	7,150,000 g	0 g		
		5111 Revenue from sales of merchandises	[FLUP 5] JBL FLUP 5 Portable Speaker IPX7 ...	0 g	6,500,000 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	650,000 g	B: 6,500,000 g	+VAT on sales of goods and services 10%
INV/2024/00006	2024-06-01	1310 Aaron Page	INV/2024/00006	185 g	0 g		
		5111 Revenue from sales of merchandises	[MC18] Portable Swamp Coolers - 1300 CFM M ...	0 g	168 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	17 g	B: 168 g	+VAT on sales of goods and services 10%
INV/2024/00007	2024-06-01	1310 Adam Glover	INV/2024/00007	7,150,000 g	0 g		
		5111 Revenue from sales of merchandises	[MC18] Portable Swamp Coolers - 1300 CFM M ...	0 g	6,500,000 g	T: 10%	+Untaxed sales of goods and services taxed 10%
		33311 Output VAT		0 g	650,000 g	B: 6,500,000 g	+VAT on sales of goods and services 10%

Taxes Applied Audit			
Name	Base Amount	Tax Amount	Due
10%	23,023,582 g	2,302,358 g	2,302,358 g

Impacted Tax Grids Audit			
Grid	+	-	Impact On Grid
Untaxed sales of goods and services taxed 10%	23,023,582 g	0 g	23,023,582 g
VAT on sales of goods and services 10%	2,302,358 g	0 g	2,302,358 g

Figure 73: Image 3.6.3.13: Customer Invoices

Vendor Bills (BILL) Journal Items							
Name	Invoice Date	Account	Label	Debit	Credit	Taxes	Tax Grids
BILL/2024/05/0001	2024-05-24	3310 JBL		0 g	1,400 g		
		1561 Purchase costs	P0001: (FBA_JBLFLIP3GRAV) JBL Flip3 Splas ---	1,273 g	0 g	T: 10%	+Untaxed Purchase of Goods and Services taxed 10%
		1331 Deductible VAT of goods and services	10%	127 g	0 g	B: 1,273 g	+VAT on purchase of goods and services 10%
BILL/2024/06/0002	2024-06-01	3310 Oppo		0 g	12,863 g		
		1561 Purchase costs	P0008: (Beosound A1 2nd Gen) Bang & Olufsen ---	11,694 g	0 g	T: 10%	+Untaxed Purchase of Goods and Services taxed 10%
		1331 Deductible VAT of goods and services	10%	1,169 g	0 g	B: 11,694 g	+VAT on purchase of goods and services 10%
BILL/2024/06/0003	2024-06-01	3310 CEA Technologies		0 g	2,585 g		
		1561 Purchase costs	P00015: (AEPIS2) Klein Tools AEPIS2 Blueto ---	400 g	0 g	T: 10%	+Untaxed Purchase of Goods and Services taxed 10%
		1561 Purchase costs	P00015: (40S350R) TCL 40-Inch Class S3 108 ---	1,950 g	0 g	T: 10%	+Untaxed Purchase of Goods and Services taxed 10%
		1331 Deductible VAT of goods and services	10%	235 g	0 g	B: 2,350 g	+VAT on purchase of goods and services 10%
BILL/2024/06/0004	2024-06-01	3310 Viewsonic		0 g	55,000,000 g		
		1561 Purchase costs	P00018: (MC18) Portable Swamp Coolers - 13 ---	50,000,000 g	0 g	T: 10%	+Untaxed Purchase of Goods and Services taxed 10%
		1331 Deductible VAT of goods and services	10%	5,000,000 g	0 g	B: 50,000,000 g	+VAT on purchase of goods and services 10%
BILL/2024/06/0005	2024-06-01	3310 ADInstruments		0 g	275,000,000 g		
		1561 Purchase costs	P00019: (FLIP 5) JBL FLIP 5 Portable Speak ---	250,000,000 g	0 g	T: 10%	+Untaxed Purchase of Goods and Services taxed 10%
		1331 Deductible VAT of goods and services	10%	25,000,000 g	0 g	B: 250,000,000 g	+VAT on purchase of goods and services 10%

Taxes Applied (Audit)			
Name	Base Amount	Tax Amount	Deductible
10%	300,015,317 g	30,001,531 g	30,001,531 g

Impacted Tax Grids (Audit)			
Grid	+	-	Impact On Grid
Untaxed Purchase of Goods and Services taxed 10%	300,015,317 g	0 g	300,015,317 g
VAT on purchase of goods and services 10%	30,001,531 g	0 g	30,001,531 g

Figure 74: Image 3.6.3.14: Vendor Bills

3.7. Customer service process

CRM customer care process is the process of managing and interacting with customers to build and maintain good relationships, through information collection, customer classification, interaction, and provision of support services Support to enhance customer satisfaction and loyalty.

3.7.1. BPMN

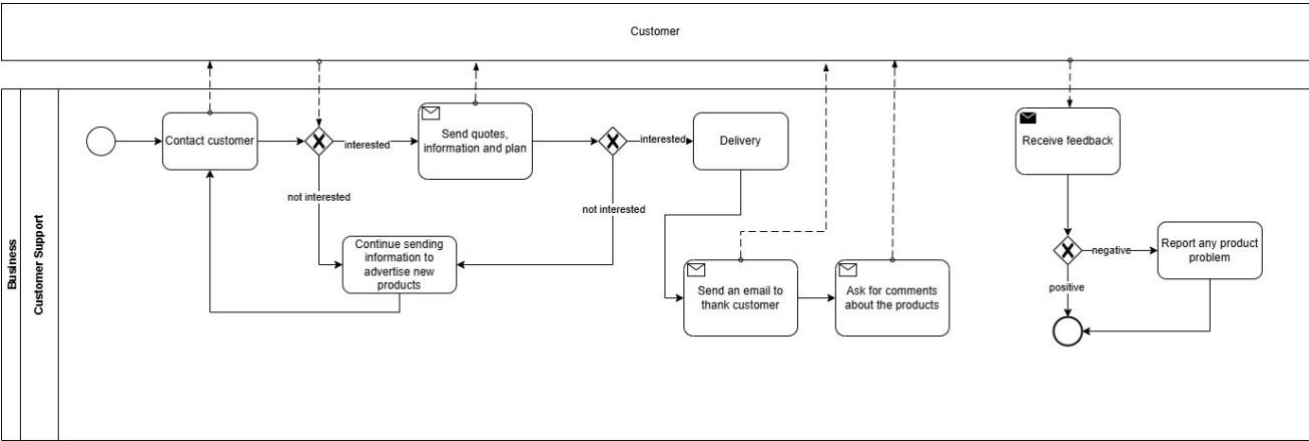


Figure 75: Image 3.7.1: BPMN model about customer service process.

3.7.2. Specification

3.7.2.1. Specification of process steps

Table 15: 3.7.2.1. Specification of process steps

No.	Actor	Action	Description
1	Staff	Contact customers	Staff contact guests to find contact information
2	Staff	Send customer information to sales staff	The employee sends customer information to the sales staff, the sales staff will continue to take care of the customer until the customer makes a purchase.
3	Staff	Send product proposal	Employees send new product proposals to customers
4	Customer	Customers respond to email	Customers respond to emails sent by employees
5	Staff	Receive feedback	Staff receives customer feedback and moves to step 8
6	Customer	Customers do not respond	The customer did not respond to the quote email that the employee sent
7	Staff	Continue to email new product advertising	The employee continues promoting the new product and returns to step 1
8	Staff	Let customers try the product	Staff will let customers try product samples
9	Staff	Send email quotation	If customers need advice about a product, the sales staff will send an email with a quote for that product
10	Staff	Send information about contract and cost plan	Employees send information about plans, contracts, and costs to customers for review
11	Customer	Customer agrees	If the customer agrees with the contract, he will move to step 12

12	Staff	Delivery	Staff will deliver goods to customers
13	Customer	Customers disagree	If the customer does not agree, the process will end
14	Staff	Email thank you to customers	Staff send thanks to customers after successful delivery
15	Staff	Send email for product comments	Employees send emails to customers to get product feedback
16	Customer	Customer feedback	Customers respond to feedback emails sent by staff
17	Staff	Employees receive feedback	Employees receive feedback from customers and end the process
18	Customer	Customer did not respond	Customers do not respond to employee feedback emails immediately
19	Staff	Call for feedback	After 3 days of not receiving a response, the staff will call to ask for feedback and end the process

3.7.2.2. Data library specification

3.7.2.2.1. Opportunities creation module

Table 16: 3.7.2.2.1. Opportunities creation module

No.	Actor	Description	Note
1	Organization/contact	Name of the customer or company	These properties are required
2	Opportunity	The name of the opportunity created, for example: product selling price, how much to sell	These properties are required
3	Email	Enter your company or customer email	These properties are not required

4	Phone	Enter the company or customer phone number	These properties are required
5	Expected revenue	Enter the expected revenue generated by this opportunity	These properties are required
6	Star rating	Use the star level to evaluate the potential of this opportunity	These properties are not required

3.7.2.2.2. Module to create Work Schedule

Table 17: 3.7.2.2.2. Module to create Work Schedule

No.	Actor	Description	Note
1	Activity type	Select the type of activity to upload, including activities: Email, call, meeting, to-do, document upload	These properties are required.
2	Summary	Summary of the activity's content	These properties are not required.
3	Deadline	Deadline for the activity to be completed	These properties are not required.
4	Assign to	Assign staff to be in charge	These properties are not required.
5	Create notes	Assign staff to be in charge. Description of the contents of this working session	These properties are required.

3.7.3. Illustration on Odoo

Step 1: Create a new customer information.

CRM Sales Reporting Configuration

New Generate Leads Pipeline

My Pipeline X Search...

New + Collect information + Proposition + Propose + Trial + Quote

6,823k 54,000 0 1,000 0

Organization / Contact ?

Opportunity
e.g. Product Pricing

Email
e.g. "email@address.com"

Phone
e.g. "0123456789"

Expected Revenue
0 g ☆☆☆

Add Edit

Brent Murphy's opportunity
1,500,000 g
Brent Murphy
☆☆☆

Alicia Flores's opportunity
5,000,000 g
Alicia Flores
☆☆☆

Christine Carr's opportunity
323,000 g
Christine Carr
☆☆☆

Adam Casey's opportunity
54,000 g
Adam Casey
☆☆☆

Aaron Jackson's opportunity
1,000 g
Aaron Jackson
☆☆☆

Figure 76: Image 3.7.3.1: Step 1 – Customer care process.

Step 2: Fill in the customer's contact information and expected revenue then select "Add"

CRM Sales Reporting Configuration

New Generate Leads Pipeline

My Pipeline X Search...

New + Collect information + Proposition + Propose + Trial + Quote

8,223k 54,000 0 1,000 0

Organization / Contact ?
David Ali

Opportunity
David Ali's opportunity

Email
david@gmail.com

Phone
(925)384-0450

Expected Revenue
5,000,000 g ☆☆☆

Add Edit

Janice Smith's opportunity
1,400,002 g
Janice Smith
☆☆☆

Brent Murphy's opportunity
1,500,000 g
Brent Murphy
☆☆☆

Alicia Flores's opportunity
5,000,000 g
Alicia Flores
☆☆☆

Adam Casey's opportunity
54,000 g
Adam Casey
☆☆☆

Aaron Jackson's opportunity
1,000 g
Aaron Jackson
☆☆☆

Figure 77: Image 3.7.3.2: Step 2 – Customer care process.

Step 3: After creating a new customer, move the newly created opportunity to phase 2 - Collect information

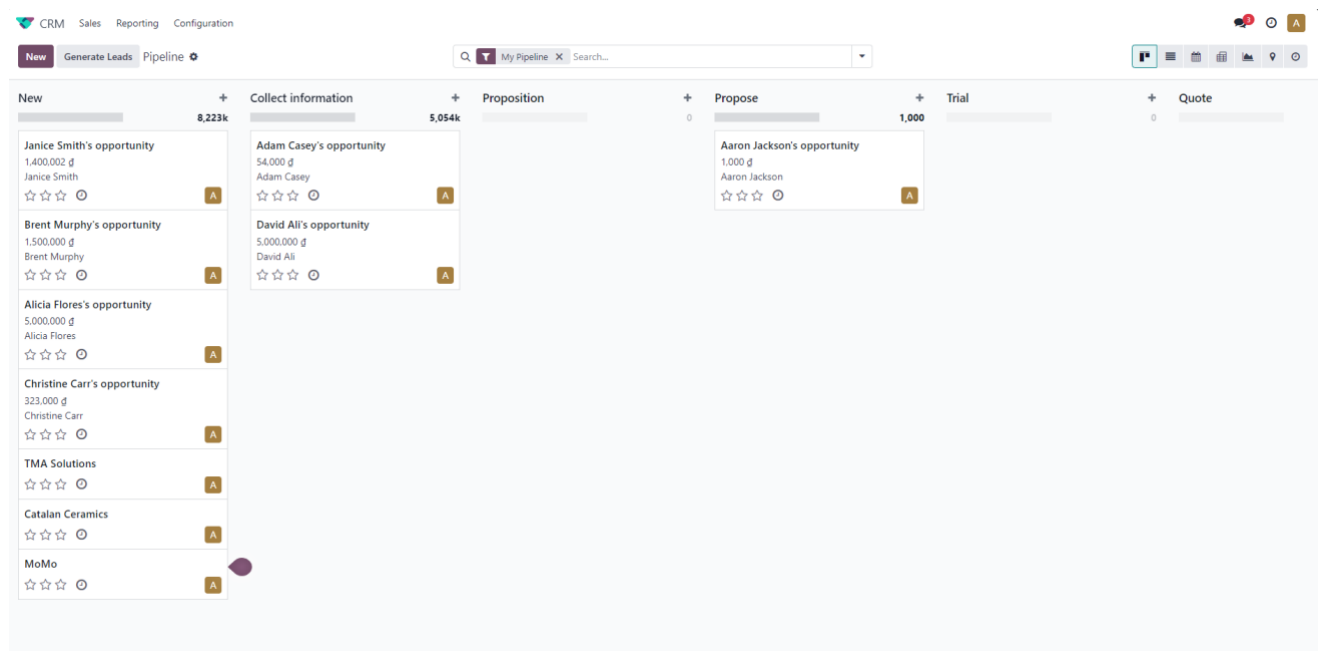


Figure 78: Image 3.7.3.3: Step 3 – Customer care process.

Step 4: After we have enough contact information about the customer, we can choose "Schedule an activity"

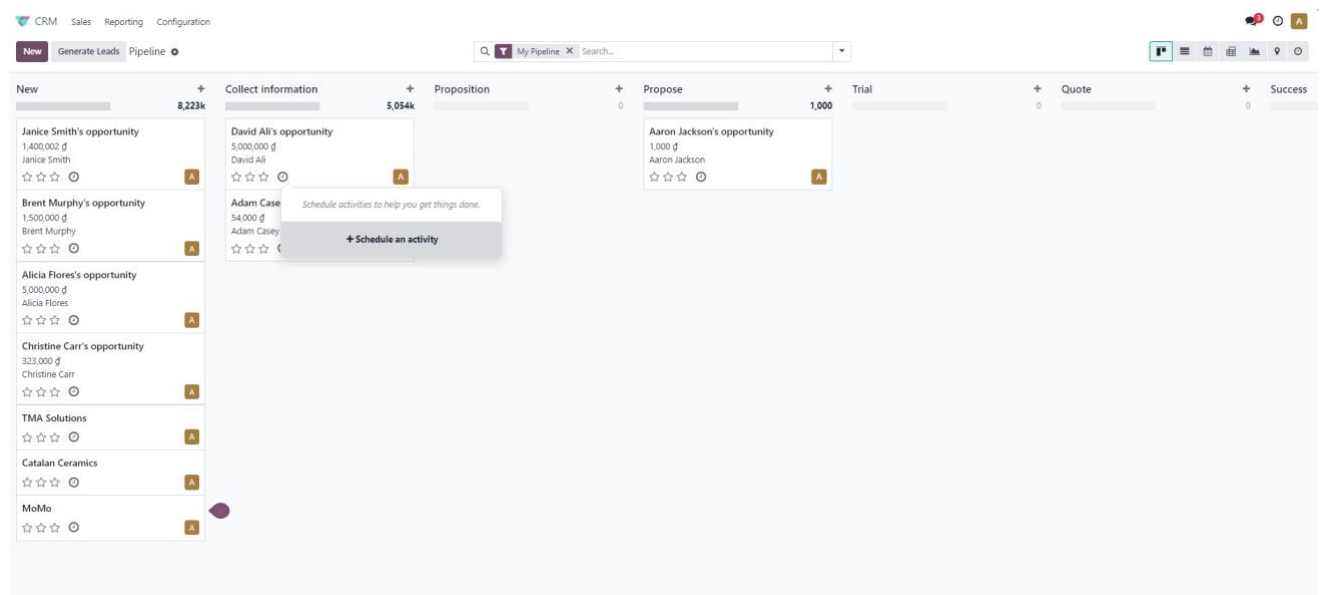


Figure 79: Image 3.7.3.4: Step 4 – Customer care process.

Step 5: Fill in the activity type, due date and assignment information and select “Save”

Schedule Activity

Activity Type

To-Do

Due Date

05/31/2024

Summary

e.g. Discuss proposal

Assigned to

A

Phạm Quốc Anh

Log a note...

Schedule

Schedule & Mark as Done

Done & Schedule Next

Cancel

Figure 80: Image 3.7.3.5: Step 5 – Customer care process.

Step 6: Once we have scheduled and discussed with the customer, we will click "Done" and a check mark will appear to mark the job as completed.

CRM Sales Reporting Configuration

My Pipeline

New

Generate Leads

Pipeline

New

Collect information

Proposition

Propose

Trial

Quote

Success

Janice Smith's opportunity

1,400,000 đ

Janice Smith

☆☆☆

A

Brent Murphy's opportunity

1,500,000 đ

Brent Murphy

☆☆☆

A

Alicia Flores's opportunity

5,000,000 đ

Alicia Flores

☆☆☆

A

Christine Carr's opportunity

323,000 đ

Christine Carr

☆☆☆

A

TMA Solutions

☆☆☆

A

Catalan Ceramics

☆☆☆

A

MoMo

☆☆☆

A

David Ali's opportunity

5,000,000 đ

David Ali

☆☆☆

A

Adam Case

54,000 đ

Adam Casey

☆☆☆

A

Aaron Jackson's opportunity

1,000 đ

Aaron Jackson

☆☆☆

A

Today

✓ To-Do

A

Phạm Quốc Anh

+ Schedule an activity

Figure 81: Image 3.7.3.6: Step 6 – Customer care process.

Step 7: After collecting enough information about customer needs, move the opportunity to the next step which is “Proposal”

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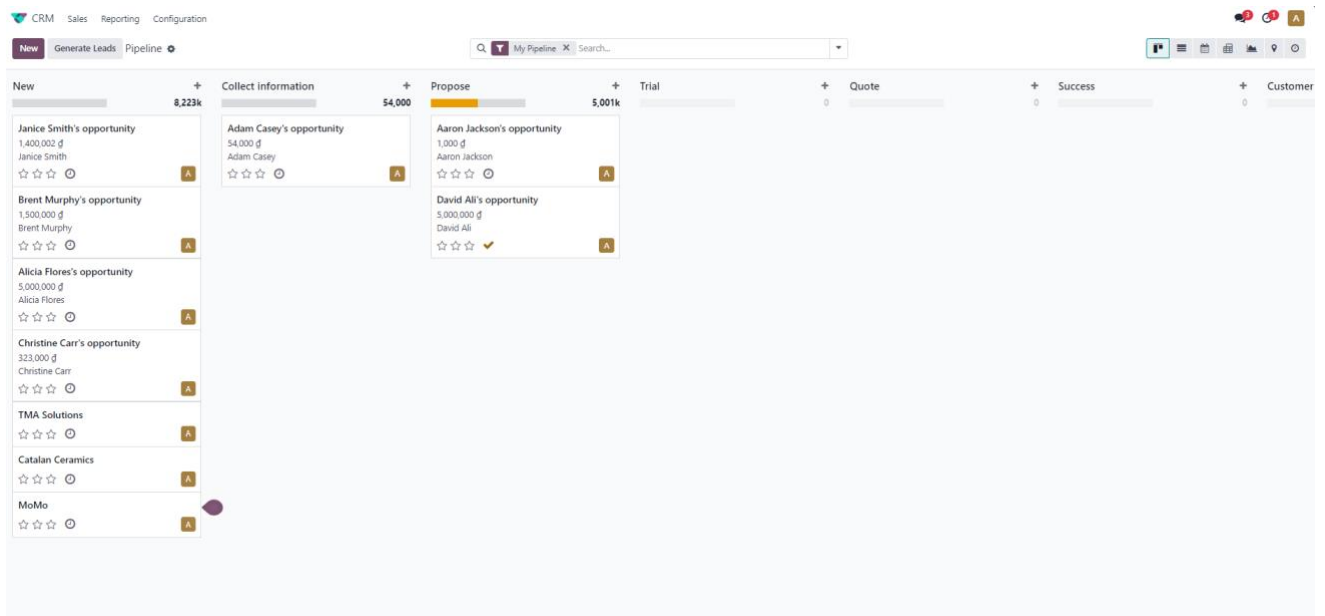


Figure 82: Image 3.7.3.7: Step 7 – Customer care process.

Step 8: After the customer agrees to try the product, move the opportunity to phase 4 - Trial.

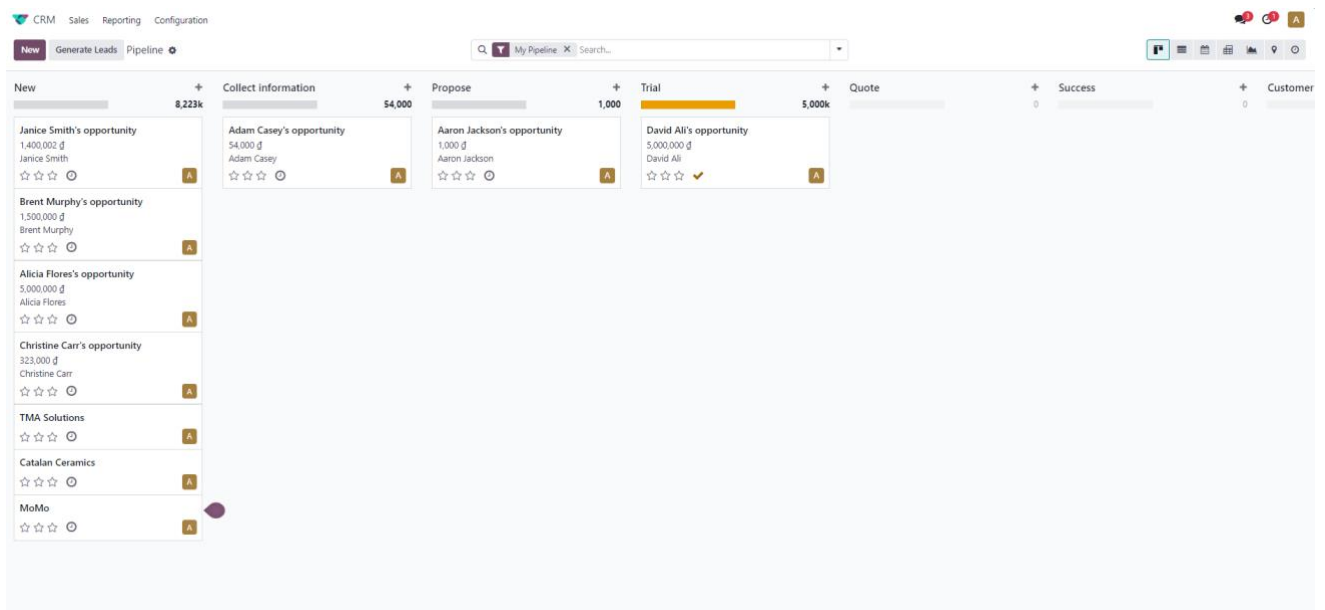


Figure 83: Image 3.7.3.8: Step 8 – Customer care process.

Step 9: After the trial, the customer requests a quote, then the opportunity moves to stage 5 - Quotation.

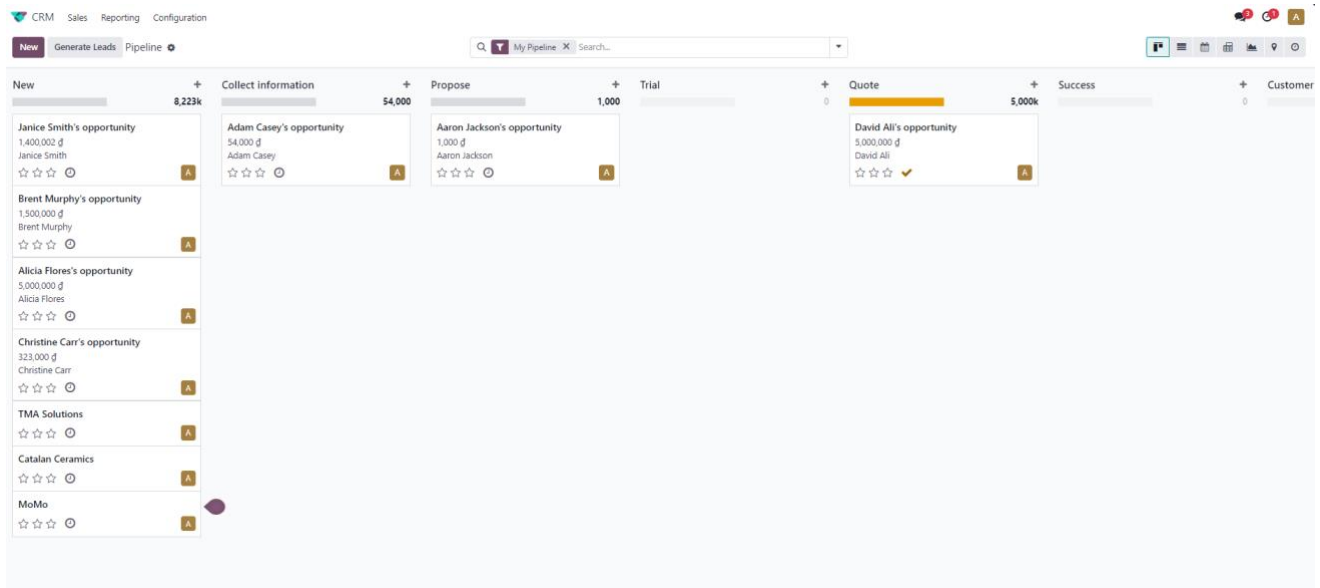


Figure 84: Image 3.7.3.9: Step 9 – Customer care process.

Step 10: If the customer agrees to quote, the opportunity moves to stage 6 - Negotiation.

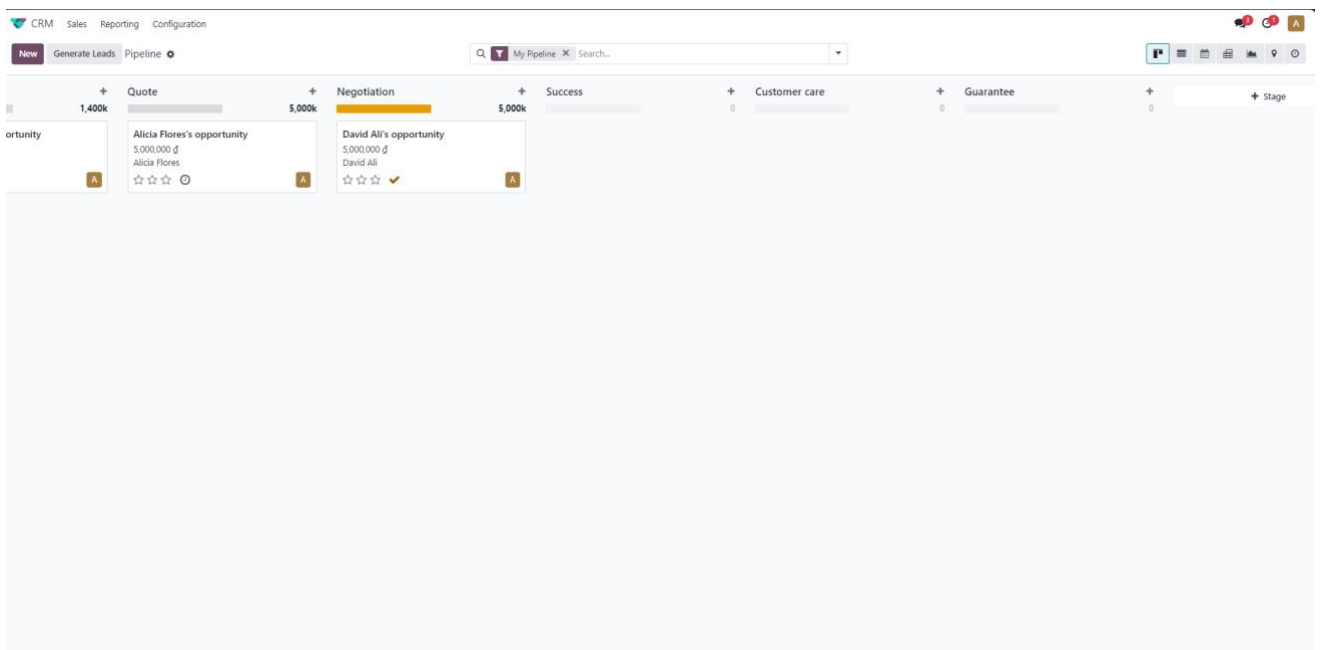


Figure 85: Image 3.7.3.10: Step 10 – Customer care process.

Step 11: If the negotiation fails, select the opportunity and select "Lose" to delete the opportunity.

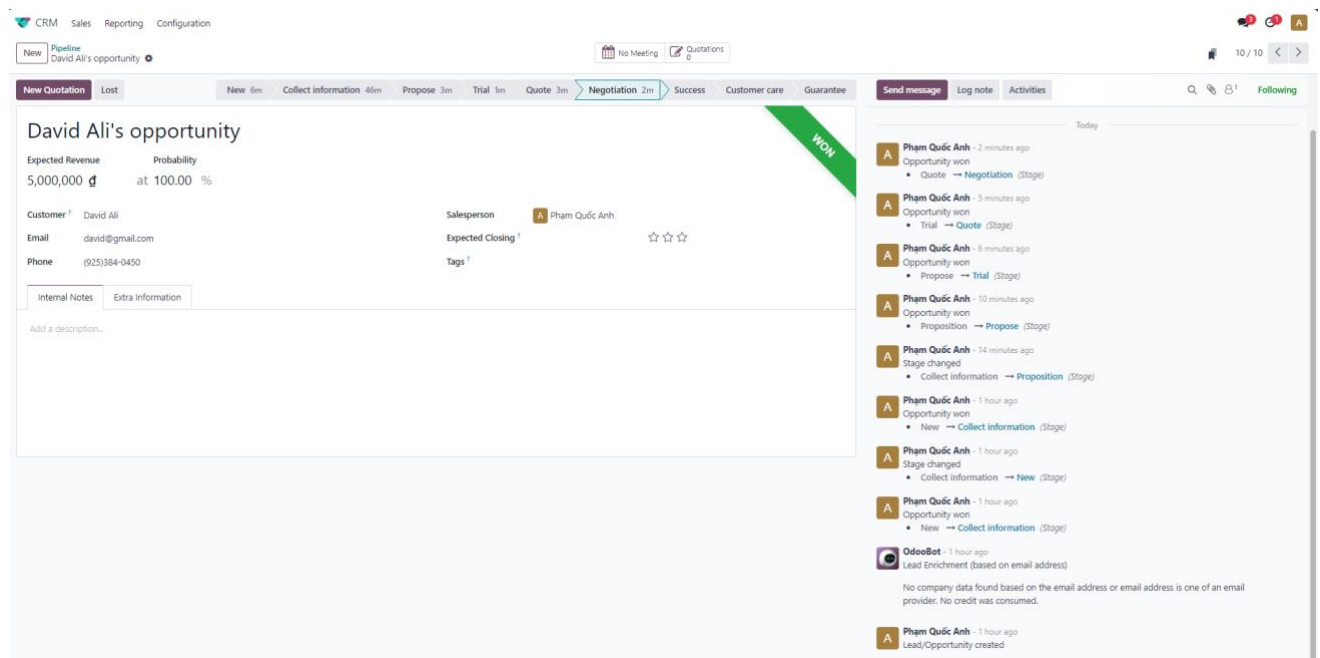


Figure 86: Image 3.7.3.11: Step 11 – Customer care process.

If the negotiation is successful, the opportunity moves to stage 7 – Success

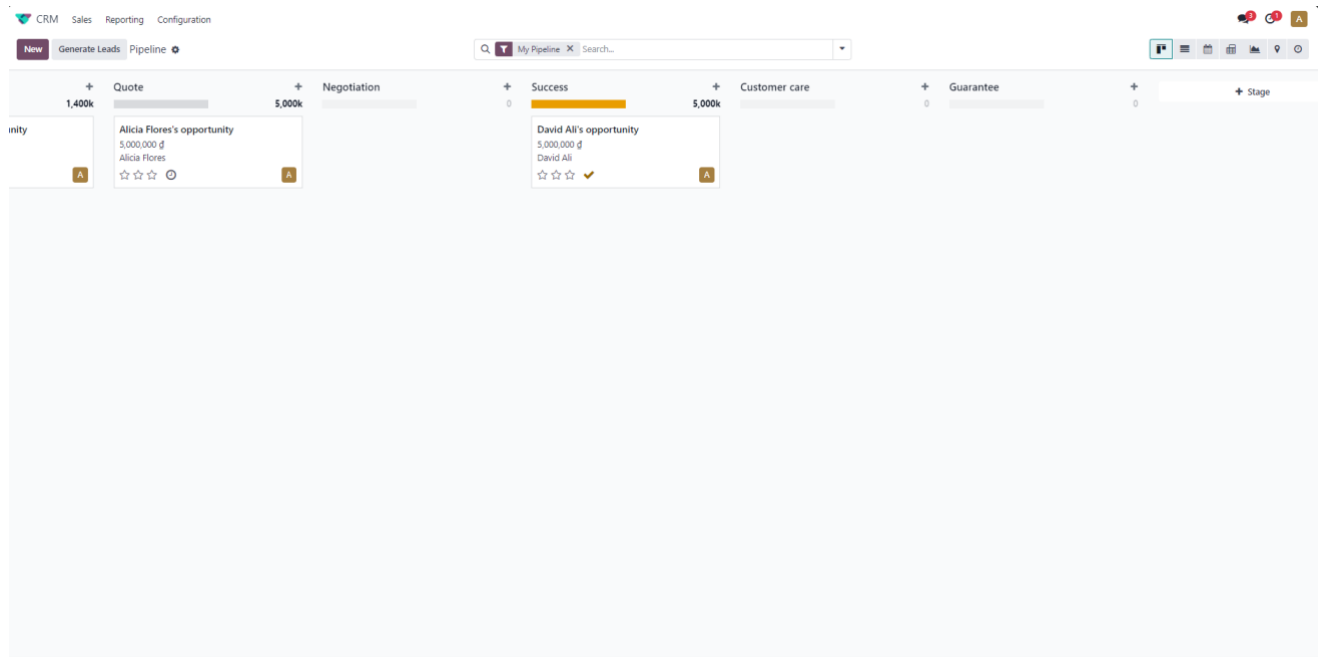


Figure 87: Image 3.7.3.12: Step 12 – Customer care process.

After moving to the successful stage, the opportunity is marked “**Won**”.

CRM Sales Reporting Configuration

New Pipeline David Ali's opportunity

No Meeting Quotations 0

New Quotation Lost New 6m Collect information 46m Propose 3m Trial 1m Quote 3m Negotiation 3m Success 3m Customer care Guarantee

David Ali's opportunity

Expected Revenue 5,000,000 Expected Closing 7 Stars

Probability at 100.00 %

Customer ? David Ali Salesperson A Phạm Quốc Anh

Email david@gmail.com Expected Closing ? Stars

Phone (925)384-0450 Tags ?

Internal Notes Extra Information

Add a description...

Figure 88: Image 3.7.3.13: Step 13 – Customer care process.

If the customer has a problem with the product and requests support, the opportunity moves to stage 9 - Customer Care.

CRM Sales Reporting Configuration

New Generate Leads Pipeline

My Pipeline Search...

Quote Negotiation Success Customer care Guarantee

400k 5,000k 323k 1,500k 5,000k

Alicia Flores's opportunity 5,000,000 Alicia Flores ☆☆☆

Christine Carr's opportunity 323,000 Christine Carr ☆☆☆

Brent Murphy's opportunity 1,500,000 Brent Murphy ☆☆☆

David Ali's opportunity 5,000,000 David Ali ☆☆☆

Figure 89: Image 3.7.3.13: Step 13 – Customer care process.

If the customer requests a warranty, move the opportunity to stage 10 – Guarantee.

CRM Sales Reporting Configuration

New Generate Leads Pipeline

My Pipeline Search...

Quote Negotiation Success Customer care Guarantee

400k 5,000k 323k 1,500k 5,000k

Alicia Flores's opportunity 5,000,000 Alicia Flores ☆☆☆

Christine Carr's opportunity 323,000 Christine Carr ☆☆☆

Brent Murphy's opportunity 1,500,000 Brent Murphy ☆☆☆

David Ali's opportunity 5,000,000 David Ali ☆☆☆

Figure 90: Image 3.7.3.13: Step 13 – Customer care process.

3.8. Human resources management process

The human resources management process within an ERP system typically involves several key components and processes designed to streamline HR activities, enhance data accuracy, and improve overall efficiency.

3.8.1. BPMN

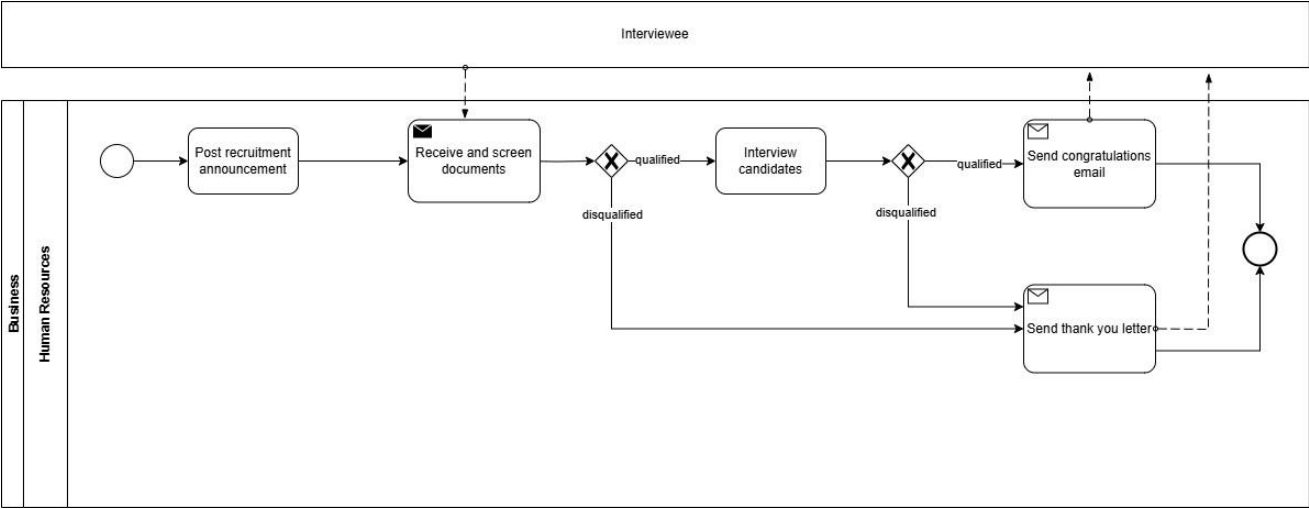


Figure 91 91: Image 3.8.1.1: BPMN model the human resources management process – Recruitment Process

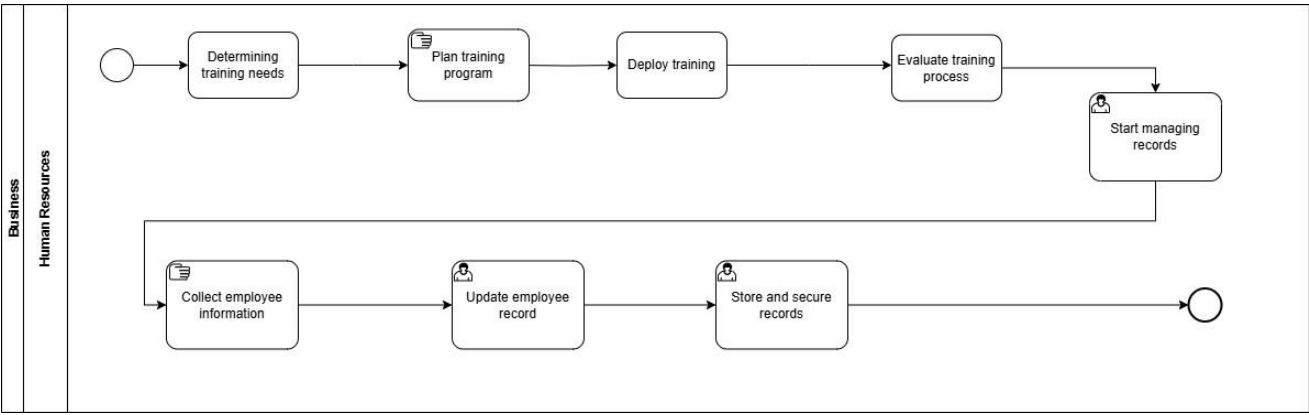


Figure 92 92: Image 3.8.1.2: BPMN model the human resources management process – Training Process

3.8.2. Specification

3.8.2.1. Specification of process steps

Table 18: 3.8.2.1. Specification of process steps

No.	Actor	Action	Description
-----	-------	--------	-------------

1	HR	Start the recruitment process	Post information about vacancies on recruitment channels
2	HR	Receive and screen documents	Collect and filter candidate profiles that match recruitment criteria
3	HR	Case: No	If not, the process stops here for that candidate
4	HR	Case: Yes	If the candidate is suitable, they will be invited for an interview
5	HR	Interview candidates	After the interview, a decision is made as to whether to hire the candidate
6	HR	Case: No	Return to the step of receiving and screening applications to find other candidates
7	HR	Case: Yes	Continue the process
8	HR	Notify results and send job offers	Notify interview results to candidates and send job offer letters to successful candidates
9	Trainer	Start the training process	The training process begins when there is a need to train employees on new skills or knowledge
10	Trainer	Determining training needs	Identify skills and knowledge that require training based on company needs and evaluate employee's current performance

11	Trainer	Plan and train training programs	Establish a detailed plan for the training program including goals, content, time, and training methods
12	Trainer	Deploy training	Carry out planned training activities, including classroom sessions, seminars, and hands-on activities
13	Trainer	Evaluate training results	Evaluate the effectiveness of the training program through feedback from employees and job performance results after training
14	Trainer	Start managing records	The performance appraisal process begins periodically or when there is a need to evaluate employee performance
15	Trainer	Collect employee information	Collect relevant data and information to evaluate employee performance, including work results, feedback from colleagues and managers
16	Trainer	Update employee records	Provide improvement plans for low-performing employees or reward high-performing employees to motivate and develop employees

17	Trainer	Store and secure records	Store employee records safely and securely
----	---------	--------------------------	--

3.8.2.2. Data library specification

3.8.2.2.1. Recruitment

Table 19: 3.8.2.2.1. Recruitment

No.	Properties	Description	Note
1	Job position	This is where the job positions the company is looking to recruit is listed	These properties are required
2	Application email	This is the email where candidates can contact and discuss job information with the company	These properties are not required.

3.8.2.2.2. Staff

Table 20: 3.8.2.2.2. Staff

No.	Properties	Description	Note
1	Subject/Application	The title represents the job being recruited	These properties are required
2	Applicant's name	Name of interviewee	These properties are required
3	Email	Email is used for communication between the company and interviewees	These properties are not required
4	Phone	Phone is used for communication between the company and interviewees	These properties are not required
5	Mobile	Mobile is used for communication between the company and interviewees	These properties are not required

6	LinkedIn Profile	LinkedIn profile is used for communication between the company and interviewees	These properties are not required
7	Degree	The qualifications the candidate must impress	These properties are not required
8	Job	The job that the candidate wants to apply for in the company	These properties are required
9	Interviewers	Interviewer	These properties are not required
10	Recruiter	The interviewer will hire if the candidate is suitable for the company	These properties are not required
11	Evaluation	Rating star	These properties are not required
12	Source	The source cites the work and achievements that the candidate has done	These properties are not required
13	Medium	Where candidates find out about posted jobs	These properties are not required
14	Availability	Job availability date	These properties are not required
15	Tags	Tag name	These properties are not required
16	Expected Salary	The salary that the candidate desires based on his or her abilities	These properties are not required
17	Proposed Salary	The salary that the company proposes to pay the candidate	These properties are not required

3.8.2.2.3. Module to create Work Schedule

Table 21: 3.8.2.2.3. Module to create Work Schedule

No.	Properties	Description	Note
-----	------------	-------------	------

1	Activity type	Select the type of activity to upload, including activities: Email, call, meeting, to-do, document upload	These properties are required.
2	Summary	Summary of the activity's content	These properties are not required.
3	Deadline	Deadline for the activity to be completed	These properties are not required.
4	Assign to	Assign staff to be in charge	These properties are not required.
5	Create notes	Assign staff to be in charge. Description of the contents of this working session	These properties are required.

3.8.3. Illustration on Odoo

Step 1: Go to the "Recruitment" application to create occupations that the company needs to recruit and contact emails

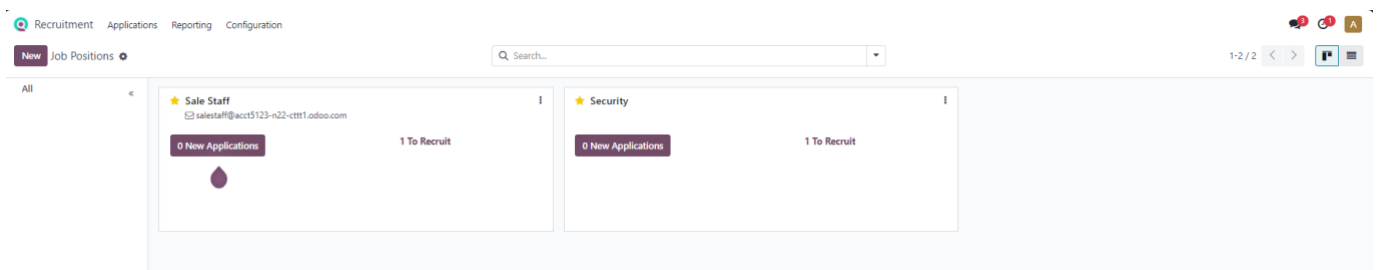


Figure 93: Image 3.8.3.1: Personnel recruitment.

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Step 2: Click “New” to create a job position that the company needs, then click “Create”

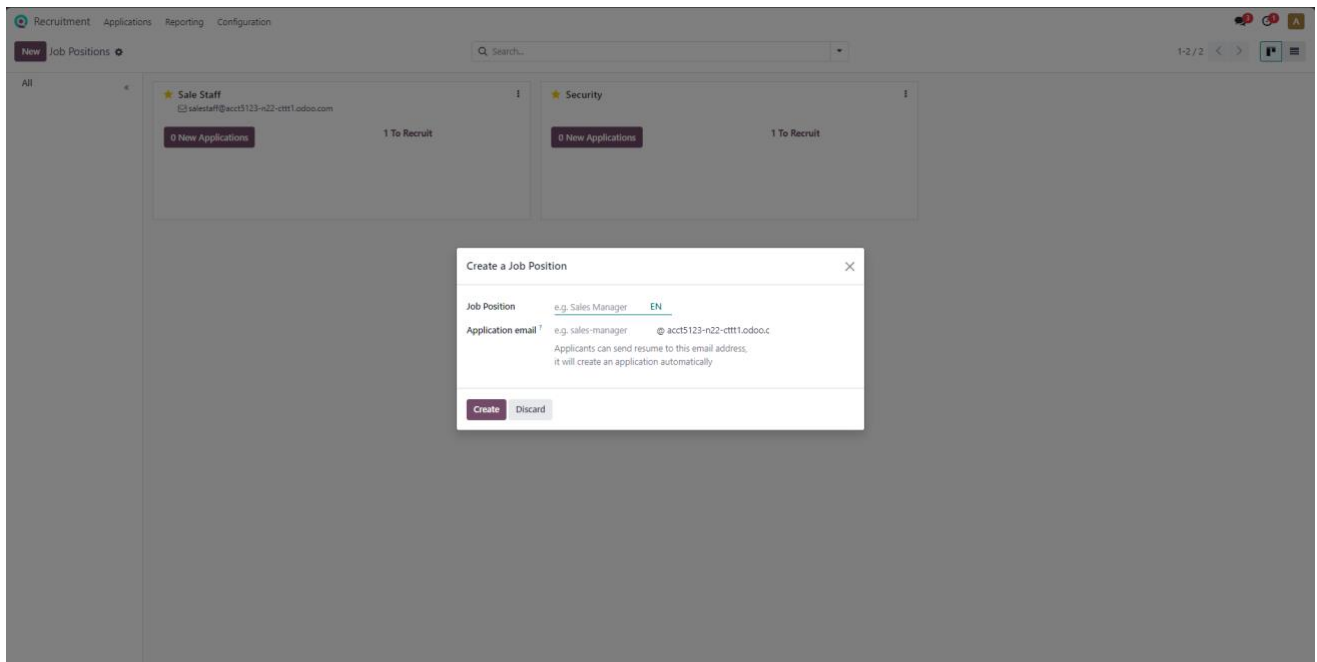


Figure 94: Image 3.8.3.2: Personnel recruitment.

Step 3: After posting the position for the company, click "New Application" to start recruiting employees

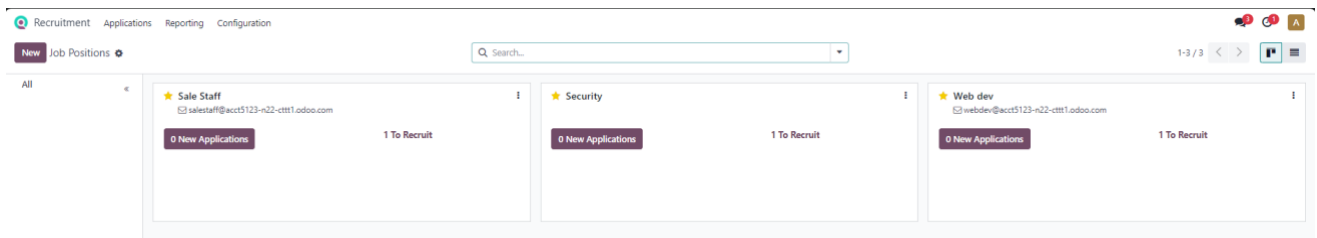


Figure 95: Image 3.8.3.3: Personnel recruitment.

Step 4: Click “New” and create a customer information

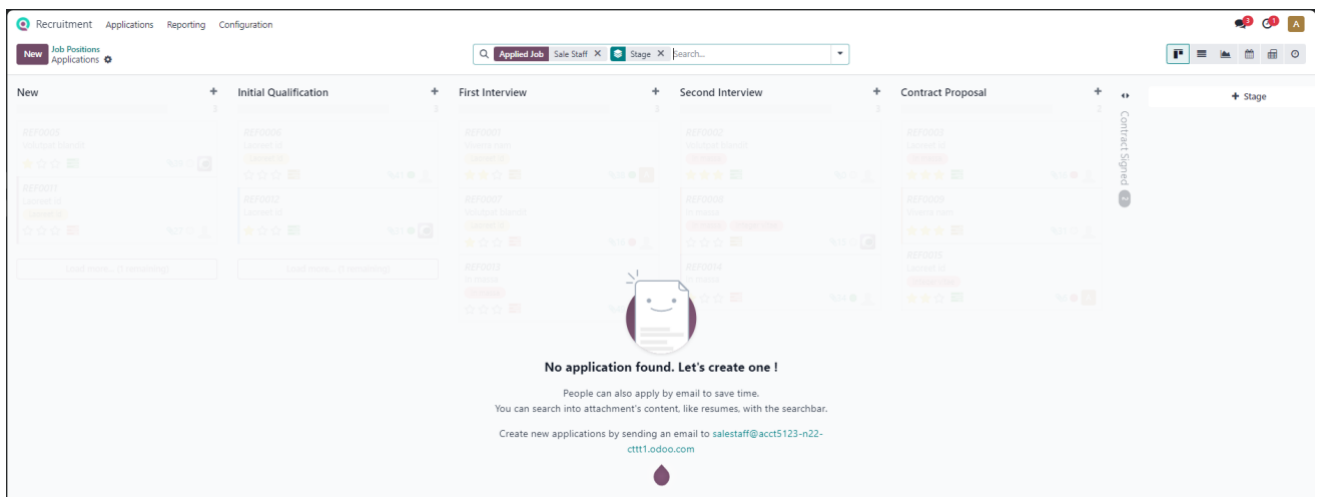


Figure 96: Image 3.8.3.4: Personnel recruitment.

Step 5: Fill in applicant information

The screenshot shows the Odoo Recruitment Application form for a 'Sale Staff' position. The form is divided into several sections. On the left, there are fields for 'Subject / Application', 'Applicant's Name' (e.g., John Doe), 'Email', 'Phone', 'Mobile', 'LinkedIn Profile', 'Degree', and 'Degree'. Below these are 'JOB' details: 'Applied Job' (Sale Staff), 'Department', and 'Motivations...'. On the right, there are sections for 'Interviewers', 'Recruiter', 'Evaluation' (with star ratings), 'Source', 'Medium', 'Availability' (Directly Available), 'Tags', and 'CONTRACT' details: 'Expected Salary' (0.00), 'Proposed Salary' (0.00), and 'Extra advantages...'. At the bottom, there are tabs for 'Application Summary' and 'Skills'. On the far right, there is a sidebar with a 'Send message' button, a 'Log note' button, and an 'Activities' section showing a recent activity by 'Phạm Quốc Anh'.

Figure 97: Image 3.8.3.5: Personnel recruitment.

Step 6: After creating candidate data, we will be transferred to the interview section, and in this stage, we have 4 rounds to interview and test candidates.

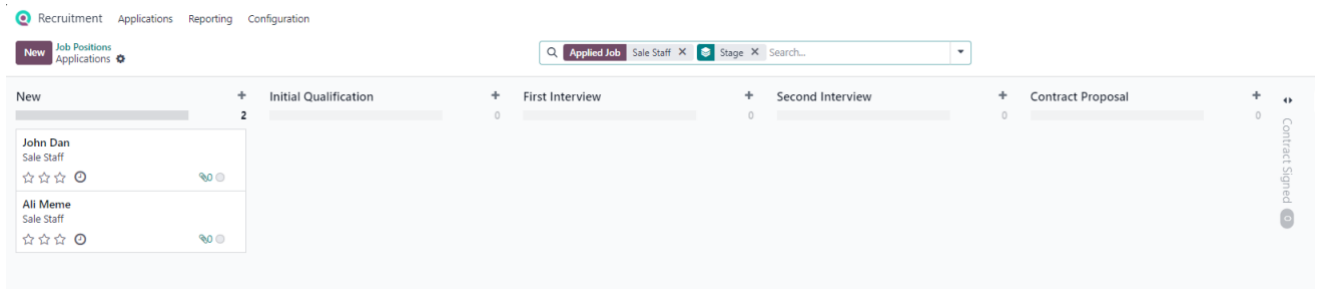


Figure 98: Image 3.8.3.6: Personnel recruitment.

We can schedule activities to start interviews with candidates

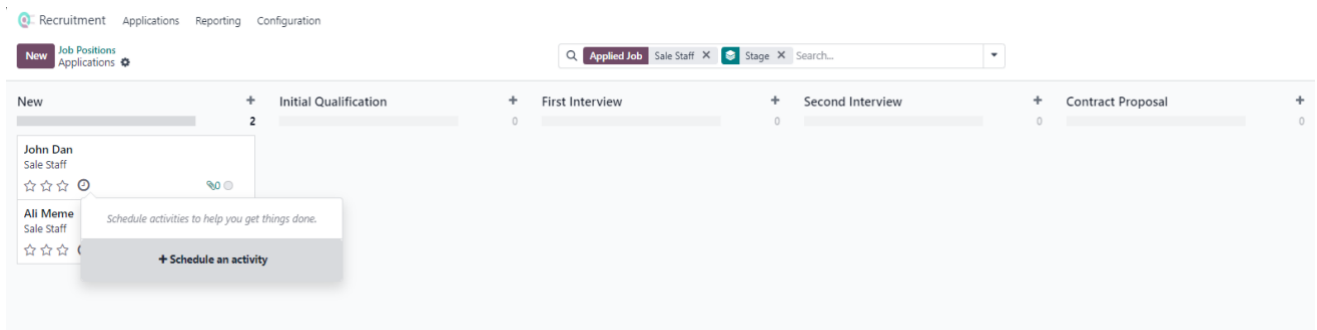


Figure 99: Image 3.8.3.7: Personnel recruitment.

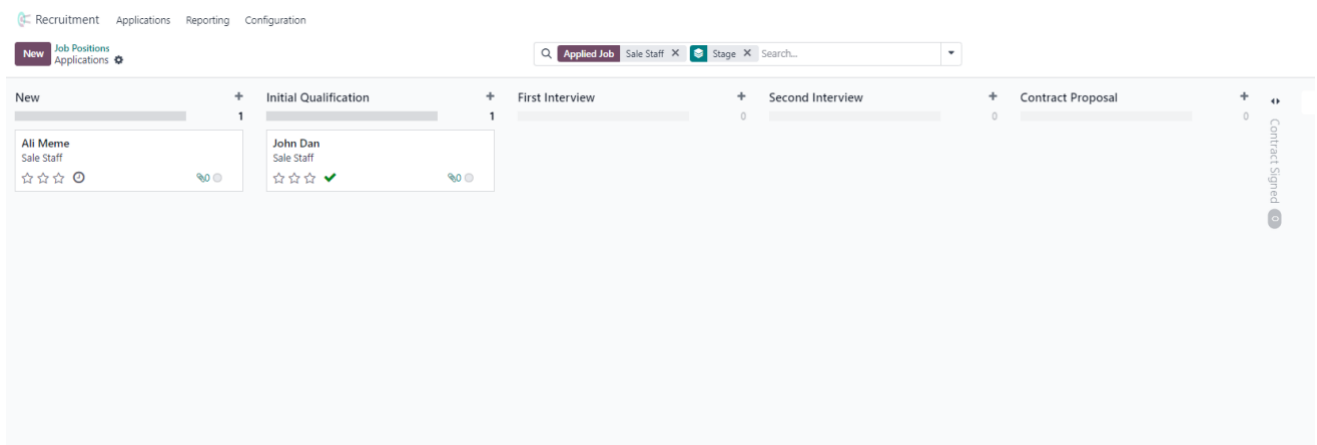


Figure 100: Image 3.8.3.8: Personnel recruitment.

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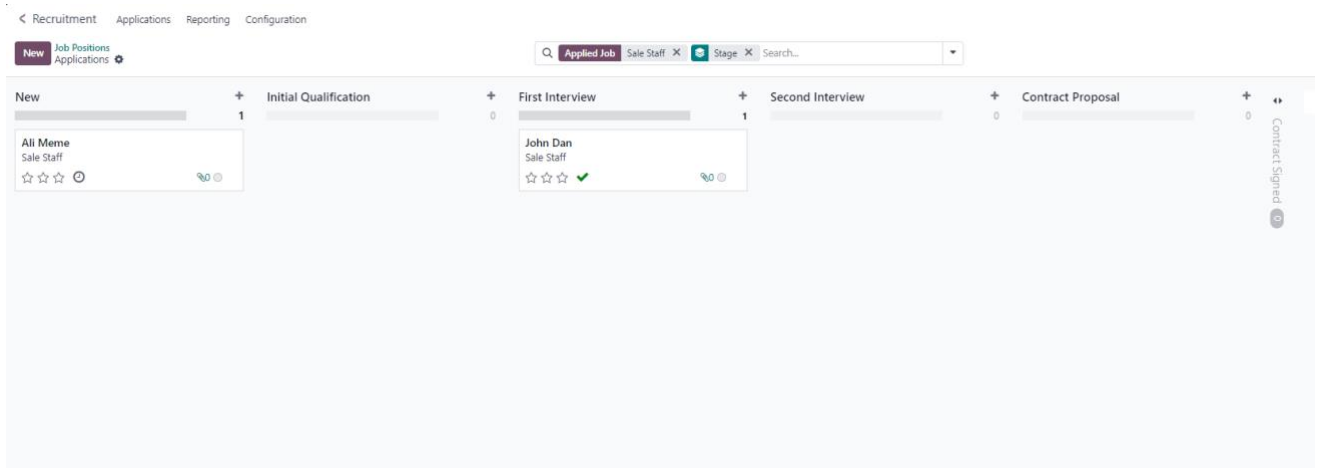


Figure 101: Image 3.8.3.9: Personnel recruitment.

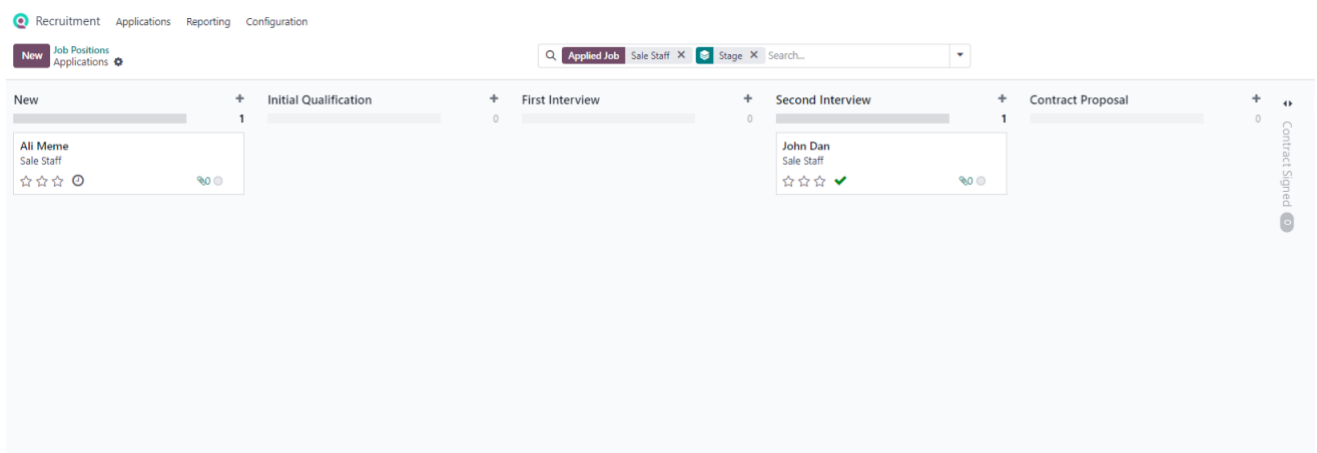


Figure 102: Image 3.8.3.10: Personnel recruitment.

And after 4 rounds of interviews, we can rate the stars to evaluate the candidate's qualifications, and then sign a contract with the candidate.

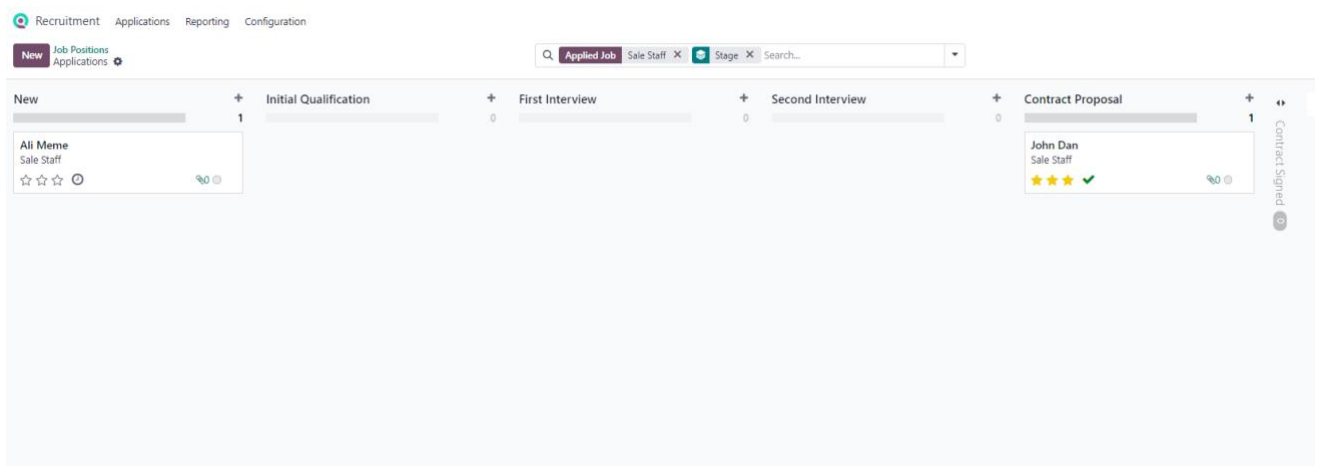


Figure 103: Image 3.8.3.11: Personnel recruitment.

Step 7: We click “Contract Signed”

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Recruitment Applications Reporting Configuration

New Job Positions / Applications Sale Staff

No Meeting

Refuse New 1h Initial Qualification 1m First Interview 1m Second Interview 1m Contract Proposal 1m ...

Subject / Application ?
Sale Staff

Contract Signed

Applicant's Name
John Dan

Email

Phone

Mobile

LinkedIn Profile

Degree Degree

Interviewers

Recruiter

Evaluation ★★

Source ?

Medium ?

Availability ? Directly Available

Tags Tags

JOB

Applied Job Sale Staff

Department

Application Summary Skills

Motivations...

Expected Salary ? 0.00 Extra advantages...

Proposed Salary ? 0.00 Extra advantages...

Figure 104: Image 3.8.3.12: Personnel recruitment.

Click “Create Employee”, and now the candidate has become an employee of the company.

Recruitment Applications Reporting Configuration

New Job Positions / Applications Sale Staff

No Meeting

Create Employee Refuse New 1h Initial Qualification 1m First Interview 1m Second Interview 1m Contract Proposal Contract Signed 4m

Subject / Application ?
Sale Staff

Applicant's Name
John Dan

Email johndan@gmail.com

Phone 0132130945

Mobile

LinkedIn Profile

Degree Degree

Interviewers

Recruiter

Hire Date 06/01/2024 20:30:29

Evaluation ★★

Source ?

Medium ?

Availability ? Directly Available

Tags Tags

JOB

Applied Job Sale Staff

Department

Application Summary Skills

type "/" for commands

Expected Salary ? 9,000,000.00 Extra advantages...

Proposed Salary ? 7,500,000.00 Extra advantages...

HIRED

Figure 105: Image 3.8.3.13: Personnel recruitment.

New Applications / Sale Staff John Dan

Contacts

Launch Plan

John Dan

Sale Staff

Tags

Work Mobile

Work Phone

Work Email johndan@gmail.com

Department

Job Position Sale Staff

Manager

Coach ?

Resume

Work Information

Private Information

HR Settings

RESUME

Experience

06/01/2024 - Current

ACCT5123 N22 CTTT

Sale Staff

ADD

SKILLS

There are no skills defined in the library.

Why not try adding some ?

Create new Skills

Figure 106: Image 3.8.3.14: Personnel recruitment.

Employees Employees Departments Reporting Configuration

New Employees

Search...

1-7/7

DEPARTMENT

All

Administration

B Bill Gates Jr

Managing Director

bgjr@gmail.com

J John Dan

Sale Staff

johndan@gmail.com

J John Doe

Financial Accounting

johndoe@gmail.com

N Nguyễn Quốc Tuấn

Production Planning

19522476@gm.uit.edu.vn

A Phạm Quốc Anh

Materials Management

20521077@gm.uit.edu.vn

P Phạm Thị Thùy Trang

Sale and Distribution

21522697@gm.uit.edu.vn

S Sean Laurent

Human Resource

seanlaur@gmail.com

Figure 107: Image 3.8.3.15: Employees in the company.

3.9. Customer debts on time handling process

The process of timely settlement of customer debts in ERP is the process of managing and resolving debts from customers in a timely and accurate manner in the ERP system. This process includes recording, classifying, checking, notifying, paying and updating debt information to ensure punctuality and good management of customer debts. The goal of this process is to optimize debt management, reduce risk and increase transparency in handling customer debts.

3.9.1. BPMN

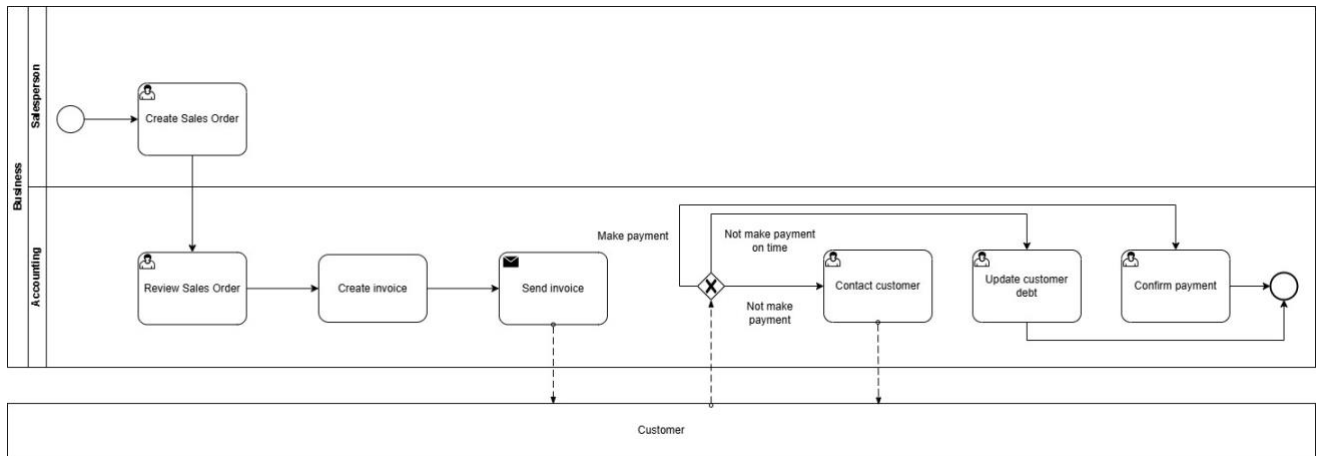


Figure 108: Image 3.9.1: BPMN model about the process of handling customer debts on time.

3.9.2. Specification

3.9.2.1. Specification of process steps

Table 22: 3.9.2.1. Specification of process steps

No	Actor	Action	Description
1	Salesperson	Create sale orders	Salesperson uses software to create sales orders for customers
2	Accounting	Check sales orders	The accounting department checks invoice information
3	Accounting	Order information bookkeeping	The accounting department shall record in the invoice information book
4	Accounting	Print or email invoice information	The accounting department will print or email invoice information to customers
5	Customer	Customers pay	Customers will proceed to pay according to the invoice
6	Salesperson	Customer contact	The salesperson will contact the customer if the customer has not paid, go back to step 4
7	Salesperson	Confirm customer has paid	The salesperson confirms the customer has paid, receives the money and

			proceeds to confirm the payment and send it to the accountant
8	Accounting	Email invoices to customers or print invoices	Resend the invoice email confirming full payment to the customer
9	Accounting	Summary of liabilities by accounting period	The corporate accounting department summarizes debts according to the accounting period

3.9.2.2. Data Library Specification

Table 23: 3.9.2.2. Data Library Specification

No	Field	Description	Note
1	Customer	Data fields describing the customer's name	Required field select
2	Invoice Date	Date of entry into the ledger	Due to the automatic generation of the system, it cannot be edited
3	Payment Reference	Data field set on journal items	Due to the automatic generation of the system, it cannot be edited
4	Due Date	The field describes the invoice's due date	Automatically generated by the system, editable
5	Journal	Data field describing the ledger type	Automatically generated by the system, editable
6	Product	Product name description data field	Required field select
7	Label	Product detail description data field	The system automatically imports (takes the data of the product when importing the product into the system), it can be edited or left blank

8	Account	Data field describing the payout account type	Required field selected, editable
8	Quantity	The data field describes the quantity to be imported of the selected product	Fields can be entered or left blank (default is 1)
9	Unit price	Data fields describing the price of each product	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
10	Taxes	Product tax description data field	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
11	Tax excl	Data field to product invoice excluding tax	The field is automatically defined when selecting the product name (defined when importing product data), which can be redefined through editing the product unit price
12	Discount	Data field describing the percentage discount for the product	Optional fields
13	Tax incl	Description data field to product invoice after tax is included	The field is automatically defined when selecting the product name (defined when importing product data), which can be

			redefined through editing unit price, taxes of the product
--	--	--	--

3.9.3. Illustration on Odoo

Step 1: Go to module “**Accounting**”, select Tab “**Customers**”, then choose “**Invoices**”.

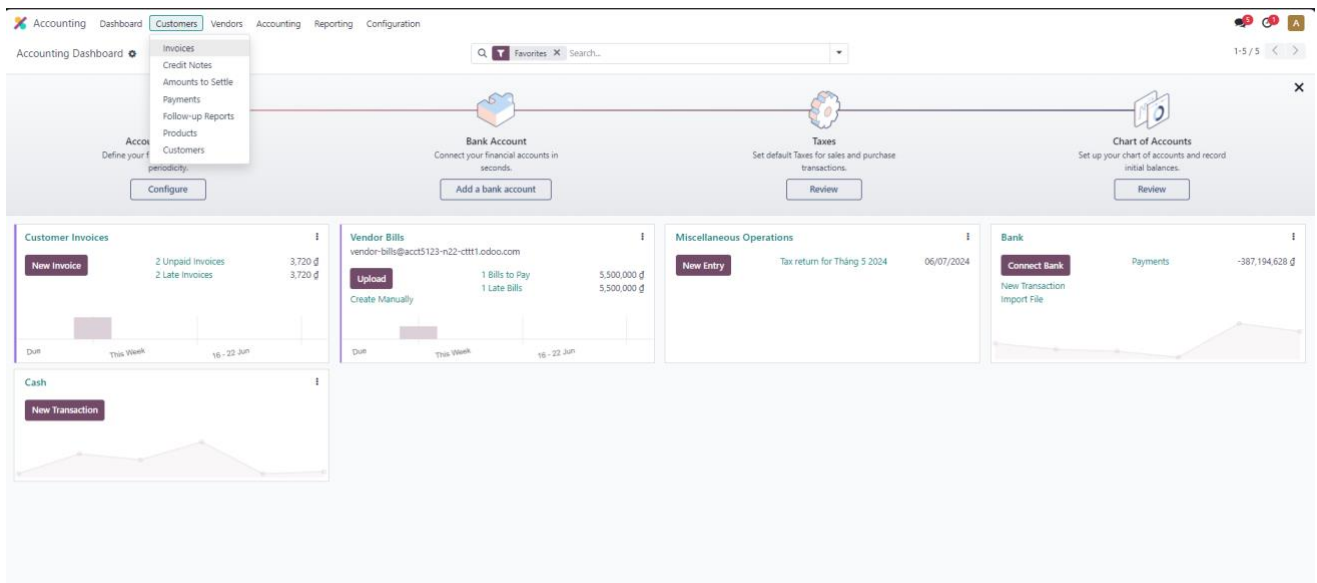


Figure 109: Image 3.9.3.1: Accounting Dashboard

Odoo will display a list of invoice details, ledgers as well as customer information. Accountants can easily manage, know information and payment status of each customer invoice.

Step 2: Select “New” to create an invoice

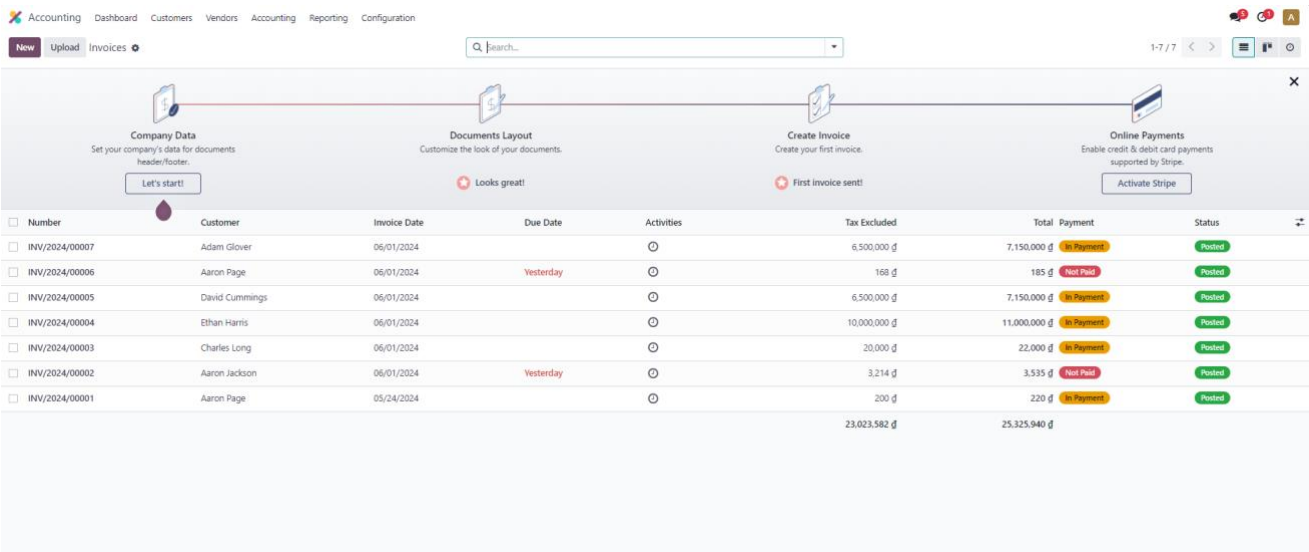


Figure 110: Image 3.9.3.2: Customer Invoices

Step 3: Fill in detailed order information as well as other's delivery information

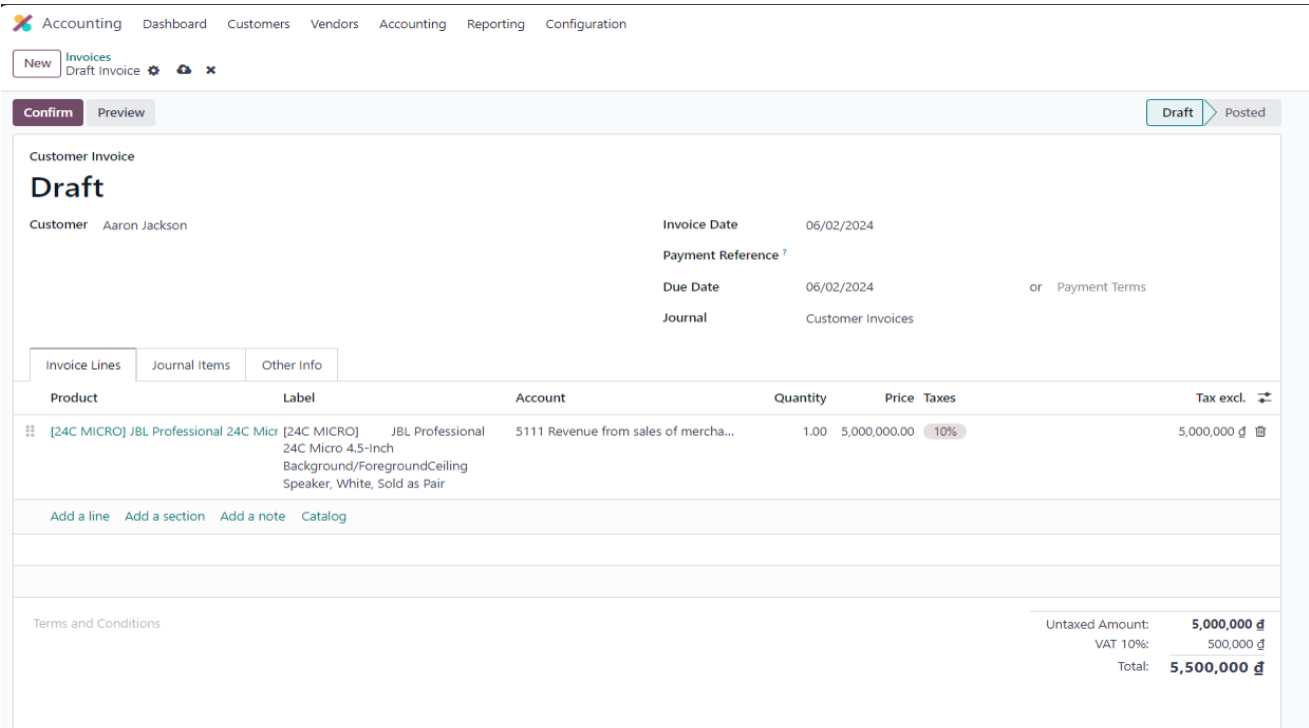


Figure 111: Image 3.9.3.3: Customer Invoices

Step 4: Click "Confirm" after verifying that the purchase information on the customer's order is correct

Accounting

Dashboard

Customers

Vendors

Accounting

Reporting

Configuration

New

Invoices

INV/2024/00008

Send & Print

Register Payment

Preview

Credit Note

Reset to Draft

Draft

Posted

Customer Invoice

INV/2024/00008

Customer Aaron Jackson

Invoice Date06/02/2024

Payment Reference INV/2024/00008

Due Date06/02/2024

JournalCustomer Invoices

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[24C MICRO] JBL Professional 24C Micro 4.5-Inch Backgr	[24C MICRO] JBL Professional 24C Micro 4.5-Inch Background/ForegroundCeiling Speaker, White, Sold as Pair	5111 Revenue from sales of merchandises	1.00	5,000,000.00	10%	5,000,000 đ

Terms and Conditions

Untaxed Amount:5,000,000 đ

VAT 10%:500,000 đ

Total:5,500,000 đ

Amount Due:5,500,000 đ

Figure 112: Image 3.9.3.4: Customer Invoices

Step 5: Click "Register Payment" to create a payment method

Register Payment

×

Journal

Bank

Amount

5,500,000 đ

Payment Method ?

Manual

Payment Date

06/02/2024

Recipient Bank Account

Memo

INV/2024/00008

Create Payment

Discard

Figure 113: Image 3.9.3.5: Payment method.

Step 6: Click “Create Payment” to make payment

Accounting

Dashboard

Customers

Vendors

Accounting

Reporting

Configuration

New

Invoices

INV/2024/00008

Send & Print

Preview

Credit Note

Reset to Draft

Draft

Posted

Customer Invoice

INV/2024/00008

Customer Aaron Jackson

Invoice Date 06/02/2024

Payment Reference INV/2024/00008

Due Date 06/02/2024

Journal Customer Invoices

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[24C MICRO] JBL Professional 24C Micro 4.5-Inch Backgr	[24C MICRO] JBL Professional 24C Micro 4.5-Inch Background/ForegroundCeiling Speaker, White, Sold as Pair	5111 Revenue from sales of merchandises	1.00	5,000,000.00	10%	5,000,000 đ

Terms and Conditions

Untaxed Amount: 5,000,000 đ

VAT 10%: 500,000 đ

Total: 5,500,000 đ

Paid on 06/02/2024 5,500,000 đ

Amount Due: 0 đ

Figure 114: Image 3.9.3.6: In payment invoice.

3.10. Customer debts delay handling process.

The delinquent customer debt settlement process is the process of managing and resolving debts from customers who do not pay on time in the ERP system. This process includes identifying debts, contacting customers, sending debt notices, taking measures to collect and updating debt information.

3.10.1. BPMN

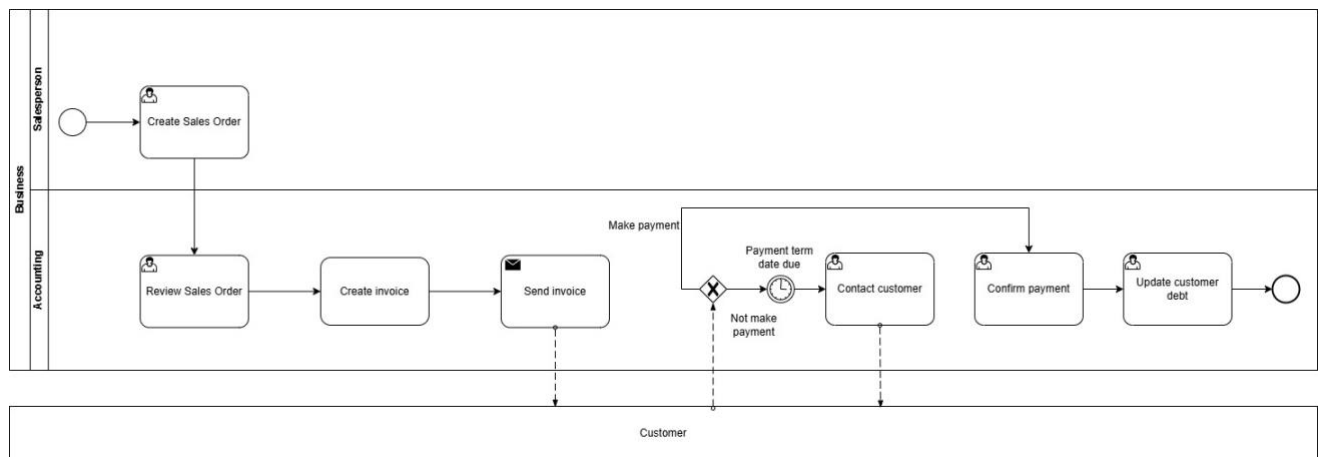


Figure 115: Image 3.10.1: BPMN model about the process of handling customer debts delay

3.10.2. Specification

3.10.2.1. Specification of process steps

Table 24: 3.10.2.1. Specification of process steps

No	Actor	Action	Description
1	Salesperson	Create sale orders	Salesperson use software to create sales orders for customers
2	Accounting	Check sales orders	The accounting department checks invoice information
3	Accounting	Order information bookkeeping	The accounting department shall record in the invoice information book
4	Accounting	Print or email invoice information	The accounting department will print or email invoice information to customers
5	Customer	Customers pay	Customers will proceed to pay according to the invoice
6	Salesperson	Customer contact	The salesperson will contact the customer if the customer has not paid, go back to step 4
7	Salesperson	Confirm customer has paid	The salesperson confirms the customer has paid, receives the money and

			proceeds to confirm the payment and send it to the accountant
8	Accounting	Email invoices to customers or print invoices	Resend the invoice email confirming full payment to the customer
9	Accounting	Summary of liabilities by accounting period	The corporate accounting department summarizes debts according to the accounting period

3.10.2.2. Data Library Specification

Table 25: 3.10.2.2. Data Library Specification

No	Field	Description	Note
1	Customer	Data fields describing the customer's name	Required field select
2	Invoice Date	Date of entry into the ledger	Due to the automatic generation of the system, it cannot be edited
3	Payment Reference	Data field set on journal items	Due to the automatic generation of the system, it cannot be edited
4	Due Date	The field describes the invoice's due date	Automatically generated by the system, editable
5	Journal	Data field describing the ledger type	Automatically generated by the system, editable
6	Product	Product name description data field	Required field select
7	Label	Product detail description data field	The system automatically imports (takes the data of the product when importing the product into the system), it can be edited or left blank

8	Account	Data field describing the payout account type	Required field selected, editable
8	Quantity	The data field describes the quantity to be imported of the selected product	Fields can be entered or left blank (default is 1)
9	Unit price	Data fields describing the price of each product	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
10	Taxes	Product tax description data field	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
11	Tax excl	Data field to product invoice excluding tax	The field is automatically defined when selecting the product name (defined when importing product data), which can be redefined through editing the product unit price
12	Discount	Data field describing the percentage discount for the product	Optional fields
13	Tax incl	Description data field to product invoice after tax is included	The field is automatically defined when selecting the product name (defined when importing product data), which can be

			redefined through editing unit price, taxes of the product
--	--	--	--

3.10.3. Illustration on Odoo

Step 1: Go to the “Accounting” module, at “Accounting information panel”, then select “Customer Invoices”.

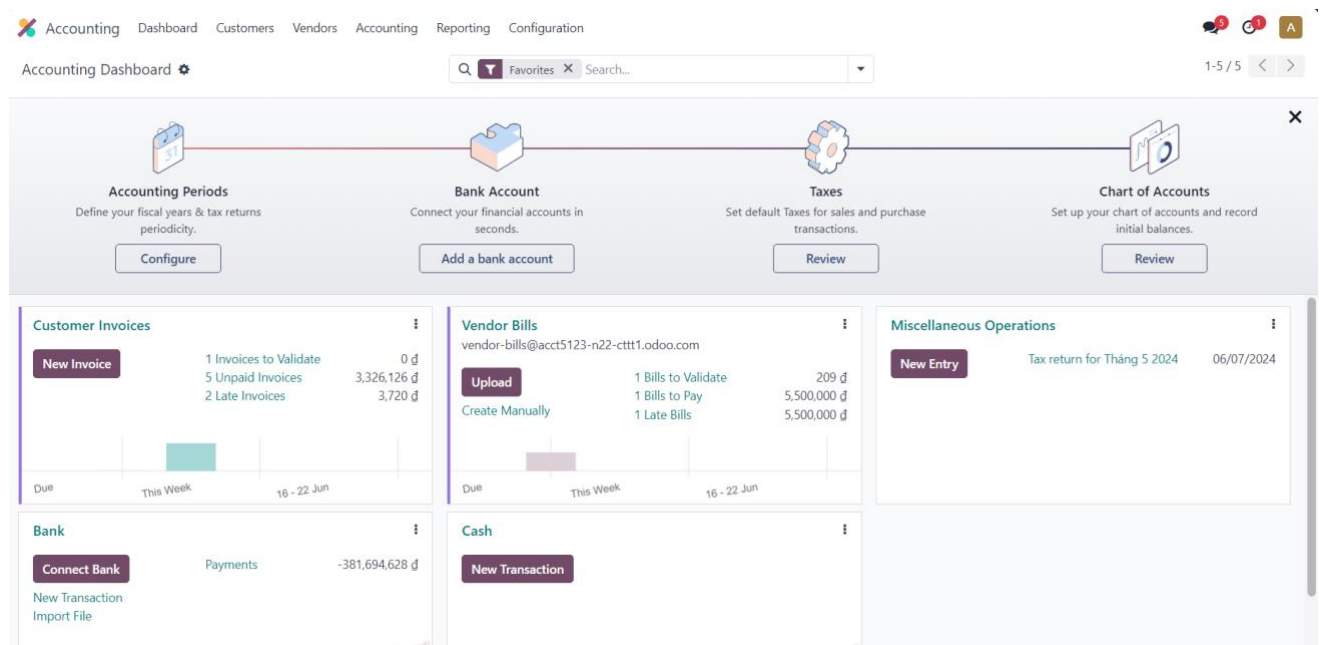


Figure 116: Image 3.10.3.1: Process for handling delinquent customer debt.

Step 2: Check invoice information and order status. Select orders with status "Unpaid"

The screenshot shows the Accounting Dashboard Invoices page. At the top, there's a navigation bar with links to Accounting, Dashboard, Customers, Vendors, Accounting, Reporting, and Configuration. Below this is a search bar and a dropdown menu. The main area features a progress bar with four steps: Company Data, Documents Layout, Create Invoice, and Online Payments. Below the progress bar is a table of customer invoices.

☐	Number	Customer	Invoice Date	Due Date	Activities	Tax Excluded	Total	Payment	Status
☐	/	#Created by: 21522697	06/03/2024	Today	🕒	0 đ	0 đ		Draft
☐	INV/2024/00011	Aaron Jackson	06/02/2024	Today	🕒	369 đ	406 đ	Not Paid	Posted
☐	INV/2024/00010	Aaron Jackson	06/02/2024	Today	🕒	3,000,000 đ	3,300,000 đ	Not Paid	Posted
☐	INV/2024/00009	Aaron Page	06/02/2024	Today	🕒	20,000 đ	22,000 đ	Not Paid	Posted
☐	INV/2024/00008	Aaron Jackson	06/02/2024		🕒	5,000,000 đ	5,500,000 đ	In Payment	Posted
☐	INV/2024/00007	Adam Glover	06/01/2024		🕒	6,500,000 đ	7,150,000 đ	In Payment	Posted
☐	INV/2024/00006	Aaron Page	06/01/2024	Yesterday	🕒	168 đ	185 đ	Not Paid	Posted
☐	INV/2024/00005	David Cummings	06/01/2024		🕒	6,500,000 đ	7,150,000 đ	In Payment	Posted
☐	INV/2024/00004	Ethan Harris	06/01/2024		🕒	10,000,000 đ	11,000,000 đ	In Payment	Posted

Figure 117: Image 3.10.3.2: Customer Invoices

Step 3: Select “Send and print” to make payment reminders

The screenshot shows the Accounting Dashboard Invoices page with the details of a specific invoice (INV/2024/00010). The top navigation bar is the same as in Figure 117. Below the navigation bar, there's a search bar and a dropdown menu. The main area features a progress bar with four steps: Company Data, Documents Layout, Create Invoice, and Online Payments. Below the progress bar is a table of customer invoices. The invoice details are shown below the table.

Customer Invoice
INV/2024/00010
Customer: Aaron Jackson

Invoice Date: 06/02/2024
Payment Reference: INV/2024/00010
Due Date: 06/02/2024
Journal: Customer Invoices

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[43UR640S9UD] LG 43UR640S9UD 43" Smart LED-LCD TV - 4K UHD TV - TAA Compliant - LG 43UR640S9UD 43" Smart LED-LCD TV - 4K UHD TV - TAA Compliant - LED Backlight - 3840 x 2160 Resolution		5111 Revenue from sales of merchandises	1.00	3,000,000.00	10%	3,000,000 đ

Figure 118: Image 3.10.3.3: Customer Invoices

ACCT5123 - Enterprise resource planning

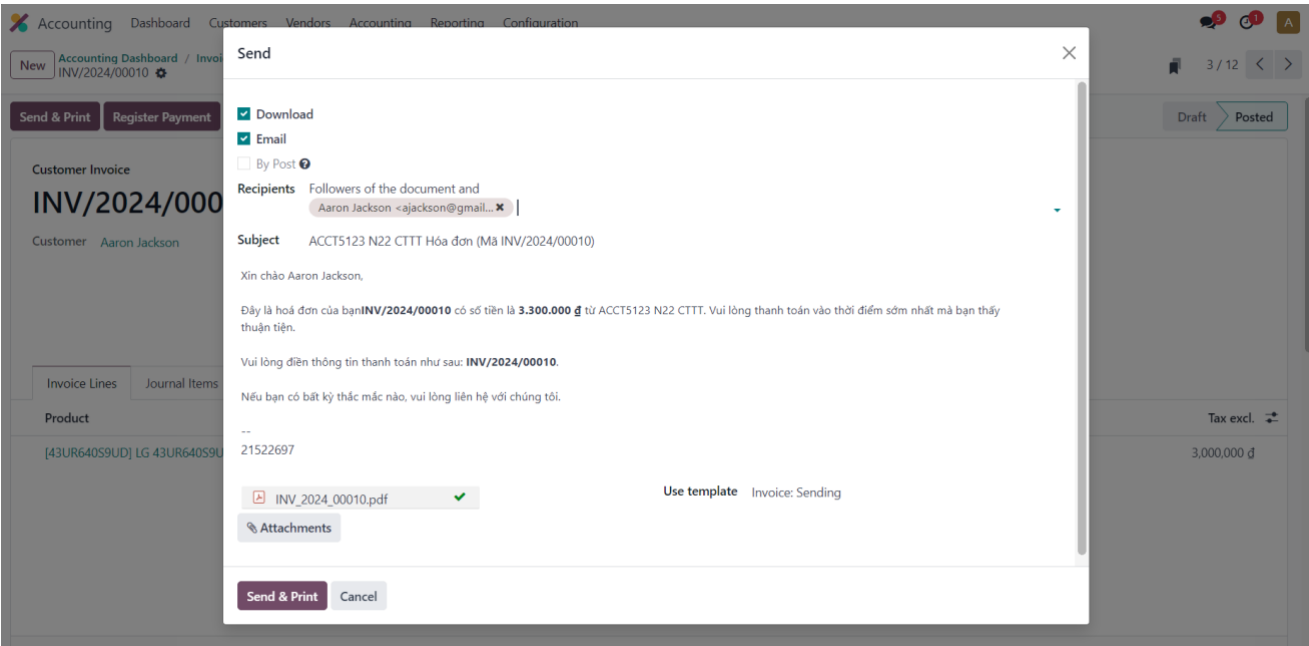


Figure 119: Image 3.10.3.4: Customer Invoices

- Send payment reminders and print them out (if necessary).
- In case it's been over 30 days, the customer still hasn't paid. The accountant will take the phone number information and send an email and call again to remind you.

Step 4: After receiving payment from the customer, the user will proceed to confirm via "Register Payment".

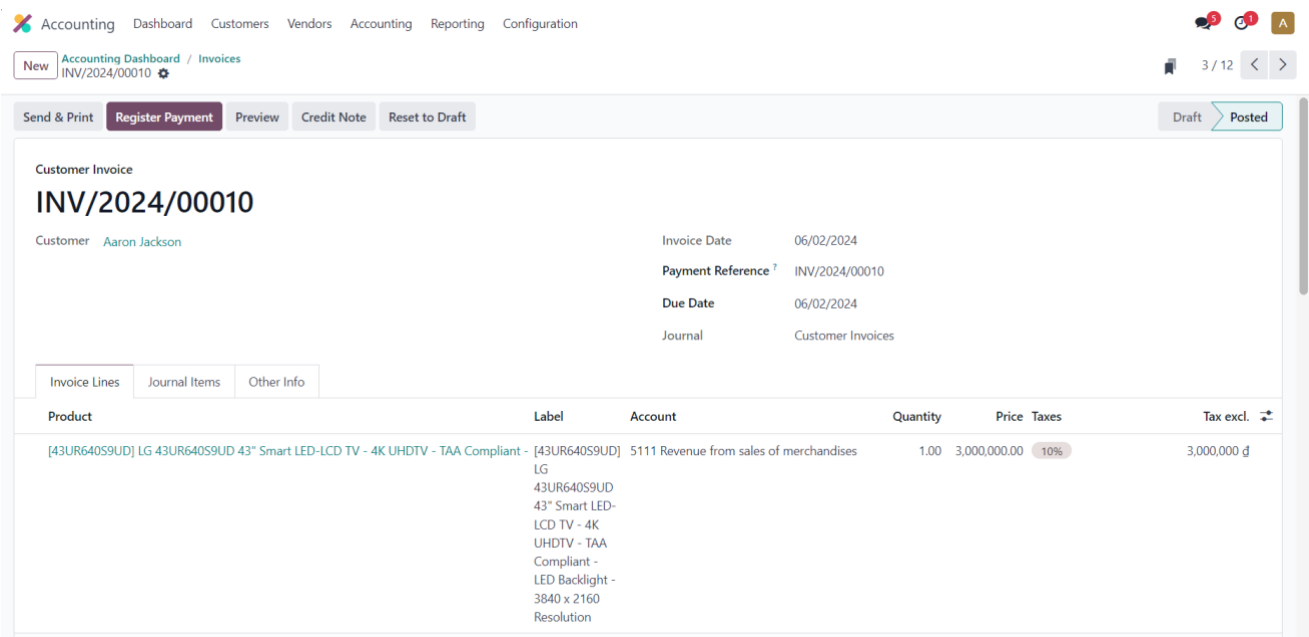


Figure 120: Image 3.10.3.5: Customer Invoices

Step 5: Choose the payment method. Select "Create payment"

Register Payment

Journal

Bank

Amount

3,300,000 đ

Payment Method ?

Manual

Payment Date

06/02/2024

Recipient Bank Account

Memo

INV/2024/00010

Create Payment

Discard

Figure 121: Image 3.10.3.6: Payment method.

Accounting Dashboard Customers Vendors Accounting Reporting Configuration

New Accounting Dashboard / Invoices INV/2024/00010

3 / 12

Send & Print Preview Credit Note Reset to Draft

Draft Posted

Customer Invoice

INV/2024/00010

Customer Aaron Jackson

Invoice Date 06/02/2024

Payment Reference ? INV/2024/00010

Due Date 06/02/2024

Journal Customer Invoices

Invoice Lines Journal Items Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[43UR640S9UD] LG 43UR640S9UD 43" Smart LED-LCD TV - 4K UHD TV - TAA Compliant -	[43UR640S9UD] LG 43UR640S9UD 43" Smart LED-LCD TV - 4K UHD TV - TAA	5111 Revenue from sales of merchandises	1.00	3,000,000.00	10%	3,000,000 đ

Figure 122: Image 3.10.3.7: Customer invoices are " In Payment "

3.11. Supplier debts handling process.

The supplier debt settlement process is the process of managing and paying debts that enterprises owe to partners to provide goods or services. This process includes identifying debts, checking and approving invoices, making payments, and updating debt information in the ERP system

3.11.1. BPMN

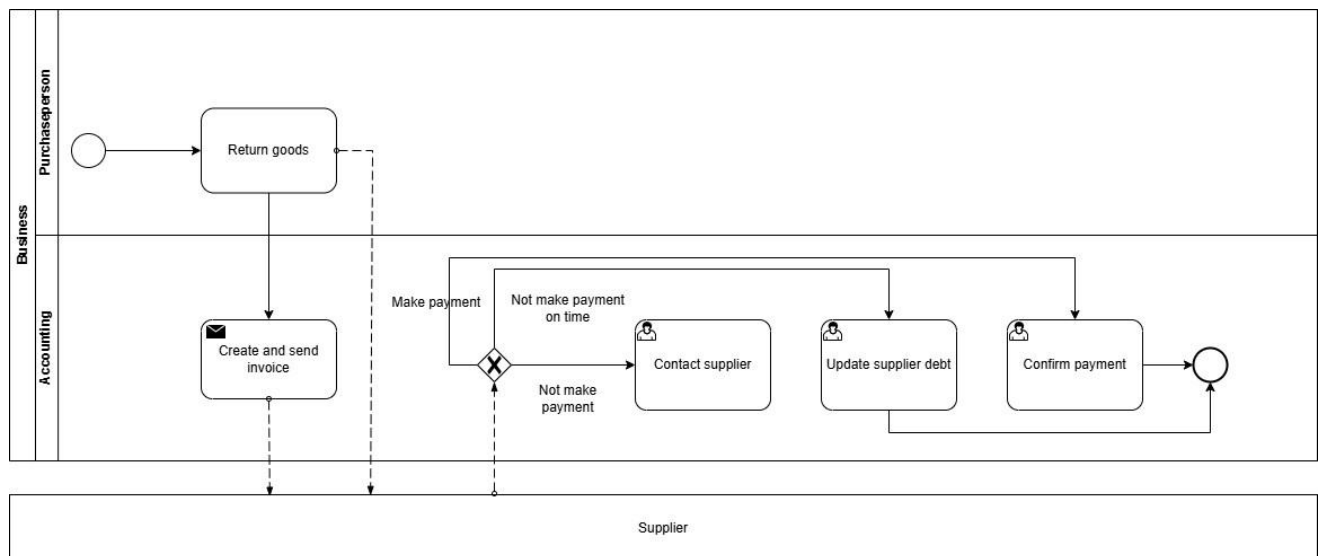


Figure 123: Image 3.10.3.7: BPMN model about the process of handling supplier debts.

3.11.2. Specification

3.11.2.1. Specification of process steps

Table 26: 3.11.2.1. Specification of process steps

No	Actor	Action	Description
1	Suppliers	Invoicing and shipment	Invoicing and shipping goods to businesses according to the content of the invoice
2	Corporate accounting	Reconcile and check purchase invoices	Reconcile and check invoices and goods received from suppliers
3	Suppliers	Check your invoice again	If the check in step 2 is not eligible, a request will be submitted and the supplier will be asked to double-check the invoice. If the invoice is faulty, repeat step 1, without error skip to step 4
4	Suppliers	Notify the business again	The supplier will notify the business of the invoice and export the goods based on the invoice after inspection

5	Corporate accounting	Bookkeeping	If the inspection invoice is sufficient, the accounting department will debit the enterprise's books
6	Corporate accounting	Proceed to payment	Proceed to pay the amount of debt
7	Suppliers	Confirm payment	Provider confirms payment from the business side
8	Corporate accounting	Summary of liabilities in the accounting period	The corporate accounting department summarizes debts according to the accounting period

3.11.2.2. Data Library Specification

Accounting Dashboard Customers Vendors Accounting Reporting Configuration

New Accounting Dashboard Draft Bill

Purchases 1

Confirm Cancel

Draft Posted

Vendor Bill

Draft

Vendor Viewsonic

Bill Date 06/02/2024

Bill Reference Accounting Date 06/02/2024

Auto-Complete ? Select a purchase order or an old bill

Payment Reference ?

Recipient Bank ?

Due Date 06/02/2024 or Payment Terms

Journal Vendor Bills

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan,	P00018: [MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan, 53.4 dB - 500 sq. ft. Coverage Evaporative Air Cooler	1561 Purchase costs	100.00	500,000.00	10%	50,000,000

Figure 124: Image 3.11.2: Vector New Bills

Table 27: 3.11.2.2. Data Library Specification

No	Field	Description	Note
----	-------	-------------	------

1	Vendor	Data fields describing the vendor's name	Required field select
2	Bill Reference	Fields describing the invoice's identity data	Optional fields
3	Auto-Complete	Data field to select autofill according to 1 old order	Optional fields
4	Bill Date	Billed date data description field	Due to the automatic generation of the system, it cannot be edited
5	Accounting Date	Accounting date data description field	Due to the automatic generation of the system, it cannot be edited
6	Payment Reference	Data field set on journal items	Due to the automatic generation of the system, it cannot be edited
7	Recipient Bank	Bank account number which invoice will be paid	Optional field
4	Due Date	The field describes the invoice's due date	Automatically generated by the system, editable
5	Journal	Data field describing the ledger type	Automatically generated by the system, editable
6	Product	Product name description data field	Required field select
7	Label	Product detail description data field	The system automatically imports (takes the data of the product when importing the product into the system), it can be edited or left blank
8	Account	Data field describing the payout account type	Required field selected, editable

8	Quantity	The data field describes the quantity to be imported of the selected product	Fields can be entered or left blank (default is 1)
9	Unit price	Data fields describing the price of each product	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
10	Taxes	Product tax description data field	The field is automatically defined when selecting the product name (defined when importing product data), can be redefined, or left blank
11	Tax excl	Data field to product invoice excluding tax	The field is automatically defined when selecting the product name (defined when importing product data), which can be redefined through editing the product unit price
12	Discount	Data field describing the percentage discount for the product	Optional fields
13	Tax incl	Description data field to product invoice after tax is included	The field is automatically defined when selecting the product name (defined when importing product data), which can be redefined through editing

			unit price, taxes of the product
--	--	--	----------------------------------

3.11.3. Illustration on Odoo

In Odoo's accounting module, there are 2 ways to record invoices from suppliers:

- Create your own craft.
- Upload invoices received from suppliers.

For how to create manual invoices:

Step 1: Go to the “Accounting” Module, in the “Vendor Bills” frame, select “Manually create”

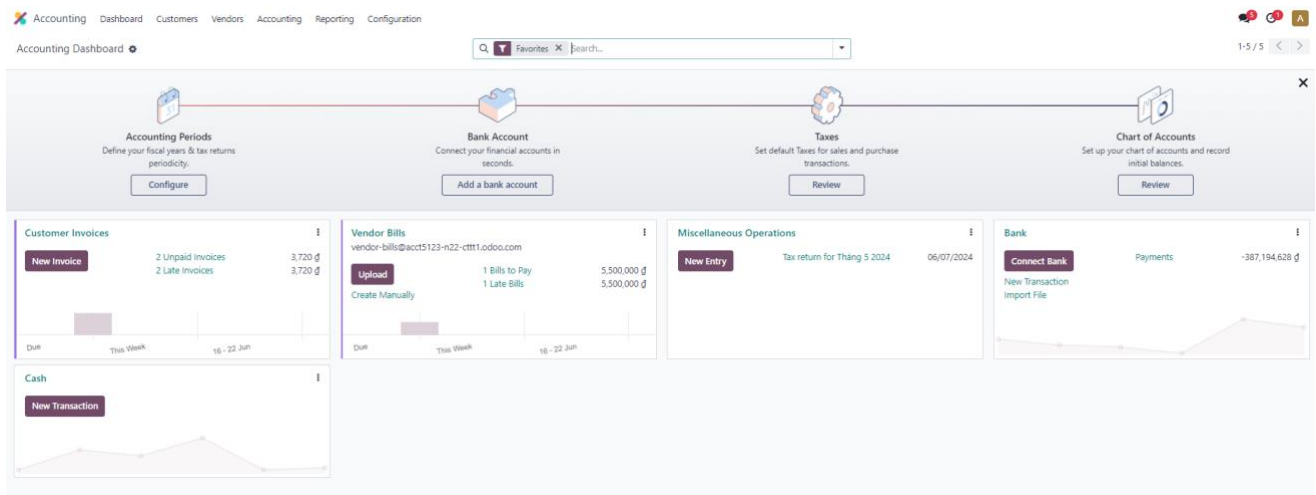


Figure 125: Image 3.11.3.1: Record debt.

Step 2: Create a supplier debt note and fill in the necessary information (supplier, address, product, price,....

Step 3: After filling in invoice information, select "Confirm"

Step 2 & 3 with the same image

ACCT5123 - Enterprise resource planning

Accounting

Dashboard

Customers

Vendors

Accounting

Reporting

Configuration

New

Accounting Dashboard

Draft Bill

Purchases

1

Confirm

Cancel

Draft

Posted

Vendor Bill

Draft

Vendor

Viewsonic

Bill Date

06/02/2024

Bill Reference

Accounting Date

06/02/2024

Auto-Complete ?

Select a purchase order or an old bill

Payment Reference ?

Recipient Bank ?

Due Date

06/02/2024

or

Payment Terms

Journal

Vendor Bills

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan,	P00018: [MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2- Speed Fan, 53.4 dB - 500 sq. ft. Coverage Evaporative Air Cooler	1561 Purchase costs	100.00	500,000.00	10%	50,000,000 g

Figure 126: Image 3.11.3.2: Record debt.

Step 4: Click “Register Payment” to create a payment method

[Accounting](#) [Dashboard](#) [Customers](#) [Vendors](#) [Accounting](#) [Reporting](#) [Configuration](#)

New

Accounting Dashboard

BILL/2024/06/0008

Purchases

1

Register Payment

Credit Note

Reset to Draft

Draft

Posted

Vendor Bill

BILL/2024/06/0008

Vendor

Viewsonic

Bill Reference

Bill Date

06/02/2024

Accounting Date

06/02/2024

Payment Reference ?

Recipient Bank ?

Due Date

06/02/2024

Journal

Vendor Bills

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan, 53.4 dB - 500 sq. ft. Coverage	P00018: [MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan, 53.4 dB - 500 sq. ft. Coverage	1561 Purchase costs	100.00	500,000.00	10%	50,000,000 đ

Figure 127: Image 3.9.3.3: Record debt.

Step 5: Click “Create Payment” to complete the payment procedure

Register Payment

X

Journal

Bank

Amount

55,000,000 đ

Payment Method ?

Manual

Payment Date

06/02/2024

Recipient Bank Account

Memo

BILL/2024/06/0008

Create Payment

Discard

Figure 128: Image 3.9.3.4: Payment method.

ACCT5123 - Enterprise resource planning

Accounting

Dashboard

Customers

Vendors

Accounting

Reporting

Configuration

New

Accounting Dashboard

BILL/2024/06/0008

Purchases

1

Credit Note

Reset to Draft

Draft

Posted

Vendor Bill

BILL/2024/06/0008

VendorViewsonic

Bill Date06/02/2024

Accounting Date06/02/2024

Payment Reference?

Recipient Bank?

Due Date06/02/2024

JournalVendor Bills

Invoice Lines

Journal Items

Other Info

Product	Label	Account	Quantity	Price	Taxes	Tax excl.
[MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan, 53.4 dB - 500 sq. ft. Coverage Evaporative Air Cooler Portable High Velocity Outdoor Cooling Fan by Hessaire -	P00018: [MC18] Portable Swamp Coolers - 1300 CFM MC18M Evaporative Air Cooler with 2-Speed Fan, 53.4 dB - 500 sq. ft. Coverage Evaporative Air Cooler Portable High Velocity Outdoor Cooling Fan by Hessaire -	1561 Purchase costs	100.00	500,000.00	10%	50,000,000 đ

Figure 129: Image 3.9.3.5: Record debt.

Step 6: Go to the “Accounting” module, select “Vendor Bills”

Accounting

Dashboard

Customers

Vendors

Accounting

Reporting

Configuration

Accounting Dashboard

Accounting Periods

Define your fiscal years & tax returns periodically.

Configure

Bank Account

Connect your financial accounts in seconds.

Add a bank account

Taxes

Set default Taxes for sales and purchase transactions.

Review

Chart of Accounts

Set up your chart of accounts and record initial balances.

Review

Customer Invoices

New Invoice

2 Unpaid Invoices 3,720 đ

2 Late Invoices 3,720 đ

Due This Week 16 - 22 Jun

Vendor Bills

vendor-bills@acct5123-n22-cttt1.odoo.com

Upload

1 Bills to Pay 5,500,000 đ

1 Late Bills 5,500,000 đ

Create Manually

Due This Week 16 - 22 Jun

Miscellaneous Operations

New Entry

Tax return for Tháng 5 2024 06/07/2024

Bank

Connect Bank

New Transaction Import File

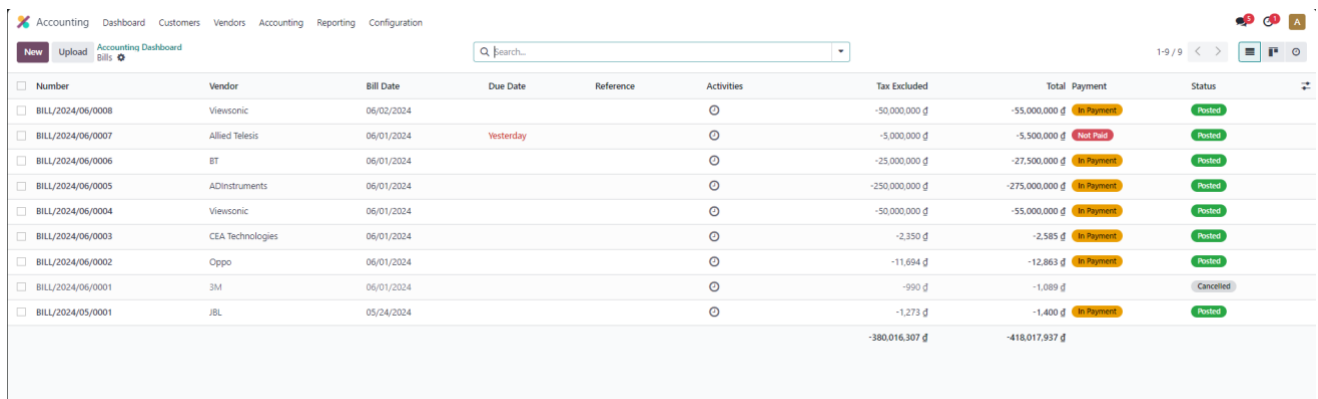
Payments -387,194,628 đ

Cash

New Transaction

Figure 130: Image 3.9.3.6: Accounting dashboard.

Step 7: At the accounting information panel, users can make statistics and observe information and invoice status from suppliers.



Number	Vendor	Bill Date	Due Date	Reference	Activities	Tax Excluded	Total Payment	Status
☐ BILL/2024/06/0008	Viewsonic	06/02/2024			○	-50,000,000 ₡	-55,000,000 ₡ In Payment	Posted
☐ BILL/2024/06/0007	Allied Telesis	06/01/2024	Yesterday		○	-5,000,000 ₡	-5,500,000 ₡ Not Paid	Posted
☐ BILL/2024/06/0006	BT	06/01/2024			○	-25,000,000 ₡	-27,500,000 ₡ In Payment	Posted
☐ BILL/2024/06/0005	ADInstruments	06/01/2024			○	-250,000,000 ₡	-275,000,000 ₡ In Payment	Posted
☐ BILL/2024/06/0004	Viewsonic	06/01/2024			○	-50,000,000 ₡	-55,000,000 ₡ In Payment	Posted
☐ BILL/2024/06/0003	CEA Technologies	06/01/2024			○	-2,350 ₡	-2,585 ₡ In Payment	Posted
☐ BILL/2024/06/0002	Oppo	06/01/2024			○	-11,694 ₡	-12,863 ₡ In Payment	Posted
☐ BILL/2024/06/0001	JM	06/01/2024			○	-990 ₡	-1,089 ₡ Cancelled	
☐ BILL/2024/05/0001	JBL	05/24/2024			○	-1,273 ₡	-1,400 ₡ In Payment	Posted
						-380,016,307 ₡	-418,017,937 ₡	

Figure 131: Image 3.9.3.1: Vendor record debt.

CHAPTER 4: CONCLUSIONS AND FUTURE DIRECTIONS

The final chapter summarizes the advantages and limitations of implementing an ERP system using Odoo. It also provides insights into potential future directions for further improving the system and expanding its capabilities.

4.1. Advantages

- Popular and practical topics
- Provide a business management system that integrates many different modules: CRM, Sales and Purchasing Management, Warehouse Management, ...
- Allows you to create and manage customer information, track the sales process from creating quotes, orders to payments, ...
- Having handled exceptions, it is possible to create automated rules and processes to handle special cases, such as processing cancellation orders, managing insufficient inventory, or processing payment status delays, etc.
- Using Odoo open-source software also saves investment costs compared to using commercial software.
- Provide automation functions and intelligent workflows, which saves time and effort for businesses.
- Meet all the Master's requirements for the project.
- The effort of team members to understand and unite during the project implementation.
- Members understand the knowledge of ERP in general and Odoo in particular.

4.2. Limitations

- Installing and setting up Odoo can be time-consuming and difficult, especially if you don't have much experience in system implementation.
- Due to time constraints, the team was unable to implement and test the Odoo system in action.
- There are important features in Odoo that the team has not explored or does not know how to use effectively.

- The project is only small and medium-sized projects, the data used may not reflect the reality of the business. This can lead to incorrect evaluation and analysis.
- Due to lack of experimentation, the processes in the Odoo system may have some flaws. Process testing and repair in a real-world environment is necessary to ensure efficient operation.

4.3. Future directions

In the future, Electronics Store – Kitchen Furniture Golden Star will focus on developing the online business segment to adapt to the fast growth trend of the current eCommerce market:

- To stand out and attract the attention of customers, Thanh Ty Electronics Store needs to carry out promotional programs to increase brand awareness. Online promotional activities, celebrity collaborations or media events can help strengthen a company's brand presence and recognition.
- To ensure customer satisfaction and loyalty, Thanh Ty Electronics Store needs to focus on improving customer service. This may include responding quickly to customer requests and complaints, providing technical support.
- Cooperate with e-commerce platforms to put products on the floors. This helps expand the market to reach customers, enhance sales capabilities and increase sales.
- To attract and retain customers, Thanh Ty Electronics Store can launch attractive promotions. Discounts, promotions, gifts, and rewards points can motivate customers to shop and increase sales.

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